Assessments for the year

Organisation. Each exam has two specimen papers for practice. They indicate the expected methods; they can be given at the start of the sequence, even if not all the notions have been taught yet. The table shows the planned time of distribution. Versions A and B are reserved for assessment.

Marks. The specimen papers keep four exercises worth 5 points. Each A/B has two new exercises worth 4 and 6 points and two retained exercises worth 5 points: total 20. The values are given question by question. For a 1-point question requiring a proof, award up to 0.5 points for a relevant method and up to 0.5 points for the justification and conclusion. For an induction proof marked out of 2 points: property and base case 0.5; hypothesis at a fixed index and inductive step calculation 1; quantified conclusion 0.5. For another value, adjust these proportions. Do not penalise a carried-forward error twice. The solutions write out the reasoning and the expected calculations; accept any correct method within the scope taught.

Duration. The 2 h written tests contain four short exercises; the 4 h mock exams contain four longer exercises. Adjust only after real feedback from the class. The distribution of points varies between exercises; the total remains 20. Timings should be adjusted according to observations in class.

Due date	Scheduled for	Specimen papers
DS1	S9 · week of 21 September	From S2
DS2	S18 · week of 12 October	From S14
Common test	S28 · week of 16 November; time slot to be coordinated	From S14
DS3	S34 · week of 30 November	From S21
Mock 1	S45 · 14-16 January, 4-hour slot to be scheduled	From S36
DS4	S57 · week of 8 February	From S44
DS5	S68 · week of 22 March	From S52
Mock 2	S71 · 1-3 April, 4-hour slot to be scheduled	From S59
Final mock exam	S90 · week of 31 May, 4-hour slot to be scheduled	From S75

DS1 · Specimen 1

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A discrete evolution — 5 points

Consider the sequence (u_n) defined by $u_0 = 1$ and, for every natural number n, $u_{n+1} = \frac{1}{2}u_n + 4$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Prove by induction that, for every $n \in \mathbb{N}$, $u_n = 8 - 7\left(\frac{1}{2}\right)^n$. (2 pt)

3. Deduce that, for every natural number n, $1 \leq u_n < 8$. (1 pt)

4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. A student claims: "There exists an integer n such that $u_n = 8$." Is this statement true? Justify. (0.5 pt)

Exercise 2 · Induction and bounds — 5 points

Consider the sequence (v_n) defined by $v_0 = 0$ and, for every natural number n, $v_{n+1} = \sqrt{72 + v_n}$.

1. Calculate v_1 and v_2. Give exact values. (0.5 pt)

2. Prove by induction that all terms are defined and that, for every $n \in \mathbb{N}$, $0 \leq v_n \leq 9$. (1.5 pt)

3. Prove by induction that, for every $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (1.5 pt)

4. Prove that, for every $n \in \mathbb{N}$, $v_n < 9$. (1 pt)

5. Consider a property $P(n)$, defined for $n \in \mathbb{N}$. We have proved $P(1)$ and, for all $n \geq 1$, the implication $P(n) \Rightarrow P(n+1)$. Do these two results allow us to state that $P(0)$ is true? Justify. (0.5 pt)

DS1 · Specimen 1 · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=4\vec{\imath}+5\vec{\jmath}+9\vec{k}, \quad \vec{z}=4\vec{\imath}+5\vec{\jmath}+10\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (1.5 pt)

4. Express the vector $5\vec{u}-4\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped ABCDEFGH non-degenerate. The vertices E, F, G and H are the images respectively of A, B, C and D under the translation by vector $\overrightarrow{AE}$. Set $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{11}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{11}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{11}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Prove that the lines (IJ) and (BD) are parallel. (1 pt)

3. Express $\overrightarrow{CG}$ in the basis, then determine whether $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are collinear. Justify. (1 pt)

4. Let O the centre of the parallelepiped. Prove that O lies on the line (AG). (1 pt)

5. Prove that the points I, J and K are not collinear. (1 pt)

Solutions · DS1 · Specimen 1

Exercise 1 · A discrete evolution

1. [0.5 pt] The relation is applied successively at indices 0 and 1:

$$u_1 = \frac{1}{2} \times 1 + 4 = \frac{9}{2}, \qquad u_2 = \frac{1}{2} \times \frac{9}{2} + 4 = \frac{25}{4}.$$

2. [2 pt] For $n \in \mathbb{N}$, let $P(n)$ the property $u_n = 8 - 7\left(\frac{1}{2}\right)^n$.

Base case. For $n = 0$, $8 - 7 \times 1 = 1 = u_0$. Hence $P(0)$ is true.

Inductive step. Fix an integer $n \geq 0$ and suppose $P(n)$ true. By the relation defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{2}u_n + 4 \\ &= \frac{1}{2}\left[8 - 7\left(\frac{1}{2}\right)^n\right] + 4 \\ &= 4 + 4 - 7\left(\frac{1}{2}\right)^{n+1} \\ &= 8 - 7\left(\frac{1}{2}\right)^{n+1}. \end{aligned}$$

This is $P(n+1)$: the inductive step holds.

Conclusion. The principle of induction gives $u_n = 8 - 7\left(\frac{1}{2}\right)^n$ for all $n \in \mathbb{N}$.

3. [1 pt] Since $0 < \frac{1}{2} < 1$, for every $n \in \mathbb{N}$ we have $0 < (\frac{1}{2})^n \leq 1$. Multiplication by $-7 < 0$ reverses the inequalities:

$$-7 \leq -7(\frac{1}{2})^n < 0.$$

Adding 8 and using the previous formula, we obtain $1 \leq u_n < 8$.

4. [1 pt] For every $n \in \mathbb{N}$, the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -7(\frac{1}{2})^{n+1} + 7(\frac{1}{2})^n \\ &= 7(1 - \frac{1}{2})(\frac{1}{2})^n = \frac{7}{2}(\frac{1}{2})^n > 0. \end{aligned}$$

The sequence is therefore strictly increasing.

5. [0.5 pt] The statement is false. For each integer n, $8 - u_n = 7(\frac{1}{2})^n > 0$. No term is therefore equal to 8, regardless of the number of steps.

Exercise 2 · Induction and bounds

1. [0.5 pt] v $_1 = \sqrt{72 + 0} = \sqrt{72}$ and $v_2 = \sqrt{72 + \sqrt{72}}$.

2. [1.5 pt] Let $P(n)$: "v_n is defined and $0 \leq v_n \leq 9$" .

Base case. $v_0 = 0$ is defined and belongs to $[0, 9]$; $P(0)$ is true.

Inductive step. Fix $n \in \mathbb{N}$ and suppose $P(n)$ true. Then

$$0 \leq 72 \leq 72 + v_n \leq 72 + 9 = 81.$$

The square root is therefore defined and, since this function is increasing on $[0, +\infty[$,

$$0 \leq v_{n+1} = \sqrt{72 + v_n} \leq \sqrt{81} = 9.$$

Thus $P(n+1)$ is true.

Conclusion. By induction, all terms are defined and belong to $[0, 9]$.

3. [1.5 pt] For $n \in \mathbb{N}$, let $R(n)$: "$v_{n+1} \geq v_n$" .

Base case. $v_1 = \sqrt{72} > 0 = v_0$; $R(0)$ is true.

Inductive step. Fix $n \geq 0$ and suppose $v_{n+1} \geq v_n$. Adding 72 and then using the fact that the square root is increasing,

$$v_{n+2} = \sqrt{72 + v_{n+1}} \geq \sqrt{72 + v_n} = v_{n+1}.$$

This is $R(n+1)$.

Conclusion. By induction, $v_{n+1} \geq v_n$ for all $n \geq 0$: (v_n) is increasing.

4. [1 pt] Let $S(n)$: "$v_n < 9$". At index 0, $v_0 = 0 < 9$. Fix $n \geq 0$ and suppose $S(n)$ true. Since radicands are positive, the strict increase of the square root gives

$$v_{n+1} = \sqrt{72 + v_n} < \sqrt{72 + 9} = \sqrt{81} = 9.$$

Hence $S(n+1)$ is true. The base case and the inductive step establish by induction that $v_n < 9$ for every $n \in \mathbb{N}$.

5. [0.5 pt] No. For example $P(n) : n \geq 1$ is true for all integers $n \geq 1$ but false at 0.

Exercise 3 · Vector calculation in a basis

1. [1 pt] In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, contradicting the uniqueness of the coordinates in a basis.

2. [0.5 pt] Expanding, $4\vec{u} + 5\vec{v} = 4(\vec{\imath} + \vec{k}) + 5(\vec{\jmath} + \vec{k}) = 4\vec{\imath} + 5\vec{\jmath} + 9\vec{k} = \vec{w}$.

3. [1.5 pt] Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis forces

$$\lambda = 4, \qquad \mu = 5, \qquad \lambda + \mu = 10.$$

The first two equalities give $\lambda + \mu = 9$, contradicting the third. The three vectors are therefore not coplanar.

4. [1 pt] $5\vec{u} - 4\vec{v} = 5(\vec{\imath} + \vec{k}) - 4(\vec{\jmath} + \vec{k}) = 5\vec{\imath} - 4\vec{\jmath} + \vec{k}$.

5. [1 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

Exercise 4 · Geometry of a parallelepiped

1. [1 pt] By Chasles, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. [1 pt] By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{11} = \frac{\overrightarrow{BD}}{11}$. These vectors are nonzero and collinear: (IJ) and (BD) are parallel.

3. [1 pt] We have $\overrightarrow{CG} = \overrightarrow{AE} = \vec{k}$ and $\overrightarrow{AB} = \vec{\imath}$. These two basis vectors are not collinear: a equality $\vec{k} = t\vec{\imath}$ would contradict the uniqueness of the decomposition in this basis.

4. [1 pt] The diagonals of a parallelepiped bisect each other. Thus $\overrightarrow{AO} = \frac{1}{2}\overrightarrow{AG}$, which proves $O \in (AG)$.

5. [1 pt] The points satisfy $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{11}$ & $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{11}$. If $\overrightarrow{IK} = t\,\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{11} = 0$, impossible. The vectors are not collinear, so I, J, K are not collinear.

DS1 · Specimen 2

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A discrete evolution — 5 points

Consider the sequence (u_n) defined by $u_0 = 2$ and, for every natural number n, $u_{n+1} = \frac{1}{5}u_n + \frac{36}{5}$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Prove by induction that, for every $n \in \mathbb{N}$, $u_n = 9 - 7\left(\frac{1}{5}\right)^n$. (2 pt)

3. Deduce that, for every natural number n, $2 \leq u_n < 9$. (1 pt)

4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. A student claims: "There exists an integer n such that $u_n = 9$." Is this statement true? Justify. (0.5 pt)

Exercise 2 · Induction and bounds — 5 points

Consider the sequence (v_n) defined by $v_0 = 0$ and, for every natural number n, $v_{n+1} = \sqrt{90 + v_n}$.

1. Calculate v_1 and v_2. Give exact values. (0.5 pt)

2. Prove by induction that all terms are defined and that, for every $n \in \mathbb{N}$, $0 \leq v_n \leq 10$. (1.5 pt)

3. Prove by induction that, for every $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (1.5 pt)

4. Prove that, for every $n \in \mathbb{N}$, $v_n < 10$. (1 pt)

5. Consider a property $P(n)$, defined for $n \in \mathbb{N}$. We have proved $P(1)$ and, for all $n \geq 1$, the implication $P(n) \Rightarrow P(n+1)$. Do these two results allow us to state that $P(0)$ is true? Justify. (0.5 pt)

DS1 · Specimen 2 · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=3\vec{\imath}+7\vec{\jmath}+10\vec{k}, \quad \vec{z}=3\vec{\imath}+7\vec{\jmath}+11\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (1.5 pt)

4. Express the vector $\vec{u}-3\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped ABCDEFGH non-degenerate. The vertices E, F, G and H are the images respectively of A, B, C and D under the translation by vector $\overrightarrow{AE}$. Set $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{18}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{18}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{18}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Prove that the lines (IJ) and (BD) are parallel. (1 pt)

3. Express $\overrightarrow{CG}$ in the basis, then determine whether $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are collinear. Justify. (1 pt)

4. Let O the centre of the parallelepiped. Prove that O lies on the line (AG). (1 pt)

5. Prove that the points I, J and K are not collinear. (1 pt)

Solutions · DS1 · Specimen 2

Exercise 1 · A discrete evolution

1. [0.5 pt] The relation is applied successively at indices 0 and 1:

$$u_1 = \frac{1}{5} \times 2 + \frac{36}{5} = \frac{38}{5}, \qquad u_2 = \frac{1}{5} \times \frac{38}{5} + \frac{36}{5} = \frac{218}{25}.$$

2. [2 pt] For $n \in \mathbb{N}$, let $P(n)$ the property $u_n = 9 - 7\left(\frac{1}{5}\right)^n$.

Base case. For $n = 0$, $9 - 7 \times 1 = 2 = u_0$. Hence $P(0)$ is true.

Inductive step. Fix an integer $n \geq 0$ and suppose $P(n)$ true. By the relation defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{5}u_n + \frac{36}{5} \\ &= \frac{1}{5}\left[9 - 7\left(\frac{1}{5}\right)^n\right] + \frac{36}{5} \\ &= \frac{9}{5} + \frac{36}{5} - 7\left(\frac{1}{5}\right)^{n+1} \\ &= 9 - 7\left(\frac{1}{5}\right)^{n+1}. \end{aligned}$$

This is $P(n+1)$: the inductive step holds.

Conclusion. The principle of induction gives $u_n = 9 - 7\left(\frac{1}{5}\right)^n$ for all $n \in \mathbb{N}$.

3. [1 pt] Since $0 < \frac{1}{5} < 1$, for every $n \in \mathbb{N}$ we have $0 < (\frac{1}{5})^n \leq 1$. Multiplication by $-7 < 0$ reverses the inequalities:

$$-7 \leq -7(\frac{1}{5})^n < 0.$$

Adding 9 and using the previous formula, we obtain $2 \leq u_n < 9$.

4. [1 pt] For every $n \in \mathbb{N}$, the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -7(\frac{1}{5})^{n+1} + 7(\frac{1}{5})^n \\ &= 7(1 - \frac{1}{5})(\frac{1}{5})^n = \frac{28}{5}(\frac{1}{5})^n > 0. \end{aligned}$$

The sequence is therefore strictly increasing.

5. [0.5 pt] The statement is false. For each integer n, $9 - u_n = 7(\frac{1}{5})^n > 0$. No term is therefore equal to 9, regardless of the number of steps.

Exercise 2 · Induction and bounds

1. [0.5 pt] v $_1 = \sqrt{90 + 0} = \sqrt{90}$ and $v_2 = \sqrt{90 + \sqrt{90}}$.

2. [1.5 pt] Let $P(n)$: "v_n is defined and $0 \leq v_n \leq 10$" .

Base case. $v_0 = 0$ is defined and belongs to $[0, 10]$; $P(0)$ is true.

Inductive step. Fix $n \in \mathbb{N}$ and suppose $P(n)$ true. Then

$$0 \leq 90 \leq 90 + v_n \leq 90 + 10 = 100.$$

The square root is therefore defined and, since this function is increasing on $[0, +\infty[$,

$$0 \leq v_{n+1} = \sqrt{90 + v_n} \leq \sqrt{100} = 10.$$

Thus $P(n+1)$ is true.

Conclusion. By induction, all terms are defined and belong to $[0, 10]$.

3. [1.5 pt] For $n \in \mathbb{N}$, let $R(n)$: "$v_{n+1} \geq v_n$" .

Base case. $v_1 = \sqrt{90} > 0 = v_0$; $R(0)$ is true.

Inductive step. Fix $n \geq 0$ and suppose $v_{n+1} \geq v_n$. Adding 90 and then using the fact that the square root is increasing,

$$v_{n+2} = \sqrt{90 + v_{n+1}} \geq \sqrt{90 + v_n} = v_{n+1}.$$

This is $R(n+1)$.

Conclusion. By induction, $v_{n+1} \geq v_n$ for all $n \geq 0$: (v_n) is increasing.

4. [1 pt] Let $S(n)$: "$v_n < 10$". At index 0, $v_0 = 0 < 10$. Fix $n \geq 0$ and suppose $S(n)$ true. Since the radicands are positive and the square root is strictly increasing, we get

$$v_{n+1} = \sqrt{90 + v_n} < \sqrt{90 + 10} = \sqrt{100} = 10.$$

Hence $S(n+1)$ is true. The base case and the inductive step establish by induction that $v_n < 10$ for every $n \in \mathbb{N}$.

5. [0.5 pt] No. For example $P(n) : n \geq 1$ is true for all integers $n \geq 1$ but false at 0.

Exercise 3 · Vector calculation in a basis

1. [1 pt] In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, contradicting the uniqueness of the coordinates in a basis.

2. [0.5 pt] Expanding, 3 $\vec{u} + 7\vec{v} = 3(\vec{\imath} + \vec{k}) + 7(\vec{\jmath} + \vec{k}) = 3\vec{\imath} + 7\vec{\jmath} + 10\vec{k} = \vec{w}$.

3. [1.5 pt] Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis forces

$$\lambda = 3, \qquad \mu = 7, \qquad \lambda + \mu = 11.$$

The first two equalities give $\lambda + \mu = 10$, contradicting the third. The three vectors are therefore not coplanar.

4. [1 pt] $7\vec{u} - 3\vec{v} = 7(\vec{\imath} + \vec{k}) - 3(\vec{\jmath} + \vec{k}) = 7\vec{\imath} - 3\vec{\jmath} + 4\vec{k}$.

5. [1 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

Exercise 4 · Geometry of a parallelepiped

1. [1 pt] By Chasles, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. [1 pt] By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{18} = \frac{\overrightarrow{BD}}{18}$. These vectors are nonzero and collinear: (IJ) and (BD) are parallel.

3. [1 pt] We have $\overrightarrow{CG} = \overrightarrow{AE} = \vec{k}$ and $\overrightarrow{AB} = \vec{\imath}$. These two basis vectors are not collinear: a equality $\vec{k} = t\vec{\imath}$ would contradict the uniqueness of the decomposition in this basis.

4. [1 pt] The diagonals of a parallelepiped bisect each other. Thus $\overrightarrow{AO} = \frac{1}{2}\overrightarrow{AG}$, which proves $O \in (AG)$.

5. [1 pt] The points satisfy $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{18}$ & $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{18}$. If $\overrightarrow{IK} = t\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{18} = 0$, impossible. The vectors are not collinear, so I, J, K are not collinear.

DS1 · A

PAPER RESERVED FOR ASSESSMENT

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A distribution to recover — 4 points

A quantity is modelled by $u_{n+1} = au_n + b$, for $n \in \mathbb{N}$. We observe $u_0 = 2$, $u_1 = 5$ and $u_2 = \frac{13}{2}$.

1. Determine a and b. (1 pt)

2. Prove by induction that $u_n = 8 - 6\left(\frac{1}{2}\right)^n$. (2 pts)

3. Deduce the variation of the sequence and an upper bound for it. (1 pt)

Exercise 2 · Induction and bounds — 5 points

Consider the sequence (v_n) defined by $v_0 = 0$ and, for every natural number n, $v_{n+1} = \sqrt{110 + v_n}$.

1. Calculate v_1 and v_2. Give exact values. (0.5 pt)

2. Prove by induction that all terms are defined and that, for every $n \in \mathbb{N}$, $0 \leq v_n \leq 11$. (1.5 pt)

3. Prove by induction that, for every $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (1.5 pt)

4. Prove that, for every $n \in \mathbb{N}$, $v_n < 11$. (1 pt)

5. Consider a property $P(n)$, defined for $n \in \mathbb{N}$. We have proved $P(1)$ and, for all $n \geq 1$, the implication $P(n) \Rightarrow P(n+1)$. Do these two results allow us to state that $P(0)$ is true? Justify. (0.5 pt)

DS1 · A · continued

Exercise 3 · A coefficient that changes coplanarity — 6 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$, let $u=\vec{\imath}+\vec{\jmath}$, $v=\vec{\jmath}+\vec{k}$ and $w_m=\vec{\imath}+m\vec{\jmath}+2\vec{k}$, for $m\in\mathbb{R}$.

1. Justify that u, v are not collinear. (1 pt)

2. Determine the values of m for which w_m is a combination of u, v. (2 pts)

3. A student claims that matching two coefficients is always enough. Explain the error by taking $m=0$. (2 pts)

4. Give the combination obtained for the admissible value of m. (1 pt)

Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped ABCDEFGH non-degenerate. The vertices E, F, G and H are the images respectively of A, B, C and D under the translation by vector $\overrightarrow{AE}$. Set $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{8}\vec{\imath},\qquad \overrightarrow{AJ}=\frac{1}{8}\vec{\jmath},\qquad \overrightarrow{AK}=\frac{1}{8}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Prove that the lines (IJ) and (BD) are parallel. (1 pt)

3. Express $\overrightarrow{CG}$ in the basis, then determine whether $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are collinear. Justify. (1 pt)

4. Let O the centre of the parallelepiped. Prove that O lies on the line (AG). (1 pt)

5. Prove that the points I, J and K are not collinear. (1 pt)

Solutions · DS1 · A

Exercise 1 · A distribution to recover

1. $2a+b=5$ and $5a+b=13/2$; the difference gives $a=1/2$, then $b=4$.

2. Let $P(n): u_n = 8-6(1/2)^n$. Base case: $8-6=2=u_0$. Fix $n\geq 0$ and suppose $P(n)$. Then $u_{n+1}=\frac{1}{2}[8-6(1/2)^n]+4=8-6(1/2)^{n+1}$. This is $P(n+1)$. By induction, the formula holds for all $n\in\mathbb{N}$. Marking: base case and statement 0.5; inductive step 1; conclusion 0.5.

3. $u_{n+1}-u_n=3(1/2)^n>0$; $u_n<8$ since $6(1/2)^n>0$. Two justifications, 0.5 point each.

Exercise 2 · Induction and bounds

1. [0.5 pt] v $_1=\sqrt{110+0}=\sqrt{110}$ and $v_2=\sqrt{110+\sqrt{110}}$.

2. [1.5 pt] Let $P(n)$: "v_n is defined and $0\leq v_n\leq 11$" .

   Base case. $v_0=0$ is defined and belongs to $[0,11]$; $P(0)$ is true.

   Inductive step. Fix $n\in\mathbb{N}$ and suppose $P(n)$ true. Then

   $$0\leq 110\leq 110+v_n\leq 110+11=121.$$

   The square root is therefore defined and, since this function is increasing on $[0,+\infty[$,

   $$0\leq v_{n+1}=\sqrt{110+v_n}\leq\sqrt{121}=11.$$

   Thus $P(n+1)$ is true.

   Conclusion. By induction, all terms are defined and belong to $[0,11]$.

3. [1.5 pt] For $n\in\mathbb{N}$, let $R(n)$: "$v_{n+1}\geq v_n$" .

   Base case. $v_1=\sqrt{110}>0=v_0$; $R(0)$ is true.

   Inductive step. Fix $n\geq 0$ and suppose $v_{n+1}\geq v_n$. Adding 110 and then using the fact that the square root is increasing,

   $$v_{n+2}=\sqrt{110+v_{n+1}}\geq\sqrt{110+v_n}=v_{n+1}.$$

   This is $R(n+1)$.

   Conclusion. By induction, $v_{n+1}\geq v_n$ for all $n\geq 0$: (v_n) is increasing.

4. [1 pt] Let $S(n)$: "$v_n<11$". At index 0, $v_0=0<11$. Fix $n\geq 0$ and suppose $S(n)$ true. Since the radicands are positive and the square root is strictly increasing, we get

   $$v_{n+1}=\sqrt{110+v_n}<\sqrt{110+11}=\sqrt{121}=11.$$

   Hence $S(n+1)$ is true. The base case and the inductive step establish by induction that $v_n<11$ for every $n\in\mathbb{N}$.

5. [0.5 pt] No. For example $P(n): n\geq 1$ is true for all integers $n\geq 1$ but false at 0.

Exercise 3 · A coefficient that changes coplanarity

1. In $v=tu$, the coefficient of $\vec{k}$ would give $1=0$.

2. $w_m=\alpha u+\beta v$ forces $\alpha=1$, $\beta=2$ and $m=\alpha+\beta=3$. This value works for all three coefficients. System 1 point, solving and checking 1 point.

3. For $m = 0$, the coefficients of $\vec{\imath}$ and $\vec{k}$ still impose 1 and 2, but the coefficient of $\vec{\jmath}$ would be 3 instead of 0. A contradiction in a single equation invalidates the combination. Calculation 1 point, explanation 1 point.

4. $w_3 = u + 2v$.

Exercise 4 · Geometry of a parallelepiped

1. [1 pt] By Chasles, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. [1 pt] By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{8} = \frac{\overrightarrow{BD}}{8}$. These vectors are nonzero and collinear: (IJ) and (BD) are parallel.

3. [1 pt] We have $\overrightarrow{CG} = \overrightarrow{AE} = \vec{k}$ and $\overrightarrow{AB} = \vec{\imath}$. These two basis vectors are not collinear: a equality $\vec{k} = t\vec{\imath}$ would contradict the uniqueness of the decomposition in this basis.

4. [1 pt] The diagonals of a parallelepiped bisect each other. Thus $\overrightarrow{AO} = \frac{1}{2}\overrightarrow{AG}$, which proves $O \in (AG)$.

5. [1 pt] The points satisfy $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{8}$ & $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{8}$. If $\overrightarrow{IK} = t\,\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{8} = 0$, impossible. The vectors are not collinear, so I, J, K are not collinear.

DS1 · B

PAPER RESERVED FOR ASSESSMENT

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Fixing an induction proof — 4 points

We define $u_0 = 1$ and $u_{n+1} = 3u_n + 2$ for $n \in \mathbb{N}$. A student proposes: " $u_n = 3^n$. It is true at index 0. If it is true at index n, then $u_{n+1} = 3^{n+1} + 2$, so it is true at the next index."

1. Identify the error precisely and give a term that refutes the formula. (1 pt)

2. Prove by induction that $u_n = 2 \times 3^n - 1$. (2 pts)

3. Determine the first index for which $u_n > 100$. (1 pt)

Exercise 2 · Induction and bounds — 5 points

Consider the sequence (v_n) defined by $v_0 = 0$ and, for every natural number n, $v_{n+1} = \sqrt{42 + v_n}$.

1. Calculate v_1 and v_2. Give exact values. (0.5 pt)

2. Prove by induction that all terms are defined and that, for every $n \in \mathbb{N}$, $0 \leq v_n \leq 7$. (1.5 pt)

3. Prove by induction that, for every $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (1.5 pt)

4. Prove that, for every $n \in \mathbb{N}$, $v_n < 7$. (1 pt)

5. Consider a property $P(n)$, defined for $n \in \mathbb{N}$. We have proved $P(1)$ and, for all $n \geq 1$, the implication $P(n) \Rightarrow P(n+1)$. Do these two results allow us to state that $P(0)$ is true? Justify. (0.5 pt)

DS1 · B · continued

Exercise 3 · Describe a vector construction — 6 points

We are given a basis $(\vec{\imath},\vec{\jmath},\vec{k})$. A tetrahedron $ABCD$ is defined by $\overrightarrow{AB}=\vec{\imath}$, $\overrightarrow{AC}=\vec{\jmath}$, $\overrightarrow{AD}=\vec{k}$. We place I, J by $\overrightarrow{AI}=2\vec{\imath}+\vec{\jmath}$ and $\overrightarrow{AJ}=\vec{\jmath}+\vec{k}$.

1. Calculate $\overrightarrow{IJ}$. (1 pt)

2. Are the vectors $\overrightarrow{AI}, \overrightarrow{AJ}, \overrightarrow{AD}$ coplanar? (2 pts)

3. Define K by $\overrightarrow{AK}=2\overrightarrow{AJ}-\overrightarrow{AI}$. Show that I, J, K are collinear. (2 pts)

4. State which point is the midpoint of the other two. (1 pt)

Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped ABCDEFGH non-degenerate. The vertices E, F, G and H are the images respectively of A, B, C and D under the translation by vector $\overrightarrow{AE}$. Set $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{15}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{15}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{15}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Prove that the lines (IJ) and (BD) are parallel. (1 pt)

3. Express $\overrightarrow{CG}$ in the basis, then determine whether $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are collinear. Justify. (1 pt)

4. Let O the centre of the parallelepiped. Prove that O lies on the line (AG). (1 pt)

5. Prove that the points I, J and K are not collinear. (1 pt)

Solutions · DS1 · B

Exercise 1 · Fixing an induction proof

1. The result of the inductive step is not the statement at index $n+1$: the term $+2$ remains. $u_1 = 5 \neq 3$.

2. Let $P(n)$: $u_n = 2 \times 3^n - 1$. At index 0, $2 \quad -1 = 1$. Suppose $P(n)$ for some $n \geq 0$; $u_{n+1} = 3(2 \times 3^n - 1) + 2 = 2 \times 3^{n+1} - 1$. By induction, $P(n)$ holds for every index. Marking scheme: 0.5; 1; 0.5 for the three steps.

3. $u_3 = 53 \leq 100 < u_4 = 161$. The formula shows that u_n is strictly increasing, so the first index is 4.

Exercise 2 · Induction and bounds

1. [0.5 pt] v $_1 = \sqrt{42+0} = \sqrt{42}$ and $v_2 = \sqrt{42+\sqrt{42}}$.

2. [1.5 pt] Let $P(n)$: "v_n is defined and $0 \leq v_n \leq 7$" .

   Base case. $v_0 = 0$ is defined and belongs to $[0,7]$; $P(0)$ is true.

   Inductive step. Fix $n \in \mathbb{N}$ and suppose $P(n)$ true. Then

   $$0 \leq 42 \leq 42 + v_n \leq 42 + 7 = 49.$$

   The square root is therefore defined and, since this function is increasing on $[0, +\infty[$,

   $$0 \leq v_{n+1} = \sqrt{42+v_n} \leq \sqrt{49} = 7.$$

   Thus $P(n+1)$ is true.

   Conclusion. By induction, all terms are defined and belong to $[0,7]$.

3. [1.5 pt] For $n \in \mathbb{N}$, let $R(n)$: "$v_{n+1} \geq v_n$" .

   Base case. $v_1 = \sqrt{42} > 0 = v_0$; $R(0)$ is true.

   Inductive step. Fix $n \geq 0$ and suppose $v_{n+1} \geq v_n$. Adding 42 and then using the fact that the square root is increasing,

   $$v_{n+2} = \sqrt{42+v_{n+1}} \geq \sqrt{42+v_n} = v_{n+1}.$$

   This is $R(n+1)$.

   Conclusion. By induction, $v_{n+1} \geq v_n$ for all $n \geq 0$: (v_n) is increasing.

4. [1 pt] Let $S(n)$: "$v_n < 7$". At index 0, $v_0 = 0 < 7$. Fix $n \geq 0$ and suppose $S(n)$ true. Since radicands are positive and the square root is strictly increasing, we get

   $$v_{n+1} = \sqrt{42+v_n} < \sqrt{42+7} = \sqrt{49} = 7.$$

   Hence $S(n+1)$ is true. The base case and the inductive step establish by induction that $v_n < 7$ for every $n \in \mathbb{N}$.

5. [0.5 pt] No. For example $P(n) : n \geq 1$ is true for all integers $n \geq 1$ but false at 0.

Exercise 3 · Describe a vector construction

1. By Chasles' relation, $\overrightarrow{IJ} = -2\vec{\imath} + \vec{k}$.

2. AI, AJ are not collinear (coefficients of $\vec{\imath}$ and $\vec{k}$). In $\vec{k} = \alpha(2\vec{\imath} + \vec{\jmath}) + \beta(\vec{\jmath} + \vec{k})$, the coefficients require $\alpha = 0$, $\beta = 1$ and $\alpha + \beta = 0$, contradiction. Non-coplanarity. 1 point for the system, 1 for the contradiction.

3. $\overrightarrow{IK} = \overrightarrow{AK} - \overrightarrow{AI} = 2(\overrightarrow{AJ} - \overrightarrow{AI}) = 2\overrightarrow{IJ}$. This vector is nonzero; the three points are collinear. Calculation 1 point, justified conclusion 1 point.

4. J is the midpoint of $[IK]$ since $\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{IK}$.

Exercise 4 · Geometry of a parallelepiped

1. [1 pt] By Chasles, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. [1 pt] By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{15} = \frac{\overrightarrow{BD}}{15}$. These vectors are nonzero and collinear: (IJ) and (BD) are parallel.

3. [1 pt] We have $\overrightarrow{CG} = \overrightarrow{AE} = \vec{k}$ and $\overrightarrow{AB} = \vec{\imath}$. These two basis vectors are not collinear: a equality $\vec{k} = t\vec{\imath}$ would contradict the uniqueness of the decomposition in this basis.

4. [1 pt] The diagonals of a parallelepiped bisect each other. Thus $\overrightarrow{AO} = \frac{1}{2}\overrightarrow{AG}$, which proves $O \in (AG)$.

5. [1 pt] The points satisfy $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{15}$ & $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{15}$. If $\overrightarrow{IK} = t\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{15} = 0$, impossible. The vectors are not collinear, so I, J, K are not collinear.

DS2 · Specimen 1

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{3n^2+4n-1}{4n^2+3}$ & $v_n = \sqrt{n^2+4n} - n$.

1. Determine the limit of (u_n) showing how its expression is transformed. (1 pt)

2. Show that, for every integer $n \geq 1$, $v_n = \frac{4}{\sqrt{1+\frac{4}{n}}+1}$. (1 pt)

3. Deduce the limit of (v_n). (1 pt)

4. Define $w_n = 3n^2 - 4n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this statement true? Justify. (1 pt)

Exercise 2 · True or false: reasoning about limits — 5 points

For each of the following statements, indicate whether it is true or false and justify the answer. An unjustified answer earns no marks. Limits of sequences are taken as n tends to $+\infty$.

1. If $u_n \to 2$ and $v_n \to -5$, then $\frac{u_n}{v_n} \to -\frac{2}{5}$. (1 pt)

2. The sequence $u_n = 2 + (-1)^n$ converges to 2. (1 pt)

3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. (1 pt)

4. The sequence $w_n = \frac{\sqrt{n^2+3n}}{n}$ for $n \geq 1$ converges to 1. (1 pt)

5. For $u_n = \frac{2n+5}{n}$, we have $|u_n - 2| < 0.01$ as soon as $n > 500$. (1 pt)

DS2 · Specimen 1 · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=2\vec{\imath}+6\vec{\jmath}+8\vec{k}, \quad \vec{z}=2\vec{\imath}+6\vec{\jmath}+9\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (1.5 pt)

4. Express the vector 6 $\vec{u}-2\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped ABCDEFGH non-degenerate. The vertices E, F, G and H are the images respectively of A, B, C and D under the translation by vector $\overrightarrow{AE}$. Set $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{18}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{18}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{18}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Prove that the lines (IJ) and (BD) are parallel. (1 pt)

3. Study the relative position of the lines (AB) and (CG) : are they parallel, intersecting or not coplanar? Justify. (1 pt)

4. Let O the parallelepiped's centre. Does point O lie in the plane (ABG) ? Justify. (1 pt)

5. Prove that the points I, J and K are not collinear. (1 pt)

Solutions · DS2 · Specimen 1

Exercise 1 · Limits calculated and justified

1. [1 pt] For $n \geq 1$, we divide by $n^2 \neq 0$:

$$u_n = \frac{3 + \frac{4}{n} - \frac{1}{n^2}}{4 + \frac{3}{n^2}}.$$

The terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 3 and the denominator to 4 $\neq 0$; by quotient, $u_n \to \frac{3}{4}$.

2. [1 pt] The conjugate is positive. For $n \geq 1$, $\sqrt{n^2} = n$, hence

$$v_n = \frac{(n^2 + 4n) - n^2}{\sqrt{n^2 + 4n} + n} = \frac{4n}{n(\sqrt{1 + \frac{4}{n}} + 1)} = \frac{4}{\sqrt{1 + \frac{4}{n}} + 1}.$$

3. [1 pt] The denominator tends to 2; limit 2.

4. [1 pt] w $_n = n^2(3 - \frac{4}{n}) \to +\infty$; the second factor tends to 3 > 0.

5. [1 pt] No: $(2n) - n = n \to +\infty$. The form $\infty - \infty$ is an indeterminate form, not a calculation rule.

Exercise 2 · True or false: reasoning about limits

1. [1 pt] True: the denominator has a non-zero limit; the limit of the quotient is $-\frac{2}{5}$.

2. [1 pt] False: the even-index terms equal 3, the odd-index terms 1. The two values remain at distance 1 of 2.

3. [1 pt] False: $u_n = \frac{1}{n}$, $v_n = n$ for $n \geq 1$ give a constant product equal to 1.

4. [1 pt] True: $w_n = \sqrt{1 + \frac{3}{n}} \to 1$.

5. [1 pt] True: the difference is $\frac{5}{n}$; $n > 500$ gives $\frac{5}{n} < 0{,}01$.

Exercise 3 · Vector calculation in a basis

1. [1 pt] In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, contradicting the uniqueness of the coordinates in a basis.

2. [0.5 pt] Expanding, $2\vec{u} + 6\vec{v} = 2(\vec{\imath} + \vec{k}) + 6(\vec{\jmath} + \vec{k}) = 2\vec{\imath} + 6\vec{\jmath} + 8\vec{k} = \vec{w}$.

3. [1.5 pt] Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis forces

$$\lambda = 2, \qquad \mu = 6, \qquad \lambda + \mu = 9.$$

The first two equalities give $\lambda + \mu = 8$, contradicting the third. The three vectors are therefore not coplanar.

4. [1 pt] $6\vec{u} - 2\vec{v} = 6(\vec{\imath} + \vec{k}) - 2(\vec{\jmath} + \vec{k}) = 6\vec{\imath} - 2\vec{\jmath} + 4\vec{k}$.

5. [1 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

Exercise 4 · Geometry of a parallelepiped

1. [1 pt] By Chasles, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. [1 pt] By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{18} = \frac{\overrightarrow{BD}}{18}$. These vectors are nonzero and collinear: (IJ) and (BD) are parallel.

3. [1 pt] Directions $\vec{\imath}$ and $\vec{k}$, not collinear. An intersection would require $t\vec{\imath} = \vec{\imath} + \vec{\jmath} + s\vec{k}$, impossible because the coefficient of $\vec{\jmath}$. They are not coplanar.

4. [1 pt] $\overrightarrow{AO} = \frac{\overrightarrow{AG}}{2}$, therefore $O \in (AG) \subset (ABG)$.

5. [1 pt] The points satisfy $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{18}$ & $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{18}$. If $\overrightarrow{IK} = t\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{18} = 0$, impossible. The vectors are not collinear, so I, J, K are not collinear.

DS2 · Specimen 2

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{2n^2+6n-1}{3n^2+1}$ & $v_n = \sqrt{n^2+6n} - n$.

1. Determine the limit of (u_n) showing how its expression is transformed. (1 pt)

2. Show that, for every integer $n \geq 1$, $v_n = \frac{6}{\sqrt{1+\frac{6}{n}}+1}$. (1 pt)

3. Deduce the limit of (v_n). (1 pt)

4. Define $w_n = 2n^2 - 6n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this statement true? Justify. (1 pt)

Exercise 2 · True or false: reasoning about limits — 5 points

For each of the following statements, indicate whether it is true or false and justify the answer. An unjustified answer earns no marks. Limits of sequences are taken as n tends to $+\infty$.

1. If $u_n \to 5$ and $v_n \to -6$, then $\frac{u_n}{v_n} \to -\frac{5}{6}$. (1 pt)

2. The sequence $u_n = 5 + (-1)^n$ converges to 5. (1 pt)

3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. (1 pt)

4. The sequence $w_n = \frac{\sqrt{n^2+n}}{n}$ for $n \geq 1$ converges to 1. (1 pt)

5. For $u_n = \frac{5n+6}{n}$, we have $|u_n - 5| < 0.01$ as soon as $n > 600$. (1 pt)

DS2 · Specimen 2 · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=4\vec{\imath}+7\vec{\jmath}+11\vec{k}, \quad \vec{z}=4\vec{\imath}+7\vec{\jmath}+12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (1.5 pt)

4. Express the vector 7 $\vec{u}-4\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped ABCDEFGH non-degenerate. The vertices E, F, G and H are the images respectively of A, B, C and D under the translation by vector $\overrightarrow{AE}$. Set $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{8}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{8}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{8}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Prove that the lines (IJ) and (BD) are parallel. (1 pt)

3. Study the relative position of the lines (AB) and (CG) : are they parallel, intersecting or not coplanar? Justify. (1 pt)

4. Let O the parallelepiped's centre. Does point O lie in the plane (ABG) ? Justify. (1 pt)

5. Prove that the points I, J and K are not collinear. (1 pt)

Solutions · DS2 · Specimen 2

Exercise 1 · Limits calculated and justified

1. [1 pt] For $n \geq 1$, we divide by $n^2 \neq 0$:

$$u_n = \frac{2 + \frac{6}{n} - \frac{1}{n^2}}{3 + \frac{1}{n^2}}.$$

The terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 2 and the denominator to 3 $\neq 0$; by quotient, $u_n \to \frac{2}{3}$.

2. [1 pt] The conjugate is positive. For $n \geq 1$, $\sqrt{n^2} = n$, hence

$$v_n = \frac{(n^2 + 6n) - n^2}{\sqrt{n^2 + 6n} + n} = \frac{6n}{n(\sqrt{1 + \frac{6}{n}} + 1)} = \frac{6}{\sqrt{1 + \frac{6}{n}} + 1}.$$

3. [1 pt] The denominator tends to 2; limit 3.

4. [1 pt] w $_n = n^2(2 - \frac{6}{n}) \to +\infty$; the second factor tends to 2 > 0.

5. [1 pt] No: $(2n) - n = n \to +\infty$. The form $\infty - \infty$ is an indeterminate form, not a calculation rule.

Exercise 2 · True or false: reasoning about limits

1. [1 pt] True: the denominator has a non-zero limit; the limit of the quotient is $-\frac{5}{6}$.

2. [1 pt] False: the even-index terms equal 6, the odd-index terms 4. The two values remain at distance 1 of 5.

3. [1 pt] False: $u_n = \frac{1}{n}$, $v_n = n$ for $n \geq 1$ give a constant product equal to 1.

4. [1 pt] True: $w_n = \sqrt{1 + \frac{1}{n}} \to 1$.

5. [1 pt] True: the difference is $\frac{6}{n}$; $n > 600$ gives $\frac{6}{n} < 0{,}01$.

Exercise 3 · Vector calculation in a basis

1. [1 pt] In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, contradicting the uniqueness of the coordinates in a basis.

2. [0.5 pt] Expanding, $4\vec{u} + 7\vec{v} = 4(\vec{\imath} + \vec{k}) + 7(\vec{\jmath} + \vec{k}) = 4\vec{\imath} + 7\vec{\jmath} + 11\vec{k} = \vec{w}$.

3. [1.5 pt] Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis forces

$$\lambda = 4, \qquad \mu = 7, \qquad \lambda + \mu = 12.$$

The first two equalities give $\lambda + \mu = 11$, contradicting the third. The three vectors are therefore not coplanar.

4. [1 pt] $7\vec{u} - 4\vec{v} = 7(\vec{\imath} + \vec{k}) - 4(\vec{\jmath} + \vec{k}) = 7\vec{\imath} - 4\vec{\jmath} + 3\vec{k}$.

5. [1 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

Exercise 4 · Geometry of a parallelepiped

1. [1 pt] By Chasles, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. [1 pt] By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{8} = \frac{\overrightarrow{BD}}{8}$. These vectors are nonzero and collinear: (IJ) and (BD) are parallel.

3. [1 pt] Directions $\vec{\imath}$ and $\vec{k}$, not collinear. An intersection would require $t\vec{\imath} = \vec{\imath} + \vec{\jmath} + s\vec{k}$, impossible because the coefficient of $\vec{\jmath}$. They are not coplanar.

4. [1 pt] $\overrightarrow{AO} = \frac{\overrightarrow{AG}}{2}$, therefore $O \in (AG) \subset (ABG)$.

5. [1 pt] The points satisfy $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{8}$ & $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{8}$. If $\overrightarrow{IK} = t\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{8} = 0$, impossible. The vectors are not collinear, so I, J, K are not collinear.

DS2 · A

PAPER RESERVED FOR ASSESSMENT

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A precision expressed as a band — 4 points

For $n \in \mathbb{N}$, set $u_n = 4 + \frac{6}{n+2}$.

1. Calculate the limit and justify why u_n never equals this limit. (1 pt)

2. Find the first index from which $|u_n - 4| < 0.01$. Check the bound. (2 pts)

3. Correctly negate the sentence "all the terms are strictly less than 5", then give a example. (1 pt)

Exercise 2 · True or false: reasoning about limits — 5 points

For each of the following statements, indicate whether it is true or false and justify the answer. An unjustified answer earns no marks. Limits of sequences are taken as n tends to $+\infty$.

1. If $u_n \to 4$ and $v_n \to -8$, then $\frac{u_n}{v_n} \to -\frac{1}{2}$. (1 pt)

2. The sequence $u_n = 4 + (-1)^n$ converges to 4. (1 pt)

3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. (1 pt)

4. The sequence $w_n = \frac{\sqrt{n^2+2n}}{n}$ for $n \geq 1$ converges to 1. (1 pt)

5. For $u_n = \frac{4n+8}{n}$, we have $|u_n - 4| < 0.01$ as soon as $n > 800$. (1 pt)

DS2 · A · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=3\vec{\imath}+4\vec{\jmath}+7\vec{k}, \quad \vec{z}=3\vec{\imath}+4\vec{\jmath}+8\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (1.5 pt)

4. Express the vector 4 $\vec{u}-3\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (1 pt)

Exercise 4 · The intersection: seen or proved — 6 points

In a coordinate system of space, $A(0;0;0)$, $B(4;0;0)$, $C(0;4;0)$, $D(0;0;4)$. Set $I(2;0;0)$, $J(0;2;0)$, $K(0;0;2)$ and $G(1;1;1)$.

1. Show that I, J, K are not collinear. (1 pt)

2. Determine whether G lies in the plane (IJK) by a linear combination. (2 pts)

3. Are the lines (IJ) and (BC) strictly parallel? (2 pts)

4. Explain why the projection of G inside the drawn triangle would not be enough to prove that it lies in the plane. (1 pt)

Solutions · DS2 · A

Exercise 1 · A precision expressed as a band

1. $6/(n+2) \to 0$, therefore $u_n \to 4$; the quotient is always strictly positive.

2. $6/(n+2) < 1/100$ is equivalent to $n+2 > 600$, i.e. $n \geq 599$. At index 598, the gap is 0 ,01 ; at index 599 it is strictly smaller, and it decreases afterwards. Solving 1 point, minimality 1 point.

3. "There exists an index n such that $u_n \geq 5$". For example $u_0 = 7$.

Exercise 2 · True or false: reasoning about limits

1. [1 pt] True: the denominator has a non-zero limit; the limit of the quotient is $-\frac{1}{2}$.

2. [1 pt] False: the even-index terms equal 5, the odd-index terms 3. The two values remain at distance 1 of 4.

3. [1 pt] False: $u_n = \frac{1}{n}$, $v_n = n$ for $n \geq 1$ give a constant product equal to 1.

4. [1 pt] True: $w_n = \sqrt{1+\frac{2}{n}} \to 1$.

5. [1 pt] True: the difference is $\frac{8}{n}$; $n > 800$ gives $\frac{8}{n} < 0{,}01$.

Exercise 3 · Vector calculation in a basis

1. [1 pt] In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, contradicting the uniqueness of the coordinates in a basis.

2. [0.5 pt] Expanding, 3 $\vec{u} + 4\vec{v} = 3(\vec{\imath}+\vec{k}) + 4(\vec{\jmath}+\vec{k}) = 3\vec{\imath}+4\vec{\jmath}+7\vec{k} = \vec{w}$.

3. [1.5 pt] Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis forces

$$\lambda = 3, \qquad \mu = 4, \qquad \lambda + \mu = 8.$$

   The first two equalities give $\lambda + \mu = 7$, contradicting the third. The three vectors are therefore not coplanar.

4. [1 pt] $4\vec{u} - 3\vec{v} = 4(\vec{\imath}+\vec{k}) - 3(\vec{\jmath}+\vec{k}) = 4\vec{\imath} - 3\vec{\jmath} + \vec{k}$.

5. [1 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

Exercise 4 · Seen or proved intersection

1. $IJ = (-2;2;0)$ and $IK = (-2;0;2)$ are not collinear, by their second and third coordinates.

2. $IG = (-1;1;1)$. An equality $IG = sIJ + tIK$ imposes $s = t = 1/2$ but the first coordinate would be $-2 \neq -1$. Hence $G \notin (IJK)$. 1 point for the system, 1 for the contradiction.

3. $BC = (-4;4;0) = 2IJ$. In addition $BI = (-2;0;0)$ is not collinear with BC : $I \notin (BC)$. Both lines are distinct and parallel. 1 point for each condition.

4. Projection loses depth; several points in space can have the same image on the sheet.

DS2 · B

PAPER RESERVED FOR ASSESSMENT

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · When two sequences become almost indistinguishable — 4 points

For $n \geq 1$, $a_n = \frac{3n+1}{n+2}$ & $b_n = \frac{3n-4}{n+2}$.

1. Calculate their limits. (1 pt)

2. Solve $|a_n - b_n| < 0.05$ and give the first admissible index. (2 pts)

3. Disprove "two sequences with the same limit have equal terms from some index onwards". (1 pt)

Exercise 2 · True or false: reasoning about limits — 5 points

For each of the following statements, indicate whether it is true or false and justify the answer. An unjustified answer earns no marks. Limits of sequences are taken as n tends to $+\infty$.

1. If $u_n \to 3$ and $v_n \to -5$, then $\frac{u_n}{v_n} \to -\frac{3}{5}$. (1 pt)

2. The sequence $u_n = 3 + (-1)^n$ converges to 3. (1 pt)

3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. (1 pt)

4. The sequence $w_n = \frac{\sqrt{n^2+3n}}{n}$ for $n \geq 1$ converges to 1. (1 pt)

5. For $u_n = \frac{3n+5}{n}$, we have $|u_n - 3| < 0.01$ as soon as $n > 500$. (1 pt)

DS2 · B · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, we define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 3\vec{\imath} + 6\vec{\jmath} + 9\vec{k}, \quad \vec{z} = 3\vec{\imath} + 6\vec{\jmath} + 10\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (1.5 pt)

4. Express the vector $\vec{u} - 3\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Two descriptions of the same plane — 6 points

In a coordinate system, $A(1;0;0)$, $B(1;1;0)$, $C(1;0;1)$, $D(1;2;-1)$, $M(2;2;-1)$.

1. Show that A, B, C define a plane. (1 pt)

2. Show that D lies in this plane and find the coefficients of the combination used. (2 pts)

3. Can one replace C by D without changing the plane? Justify. (2 pts)

4. Is point M lie in the plane? (1 pt)

Solutions · DS2 · B

Exercise 1 · When two sequences become almost indistinguishable

1. By dividing by n, each tends to 3 (0.5 point each).

2. $a_n - b_n = 5/(n+2) > 0$. The inequality is equivalent to $n+2 > 100$, so $n \geq 99$. At index 98 the gap is exactly 0.05. Calculating the gap 1 point, solving and bound 1 point.

3. These two sequences have the same limit but their difference 5 $/(n+2)$ is never zero: they form a complete counterexample.

Exercise 2 · True or false: reasoning about limits

1. [1 pt] True: the denominator has a non-zero limit; the limit of the quotient is $-\frac{3}{5}$.

2. [1 pt] False: the even-index terms equal 4, the odd-index terms 2. The two values remain at distance 1 of 3.

3. [1 pt] False: $u_n = \frac{1}{n}$, $v_n = n$ for $n \geq 1$ give a constant product equal to 1.

4. [1 pt] True: $w_n = \sqrt{1+\frac{3}{n}} \to 1$.

5. [1 pt] True: the difference is $\frac{5}{n}$; $n > 500$ gives $\frac{5}{n} < 0{,}01$.

Exercise 3 · Vector calculation in a basis

1. [1 pt] In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, contradicting the uniqueness of the coordinates in a basis.

2. [0.5 pt] Expanding, 3 $\vec{u} + 6\vec{v} = 3(\vec{\imath}+\vec{k}) + 6(\vec{j}+\vec{k}) = 3\vec{\imath} + 6\vec{j} + 9\vec{k} = \vec{w}$.

3. [1.5 pt] Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis forces

$$\lambda = 3, \qquad \mu = 6, \qquad \lambda + \mu = 10.$$

   The first two equalities give $\lambda + \mu = 9$, contradicting the third. The three vectors are therefore not coplanar.

4. [1 pt] $6\vec{u} - 3\vec{v} = 6(\vec{\imath}+\vec{k}) - 3(\vec{j}+\vec{k}) = 6\vec{\imath} - 3\vec{j} + 3\vec{k}$.

5. [1 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

Exercise 4 · Two descriptions of the same plane

1. $AB = (0;1;0)$ and $AC = (0;0;1)$ are not collinear.

2. $AD = (0;2;-1) = 2AB - AC$. The combination proves $D \in (ABC)$. Coefficients 1 point, justification tion 1 point.

3. AB, AD are not collinear since AD has a non-zero third coordinate. Moreover $AC = 2AB - AD$. The two pairs span the same plane through A. 1 point for each argument.

4. $AM = (1;2;-1)$ has a non-zero first coordinate, whereas any combination of AB, AC has first coordinate zero. So no.

Common test · Specimen 1

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A discrete evolution — 5 points

Consider the sequence (u_n) defined by $u_0 = 2$ and, for every natural number n, $u_{n+1} = \frac{1}{4}u_n + 6$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Prove by induction that, for every $n \in \mathbb{N}$, $u_n = 8 - 6\left(\frac{1}{4}\right)^n$. (2 pt)

3. Deduce that, for every natural number n, $2 \leq u_n < 8$. (1 pt)

4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. A student claims: "There exists an integer n such that $u_n = 8$." Is this statement true? Justify. (0.5 pt)

Exercise 2 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{3n^2+5n-1}{4n^2+2}$ & $v_n = \sqrt{n^2+5n} - n$.

1. Determine the limit of (u_n) showing how its expression is transformed. (1 pt)

2. Show that, for every integer $n \geq 1$, $v_n = \frac{5}{\sqrt{1+\frac{5}{n}}+1}$. (1 pt)

3. Deduce the limit of (v_n). (1 pt)

4. Define $w_n = 3n^2 - 5n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this statement true? Justify. (1 pt)

Common test · Specimen 1 · continued

Exercise 3 · Limits and asymptotes — 5 points

Consider the function f defined by $f(x) = \frac{2x+6}{x-2}$. Let $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the domain of f. Find two reals A and B such that, for all $x \neq 2$, $f(x) = A + \frac{B}{x-2}$. (1 pt)
2. Determine the limits of f at $-\infty$ and at $+\infty$. (1 pt)
3. Determine the limits of f at 2^- and at 2^+. (1 pt)
4. Give the equations of the asymptotes to $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. (1 pt)
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^2 e^{-2x}$ with justification. (1 pt)

Exercise 4 · Codes and selections — 5 points

A code is an ordered sequence of 4 symbols chosen from an alphabet that contains 15 distinct symbols. The rules concerning repetitions are specified in each question, independently of the other questions.

1. How many codes can be formed if repetitions are allowed? (1 pt)
2. How many codes can be formed if each symbol is to be used at most once? (1 pt)
3. One symbol of the alphabet is fixed in advance. With repetition allowed, how many codes contain this symbol at least once? (1 pt)
4. The first symbol of the code is fixed and repetitions are not allowed. How many codes can be formed? (1 pt)
5. Two distinct symbols of the alphabet are forbidden. The other symbols may be repeated. How many codes can be formed? (1 pt)

Solutions · Common test · Specimen 1

Exercise 1 · A discrete evolution

1. [0.5 pt] The relation is applied successively at indices 0 and 1:

$$u_1 = \frac{1}{4} \times 2 + 6 = \frac{13}{2}, \qquad u_2 = \frac{1}{4} \times \frac{13}{2} + 6 = \frac{61}{8}.$$

2. [2 pt] For $n \in \mathbb{N}$, let $P(n)$ the property $u_n = 8 - 6\left(\frac{1}{4}\right)^n$.

Base case. For $n = 0$, $8 - 6 \times 1 = 2 = u_0$. Hence $P(0)$ is true.

Inductive step. Fix an integer $n \geq 0$ and suppose $P(n)$ true. By the relation defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{4}u_n + 6 \\ &= \frac{1}{4}\left[8 - 6\left(\frac{1}{4}\right)^n\right] + 6 \\ &= 2 + 6 - 6\left(\frac{1}{4}\right)^{n+1} \\ &= 8 - 6\left(\frac{1}{4}\right)^{n+1}. \end{aligned}$$

This is $P(n+1)$: the inductive step holds.

Conclusion. The principle of induction gives $u_n = 8 - 6\left(\frac{1}{4}\right)^n$ for all $n \in \mathbb{N}$.

3. [1 pt] Since $0 < \frac{1}{4} < 1$, for every $n \in \mathbb{N}$ we have $0 < (\frac{1}{4})^n \leq 1$. Multiplication by $-6 < 0$ reverses the inequalities:

$$-6 \leq -6(\frac{1}{4})^n < 0.$$

Adding 8 and using the previous formula, we obtain $2 \leq u_n < 8$.

4. [1 pt] For every $n \in \mathbb{N}$, the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -6(\frac{1}{4})^{n+1} + 6(\frac{1}{4})^n \\ &= 6(1 - \frac{1}{4})(\frac{1}{4})^n = \frac{9}{2}(\frac{1}{4})^n > 0. \end{aligned}$$

The sequence is therefore strictly increasing.

5. [0.5 pt] The statement is false. For each integer n, $8 - u_n = 6(\frac{1}{4})^n > 0$. No term is therefore equal to 8, regardless of the number of steps.

Exercise 2 · Limits calculated and justified

1. [1 pt] For $n \geq 1$, we divide by $n^2 \neq 0$:

$$u_n = \frac{3 + \frac{5}{n} - \frac{1}{n^2}}{4 + \frac{2}{n^2}}.$$

The terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 3 and the denominator to $4 \neq 0$; by quotient, $u_n \to \frac{3}{4}$.

2. [1 pt] The conjugate is positive. For $n \geq 1$, $\sqrt{n^2} = n$, hence

$$v_n = \frac{(n^2+5n)-n^2}{\sqrt{n^2+5n}+n} = \frac{5n}{n(\sqrt{1+\frac{5}{n}}+1)} = \frac{5}{\sqrt{1+\frac{5}{n}}+1}.$$

3. [1 pt] The denominator tends to 2; limit $\frac{5}{2}$.

4. [1 pt] w $_n = n^2(3-\frac{5}{n}) \to +\infty$; the second factor tends to 3 > 0.

5. [1 pt] No: $(2n) - n = n \to +\infty$. The form $\infty - \infty$ is an indeterminate form, not a calculation rule.

Exercise 3 · Limits and asymptotes

1. [1 pt] The denominator is zero only at 2: $D_f = \mathbb{R} \setminus \{2\}$. Since 2 $x+6 = 2(x-2)+10$, we get $f(x) = 2 + \frac{10}{x-2}$.

2. [1 pt] The two limits are 2.

3. [1 pt] When $x \to 2^-$, $x-2 \to 0$ through negative values; since 10 > 0, $\frac{10}{x-2} \to -\infty$, therefore $f(x) \to -\infty$. On the right, the denominator tends to 0 through positive values: the quotient, then $f(x)$, tend to $+\infty$.

4. [1 pt] x $= 2$ and $y = 2$; curve below for $x < 2$ and above for $x > 2$.

5. [1 pt] Let $t = 2x$. Then $t \to +\infty$ & $x^2e^{-2x} = \frac{1}{4}\frac{t^2}{e^t}$. Comparing growth rates gives $\frac{t^2}{e^t} \to 0$; the required limit is therefore 0.

Exercise 4 · Codes and selections

1. [1 pt] $15^4 = 50625$ by the multiplicative principle.

2. [1 pt] The successive choices have 15, 14, then one fewer choice at each step, down to 12 possibilities. By multiplication,

$$15 \times 14 \times 13 \times 12 = 32760.$$

The order of symbols is part of the code.

3. [1 pt] $15^4 - 14^4 = 12209$ by the complement.

4. [1 pt] The symbol in the first position is fixed and can no longer be used. The next 3 positions successively give 14, . . . , 12 choices:

$$14 \times 13 \times 12 = 2184.$$

5. [1 pt] $13^4 = 28561$.

Common test · Specimen 2

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A discrete evolution — 5 points

Consider the sequence (u_n) defined by $u_0 = 3$ and, for every natural number n, $u_{n+1} = \frac{1}{3}u_n + \frac{20}{3}$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Prove by induction that, for every $n \in \mathbb{N}$, $u_n = 10 - 7\left(\frac{1}{3}\right)^n$. (2 pt)

3. Deduce that, for every natural number n, $3 \leq u_n < 10$. (1 pt)

4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. A student claims: "There exists an integer n such that $u_n = 10$." Is this statement true? Justify. (0.5 pt)

Exercise 2 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{2n^2+7n-1}{3n^2+3}$ & $v_n = \sqrt{n^2+7n} - n$.

1. Determine the limit of (u_n) showing how its expression is transformed. (1 pt)

2. Show that, for every integer $n \geq 1$, $v_n = \frac{7}{\sqrt{1+\frac{7}{n}}+1}$. (1 pt)

3. Deduce the limit of (v_n). (1 pt)

4. Define $w_n = 2n^2 - 7n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this statement true? Justify. (1 pt)

Common test · Specimen 2 · continued

Exercise 3 · Limits and asymptotes — 5 points

Consider the function f defined by $f(x) = \frac{5x+7}{x-3}$. Let $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the domain of f. Find two reals A and B such that, for all $x \neq 3$, $f(x) = A + \frac{B}{x-3}$. (1 pt)
2. Determine the limits of f at $-\infty$ and at $+\infty$. (1 pt)
3. Determine the limits of f at 3^- and at 3^+. (1 pt)
4. Give the equations of the asymptotes to $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. (1 pt)
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^5 e^{-3x}$ with justification. (1 pt)

Exercise 4 · Codes and selections — 5 points

A code is an ordered sequence of 5 symbols chosen from an alphabet that contains 16 distinct symbols. The rules concerning repetitions are specified in each question, independently of the other questions.

1. How many codes can be formed if repetitions are allowed? (1 pt)
2. How many codes can be formed if each symbol is to be used at most once? (1 pt)
3. One symbol of the alphabet is fixed in advance. With repetition allowed, how many codes contain this symbol at least once? (1 pt)
4. The first symbol of the code is fixed and repetitions are not allowed. How many codes can be formed? (1 pt)
5. Two distinct symbols of the alphabet are forbidden. The other symbols may be repeated. How many codes can be formed? (1 pt)

Solutions · Common test · Specimen 2

Exercise 1 · A discrete evolution

1. [0.5 pt] The relation is applied successively at indices 0 and 1:

$$u_1 = \frac{1}{3} \times 3 + \frac{20}{3} = \frac{23}{3}, \qquad u_2 = \frac{1}{3} \times \frac{23}{3} + \frac{20}{3} = \frac{83}{9}.$$

2. [2 pt] For $n \in \mathbb{N}$, let $P(n)$ the property $u_n = 10 - 7\left(\frac{1}{3}\right)^n$.

Base case. For $n = 0$, $10 - 7 \times 1 = 3 = u_0$. Hence $P(0)$ is true.

Inductive step. Fix an integer $n \geq 0$ and suppose $P(n)$ true. By the relation defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{3}u_n + \frac{20}{3} \\ &= \frac{1}{3}\left[10 - 7\left(\frac{1}{3}\right)^n\right] + \frac{20}{3} \\ &= \frac{10}{3} + \frac{20}{3} - 7\left(\frac{1}{3}\right)^{n+1} \\ &= 10 - 7\left(\frac{1}{3}\right)^{n+1}. \end{aligned}$$

This is $P(n+1)$: the inductive step holds.

Conclusion. The principle of induction gives $u_n = 10 - 7\left(\frac{1}{3}\right)^n$ for all $n \in \mathbb{N}$.

3. [1 pt] Since $0 < \frac{1}{3} < 1$, for every $n \in \mathbb{N}$ we have $0 < \left(\frac{1}{3}\right)^n \leq 1$. Multiplication by $-7 < 0$ reverses the inequalities:

$$-7 \leq -7\left(\frac{1}{3}\right)^n < 0.$$

Adding 10 and using the previous formula, we obtain $3 \leq u_n < 10$.

4. [1 pt] For every $n \in \mathbb{N}$, the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -7\left(\frac{1}{3}\right)^{n+1} + 7\left(\frac{1}{3}\right)^n \\ &= 7\left(1 - \frac{1}{3}\right)\left(\frac{1}{3}\right)^n = \frac{14}{3}\left(\frac{1}{3}\right)^n > 0. \end{aligned}$$

The sequence is therefore strictly increasing.

5. [0.5 pt] The statement is false. For each integer n, $10 - u_n = 7\left(\frac{1}{3}\right)^n > 0$. No term is therefore equal to 10, whatever the number of steps.

Exercise 2 · Limits calculated and justified

1. [1 pt] For $n \geq 1$, we divide by $n^2 \neq 0$:

$$u_n = \frac{2 + \frac{7}{n} - \frac{1}{n^2}}{3 + \frac{3}{n^2}}.$$

The terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 2 and the denominator to $3 \neq 0$; by quotient, $u_n \to \frac{2}{3}$.

2. [1 pt] The conjugate is positive. For $n \geq 1$, $\sqrt{n^2} = n$, hence

$$v_n = \frac{(n^2+7n)-n^2}{\sqrt{n^2+7n}+n} = \frac{7n}{n(\sqrt{1+\frac{7}{n}}+1)} = \frac{7}{\sqrt{1+\frac{7}{n}}+1}.$$

3. [1 pt] The denominator tends to 2; limit $\frac{7}{2}$.

4. [1 pt] w $_n = n^2(2-\frac{7}{n}) \to +\infty$; the second factor tends to 2 > 0.

5. [1 pt] No: $(2n) - n = n \to +\infty$. The form $\infty - \infty$ is an indeterminate form, not a calculation rule.

Exercise 3 · Limits and asymptotes

1. [1 pt] The denominator is zero only at 3: $D_f = \mathbb{R} \setminus \{3\}$. Since $5 \quad x+7 = 5(x-3)+22$, we get $f(x) = 5 + \dfrac{22}{x-3}$.

2. [1 pt] Both limits are 5.

3. [1 pt] When $x \to 3^-$, $x-3 \to 0$ through negative values; since $22 > 0$, $\frac{22}{x-3} \to -\infty$, therefore $f(x) \to -\infty$. On the right, the denominator tends to 0 through positive values: the quotient, then $f(x)$, tend to $+\infty$.

4. [1 pt] x $= 3$ and $y = 5$; curve below for $x < 3$ and above for $x > 3$.

5. [1 pt] Let $t = 3x$. Then $t \to +\infty$ & $x^5 e^{-3x} = \frac{1}{243}\frac{t^5}{e^t}$. Comparing growth rates gives $\frac{t^5}{e^t} \to 0$; the required limit is therefore 0.

Exercise 4 · Codes and selections

1. [1 pt] $16^5 = 1048576$ by the multiplicative principle.

2. [1 pt] The successive choices have 16, 15, then one fewer choice at each step, down to 12 possibilities. By multiplication,

$$16 \times 15 \times 14 \times 13 \times 12 = 524160.$$

The order of symbols is part of the code.

3. [1 pt] $16^5 - 15^5 = 289201$ by the complement.

4. [1 pt] The symbol in the first position is fixed and can no longer be used. The next 4 positions successively give 15, . . . , 12 choices:

$$15 \times 14 \times 13 \times 12 = 32760.$$

5. [1 pt] $14^5 = 537824$.

Common test · A

PAPER RESERVED FOR ASSESSMENT

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A discrete evolution — 5 points

Consider the sequence (u_n) defined by $u_0 = 1$ and, for every natural number n, $u_{n+1} = \frac{1}{2}u_n + 6$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Prove by induction that, for every $n \in \mathbb{N}$, $u_n = 12 - 11\left(\frac{1}{2}\right)^n$. (2 pt)

3. Deduce that, for every natural number n, $1 \leq u_n < 12$. (1 pt)

4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. A student claims: "There exists an integer n such that $u_n = 12$." Is this statement true? Justify. (0.5 pt)

Exercise 2 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{5n^2+8n-1}{6n^2+1}$ & $v_n = \sqrt{n^2+8n} - n$.

1. Determine the limit of (u_n) showing how its expression is transformed. (1 pt)

2. Show that, for every integer $n \geq 1$, $v_n = \frac{8}{\sqrt{1+\frac{8}{n}}+1}$. (1 pt)

3. Deduce the limit of (v_n). (1 pt)

4. Define $w_n = 5n^2 - 8n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this statement true? Justify. (1 pt)

Common test · A · continued

Exercise 3 · Identify a model from its asymptotes — 4 points

We look for f in the form $f(x) = a + \frac{b}{x-1}$, with $b \neq 0$. Its graph has the horizontal asymptote $y = 2$ and passes through $(0; -1)$.

1. Determine a, b and the domain. (1 pt)

2. Calculate the limits at 1^- and 1^+. (1 pt)

3. Study the position of the curve relative to $y = 2$ and solve $f(x) = 5$. (2 pts)

Exercise 4 · Codes whose rules change — 6 points

A code contains four digits taken from 0, 1, 2, 3, 4. Zero may be in the first position.

1. Count all codes with repetition. (1 pt)

2. Count those without repetition. (1 pt)

3. Count those containing exactly one zero, the other digits possibly repeated. (2 pts)

4. Count those containing at least one zero by two methods, or by one method explained in detail. (2 pts)

Solutions · Common test · A

Exercise 1 · A discrete evolution

1. [0.5 pt] The relation is applied successively at indices 0 and 1:

$$u_1 = \frac{1}{2} \times 1 + 6 = \frac{13}{2}, \qquad u_2 = \frac{1}{2} \times \frac{13}{2} + 6 = \frac{37}{4}.$$

2. [2 pt] For $n \in \mathbb{N}$, let $P(n)$ the property $u_n = 12 - 11\left(\frac{1}{2}\right)^n$.

Base case. For $n = 0$, $12 - 11 \times 1 = 1 = u_0$. Hence $P(0)$ is true.

Inductive step. Fix an integer $n \geq 0$ and suppose $P(n)$ true. By the relation defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{2}u_n + 6 \\ &= \frac{1}{2}\left[12 - 11\left(\frac{1}{2}\right)^n\right] + 6 \\ &= 6 + 6 - 11\left(\frac{1}{2}\right)^{n+1} \\ &= 12 - 11\left(\frac{1}{2}\right)^{n+1}. \end{aligned}$$

This is $P(n+1)$: the inductive step holds.

Conclusion. The principle of induction gives $u_n = 12 - 11\left(\frac{1}{2}\right)^n$ for all $n \in \mathbb{N}$.

3. [1 pt] Since $0 < \frac{1}{2} < 1$, for every $n \in \mathbb{N}$ we have $0 < (\frac{1}{2})^n \leq 1$. Multiplication by $-11 < 0$ reverses the inequalities:

$$-11 \leq -11(\frac{1}{2})^n < 0.$$

Adding 12 and using the previous formula, we obtain $1 \leq u_n < 12$.

4. [1 pt] For every $n \in \mathbb{N}$, the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -11(\frac{1}{2})^{n+1} + 11(\frac{1}{2})^n \\ &= 11(1 - \frac{1}{2})(\frac{1}{2})^n = \frac{11}{2}(\frac{1}{2})^n > 0. \end{aligned}$$

The sequence is therefore strictly increasing.

5. [0.5 pt] The statement is false. For each integer n, $12 - u_n = 11(\frac{1}{2})^n > 0$. No term is therefore equal to 12, whatever the number of steps.

Exercise 2 · Limits calculated and justified

1. [1 pt] For $n \geq 1$, we divide by $n^2 \neq 0$:

$$u_n = \frac{5 + \frac{8}{n} - \frac{1}{n^2}}{6 + \frac{1}{n^2}}.$$

The terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 5 and the denominator to $6 \neq 0$; by quotient, $u_n \to \frac{5}{6}$.

2. [1 pt] The conjugate is positive. For $n \geq 1$, $\sqrt{n^2} = n$, hence

$$v_n = \frac{(n^2+8n)-n^2}{\sqrt{n^2+8n}+n} = \frac{8n}{n(\sqrt{1+\frac{8}{n}}+1)} = \frac{8}{\sqrt{1+\frac{8}{n}}+1}.$$

3. [1 pt] The denominator tends to 2; limit 4.

4. [1 pt] w $_n = n^2(5-\frac{8}{n}) \to +\infty$; the second factor tends to 5 > 0.

5. [1 pt] No: $(2n)-n = n \to +\infty$. The form $\infty - \infty$ is an indeterminate form, not a calculation rule.

Exercise 3 · Identify a model from its asymptotes

1. The limit at infinity is a, therefore $a = 2$. Then $2-b=-1$ gives $b = 3$. Domain: $\mathbb{R} \setminus \{1\}$.

2. The quotient $3/(x-1)$ tends to $-\infty$ on the left and $+\infty$ on the right (0.5 point per side).

3. $f-2 = 3/(x-1)$: below for $x < 1$, above for $x > 1$, never on the asymptote (1 pt).
$f = 5 \iff 3/(x-1) = 3 \iff x = 2$, admissible (1 pt).

Exercise 4 · Codes whose rules change

1. $5^4 = 625$.

2. $5 \times 4 \times 3 \times 2 = 120$.

3. Four positions for zero, then four choices for each of the other three positions: 4 $\times 4^3 = 256$. Position 1 point, filling 1 point.

4. By complement: 5 $^4 - 4^4 = 369$; remove the codes using only 1, 2, 3, 4. Or according to the number of zeros: 256 $+ 6 \times 16 + 4 \times 4 + 1 = 369$. The coefficient 6 is found by listing the six pairs of positions, without requiring combinations. Model 1 point, calculation 1 point.

Common test · B

PAPER RESERVED FOR ASSESSMENT

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A discrete evolution — 5 points

Consider the sequence (u_n) defined by $u_0 = 2$ and, for every natural number n, $u_{n+1} = \frac{1}{5}u_n + \frac{32}{5}$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Prove by induction that, for every $n \in \mathbb{N}$, $u_n = 8 - 6\left(\frac{1}{5}\right)^n$. (2 pt)

3. Deduce that, for every natural number n, $2 \leq u_n < 8$. (1 pt)

4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. A student claims: "There exists an integer n such that $u_n = 8$." Is this statement true? Justify. (0.5 pt)

Exercise 2 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{4n^2+5n-1}{5n^2+2}$ & $v_n = \sqrt{n^2+5n} - n$.

1. Determine the limit of (u_n) showing how its expression is transformed. (1 pt)

2. Show that, for every integer $n \geq 1$, $v_n = \frac{5}{\sqrt{1+\frac{5}{n}}+1}$. (1 pt)

3. Deduce the limit of (v_n). (1 pt)

4. Define $w_n = 4n^2 - 5n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this statement true? Justify. (1 pt)

Common test · B · continued

Exercise 3 · One root and two infinities — 4 points

Consider $g(x) = \sqrt{x^2+4} - x$ on $\mathbb{R}$.

1. Show that $g(x) > 0$ on $\mathbb{R}$. (1 pt)

2. Determine the limit at $+\infty$ by a justified transformation. (2 pts)

3. Determine the limit at $-\infty$ and explain why the previous result does not carry over. (1 pt)

Exercise 4 · Build a decision tree — 6 points

An outfit combines one shirt from 3, one pair of trousers from 4 and one pair of shoes from 2. One of the shirts is red, one of the pairs of trousers is red. The items are all distinct.

1. Count outfits without restriction. (1 pt)

2. Count outfits with a red shirt and red trousers. (1 pt)

3. Count the outfits with exactly one of these two red items. (2 pts)

4. A student adds "red shirt" and "red trousers" to count the outfits with at least one red item. Correct their result and explain the double counting. (2 pts)

Solutions · Common test · B

Exercise 1 · A discrete evolution

1. [0.5 pt] The relation is applied successively at indices 0 and 1:

$$u_1 = \frac{1}{5} \times 2 + \frac{32}{5} = \frac{34}{5}, \qquad u_2 = \frac{1}{5} \times \frac{34}{5} + \frac{32}{5} = \frac{194}{25}.$$

2. [2 pt] For $n \in \mathbb{N}$, let $P(n)$ the property $u_n = 8 - 6\left(\frac{1}{5}\right)^n$.

Base case. For $n = 0$, $8 - 6 \times 1 = 2 = u_0$. Hence $P(0)$ is true.

Inductive step. Fix an integer $n \geq 0$ and suppose $P(n)$ true. By the relation defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{5}u_n + \frac{32}{5} \\ &= \frac{1}{5}\left[8 - 6\left(\frac{1}{5}\right)^n\right] + \frac{32}{5} \\ &= \frac{8}{5} + \frac{32}{5} - 6\left(\frac{1}{5}\right)^{n+1} \\ &= 8 - 6\left(\frac{1}{5}\right)^{n+1}. \end{aligned}$$

This is $P(n+1)$: the inductive step holds.

Conclusion. The principle of induction gives $u_n = 8 - 6\left(\frac{1}{5}\right)^n$ for all $n \in \mathbb{N}$.

3. [1 pt] Since $0 < \frac{1}{5} < 1$, for every $n \in \mathbb{N}$ we have $0 < \left(\frac{1}{5}\right)^n \leq 1$. Multiplication by $-6 < 0$ reverses the inequalities:

$$-6 \leq -6\left(\frac{1}{5}\right)^n < 0.$$

Adding 8 and using the previous formula, we obtain $2 \leq u_n < 8$.

4. [1 pt] For every $n \in \mathbb{N}$, the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -6\left(\frac{1}{5}\right)^{n+1} + 6\left(\frac{1}{5}\right)^n \\ &= 6\left(1 - \frac{1}{5}\right)\left(\frac{1}{5}\right)^n = \frac{24}{5}\left(\frac{1}{5}\right)^n > 0. \end{aligned}$$

The sequence is therefore strictly increasing.

5. [0.5 pt] The statement is false. For each integer n, $8 - u_n = 6\left(\frac{1}{5}\right)^n > 0$. No term is therefore equal to 8, regardless of the number of steps.

Exercise 2 · Limits calculated and justified

1. [1 pt] For $n \geq 1$, we divide by $n^2 \neq 0$:

$$u_n = \frac{4 + \frac{5}{n} - \frac{1}{n^2}}{5 + \frac{2}{n^2}}.$$

The terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 4 and the denominator to $5 \neq 0$; by quotient, $u_n \to \frac{4}{5}$.

2. [1 pt] The conjugate is positive. For $n \geq 1$, $\sqrt{n^2} = n$, hence

$$v_n = \frac{(n^2+5n)-n^2}{\sqrt{n^2+5n}+n} = \frac{5n}{n(\sqrt{1+\frac{5}{n}}+1)} = \frac{5}{\sqrt{1+\frac{5}{n}}+1}.$$

3. [1 pt] The denominator tends to 2; limit $\frac{5}{2}$.

4. [1 pt] w $_n = n^2(4-\frac{5}{n}) \to +\infty$; the second factor tends to 4 > 0.

5. [1 pt] No: $(2n) - n = n \to +\infty$. The form $\infty - \infty$ is an indeterminate form, not a calculation rule.

Exercise 3 · One root and two infinities

1. $\sqrt{x^2+4} > |x| \geq x$, therefore $g(x) > 0$.

2. By conjugation $g(x) = 4/(\sqrt{x^2+4}+x)$. The denominator tends to $+\infty$, therefore $g \to 0$. Transformation mation 1 point, limit 1 point.

3. At $-\infty$, $\sqrt{x^2+4} \to +\infty$ & $-x \to +\infty$, therefore $g \to +\infty$. Writing $\sqrt{x^2} = x$ would be false for $x < 0$.

Exercise 4 · Build a decision tree

1. $3 \times 4 \times 2 = 24$.

2. $1 \times 1 \times 2 = 2$.

3. Red shirt only: 1 $\times 3 \times 2 = 6$; red trousers only: 2 $\times 1 \times 2 = 4$. The cases are disjoint, total 10. One point per case.

4. Red shirt: 8; red trousers: 6. The 2 outfits with both are counted twice, so $8+6-2 = 12$. Explanation 1 point, result 1 point.

DS3 · Specimen 1

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits and asymptotes — 5 points

Consider the function f defined by $f(x) = \frac{5x+4}{x-1}$. Let $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the domain of f. Find two reals A and B such that, for all $x \neq 1$, $f(x) = A + \frac{B}{x-1}$. (1 pt)
2. Determine the limits of f at $-\infty$ and at $+\infty$. (1 pt)
3. Determine the limits of f at 1^- and at 1^+. (1 pt)
4. Give the equations of the asymptotes to $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. (1 pt)
5. Independently of the previous questions, determine $\lim_{x \to +\infty} x^5 e^{-1x}$ with justification. (1 pt)

Exercise 2 · Variations, convexity and equation — 5 points

Consider the function f defined on $\mathbb{R}$ by $f(x) = x^3 + 4x - 5$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f'(x)$, then study the direction of variation of f on $\mathbb{R}$. (1 pt)
2. Determine the limits of f at $-\infty$ and at $+\infty$. (0.5 pt)
3. Prove that the equation $f(x) = 0$ has a unique real solution α. Give its exact value if it is an integer; otherwise, bound it between two consecutive integers. (1.5 pt)
4. Study the convexity of f, then find the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0, then study the position of the curve relative to this tangent. (1 pt)

DS3 · Specimen 1 · continued

Exercise 3 · An exponential curve — 5 points

Consider the function f defined on $[0;+\infty[$ by $f(x)=(x+3)e^{-x}$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Calculate $f''(x)$, then study the convexity of f. (1 pt)
4. Prove that the equation $f(x)=\frac{3}{2}$ has a unique solution α in $[0;+\infty[$. (1.5 pt)
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (1 pt)

Exercise 4 · Forming a group — 5 points

A team consists of 6 teachers and 10 students, all distinct. A group of four people is chosen; the order of choice does not matter.

1. How many groups of four people can be formed? (1 pt)
2. How many of these groups include exactly two teachers? (1 pt)
3. How many of these groups include at least one teacher and at least one student? (1 pt)
4. We choose a group containing exactly two teachers, then appoint from among its members a teacher spokesperson and a student rapporteur. Count the choices of the group together with these two assignments. (1 pt)
5. Explain why the number of ordered lists of four distinct people, divided by $4!$, gives the number of groups obtained in question 1. (1 pt)

Solutions · DS3 · Specimen 1

Exercise 1 · Limits and asymptotes

1. [1 pt] The denominator is zero only at 1: $D_f = \mathbb{R} \setminus \{1\}$. Since $5 \quad x + 4 = 5(x-1) + 9$, we get $\quad f(x) = 5 + \dfrac{9}{x-1}$.

2. [1 pt] Both limits are 5.

3. [1 pt] When $\quad x \to 1^-$, $x - 1 \to 0$ through negative values; since $9 \quad > 0$, $\frac{9}{x-1} \to -\infty$, therefore $f(x) \to -\infty$. On the right, the denominator tends to 0 through positive values: the quotient, then $\quad f(x)$, tend to $\quad +\infty$.

4. [1 pt] x $= 1$ and $y = 5$; curve below for $\quad x < 1$ and above for $\quad x > 1$.

5. [1 pt] Comparing the growth of powers and the exponential directly gives $\quad x^5 e^{-x} = \frac{x^5}{e^x} \to 0$ when $\quad x \to +\infty$.

Exercise 2 · Variations, convexity and equation

1. [1 pt] f is differentiable on $\quad \mathbb{R}$ and $f'(x) = 3x^2 + 4$. Since $3 \quad x^2 \geq 0$ and $4 > 0$, $f'(x) > 0$ for every $\quad x$; f is strictly increasing on $\quad \mathbb{R}$.

2. [0.5 pt] The dominant term is $\quad x^3$: limits $\quad -\infty$ & $+\infty$.

3. [1.5 pt] $\quad f$ is a polynomial, so continuous on $\quad \mathbb{R}$. Its strict monotonicity and its limits $\quad -\infty, +\infty$ allow the corollary of the IVT to be applied: there is a unique zero $\quad \alpha$. $f(1) = 0$, so the solution is exactly 1.

4. [1 pt] f $''(x) = 6x$: concave for $\quad x \leq 0$, convex for $\quad x \geq 0$, inflection $(0, -5)$.

5. [1 pt] Tangent $\quad y = 4x - 5$; difference $\quad x^3$, negative to the left and positive to the right of 0.

Exercise 3 · An exponential curve

1. [0.5 pt] f $(0) = 3$; limit 0 by comparing growth rates.

2. [1 pt] By the product rule,

$$f'(x) = e^{-x} - (x+3)e^{-x} = (-x-2)e^{-x}.$$

The two factors are respectively strictly negative and strictly positive on $\quad [0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	3	$\searrow$	0

3. [1 pt] Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-2)e^{-x} = (x+1)e^{-x}.$$

On the domain, $\quad x + 1 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity on the interior of the domain, so no inflection point.

4. [1.5 pt] Continuity, strict decrease, $\quad f(0) = 3 > \frac{3}{2}$ and limit $0 < \frac{3}{2}$: exactly one solution.

5. [1 pt] The tangent at 0 has equation $\quad y = f(0) + f'(0)x = 3 - 2x$. A convex function is above each of its tangents, so $\quad (x+3)e^{-x} \geq 3 - 2x$ for all $\quad x \geq 0$.

Exercise 4 · Forming a group

1. [1 pt] Order does not distinguish the groups: we must choose a subset of size 4 from 16 people, i.e. $\binom{16}{4} = 1820$ groups.

2. [1 pt] Two teachers are chosen from 6 and two pupils from 10. Each pair of choices gives a single group: $\binom{6}{2}\binom{10}{2} = 675$.

3. [1 pt] The excluded groups are those made up only of teachers or only of pupils; these two categories are disjoint. The required number is $\binom{16}{4} - \binom{6}{4} - \binom{10}{4} = 1595$.

4. [1 pt] Each group produces 2 $\times$ 2 arrangements: 2700.

5. [1 pt] Each group of four distinct people corresponds to exactly 4 ! ordered lists. This multiplicity is constant, so dividing removes the overcounting.

DS3 · Specimen 2

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits and asymptotes — 5 points

Consider the function f defined by $f(x) = \frac{4x+6}{x-2}$. Let $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the domain of f. Find two reals A and B such that, for all $x \neq 2$, $f(x) = A + \frac{B}{x-2}$. (1 pt)
2. Determine the limits of f at $-\infty$ and at $+\infty$. (1 pt)
3. Determine the limits of f at 2^- and at 2^+. (1 pt)
4. Give the equations of the asymptotes to $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. (1 pt)
5. Independently of the previous questions, determine $\lim_{x \to +\infty} x^4 e^{-2x}$ with justification. (1 pt)

Exercise 2 · Variations, convexity and equation — 5 points

Consider the function f defined on $\mathbb{R}$ by $f(x) = x^3 + 3x - 7$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f'(x)$, then study the direction of variation of f on $\mathbb{R}$. (1 pt)
2. Determine the limits of f at $-\infty$ and at $+\infty$. (0.5 pt)
3. Prove that the equation $f(x) = 0$ has a unique real solution α. Give its exact value if it is an integer; otherwise, bound it between two consecutive integers. (1.5 pt)
4. Study the convexity of f, then find the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0, then study the position of the curve relative to this tangent. (1 pt)

DS3 · Specimen 2 · continued

Exercise 3 · An exponential curve — 5 points

Consider the function f defined on $[0;+\infty[$ by $f(x) = (x+2)e^{-x}$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Calculate $f''(x)$, then study the convexity of f. (1 pt)
4. Prove that the equation $f(x) = 1$ has a unique solution α in $[0;+\infty[$. (1.5 pt)
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (1 pt)

Exercise 4 · Forming a group — 5 points

A team consists of 9 teachers and 11 students, all distinct. A group of four people is chosen; the order of choice does not matter.

1. How many groups of four people can be formed? (1 pt)
2. How many of these groups include exactly two teachers? (1 pt)
3. How many of these groups include at least one teacher and at least one student? (1 pt)
4. We choose a group containing exactly two teachers, then appoint from among its members a teacher spokesperson and a student rapporteur. Count the choices of the group together with these two assignments. (1 pt)
5. Explain why the number of ordered lists of four distinct people, divided by $4!$, gives the number of groups obtained in question 1. (1 pt)

Solutions · DS3 · Specimen 2

Exercise 1 · Limits and asymptotes

1. [1 pt] The denominator is zero only at 2: $D_f = \mathbb{R} \setminus \{2\}$. Since $4 \quad x+6 = 4(x-2)+14$, we get $f(x) = 4 + \dfrac{14}{x-2}$.

2. [1 pt] Both limits are 4.

3. [1 pt] When $x \to 2^-$, $x-2 \to 0$ through negative values; since $14 > 0$, $\frac{14}{x-2} \to -\infty$, therefore $f(x) \to -\infty$. On the right, the denominator tends to 0 through positive values: the quotient, then $f(x)$, tend to $+\infty$.

4. [1 pt] x $= 2$ and $y = 4$; curve below for $x < 2$ and above for $x > 2$.

5. [1 pt] Let $t = 2x$. Then $t \to +\infty$ & $x^4 e^{-2x} = \frac{1}{16}\frac{t^4}{e^t}$. Comparing growth rates gives $\frac{t^4}{e^t} \to 0$; the required limit is therefore 0.

Exercise 2 · Variations, convexity and equation

1. [1 pt] f is differentiable on $\mathbb{R}$ and $f'(x) = 3x^2 + 3$. Since $3 \quad x^2 \geq 0$ and $3 > 0$, $f'(x) > 0$ for every x ; f is strictly increasing on $\mathbb{R}$.

2. [0.5 pt] The dominant term is x^3 : limits $-\infty$ & $+\infty$.

3. [1.5 pt] f is a polynomial, so continuous on $\mathbb{R}$. Its strict monotonicity and its limits $-\infty, +\infty$ allow the corollary of the IVT to be applied: there is a unique zero α. $f(1) = -3 < 0$ and $f(2) = 7 > 0$; therefore $1 < \alpha < 2$.

4. [1 pt] f $''(x) = 6x$: concave for $x \leq 0$, convex for $x \geq 0$, inflection $(0, -7)$.

5. [1 pt] Tangent $y = 3x - 7$; difference x^3, negative to the left and positive to the right of 0.

Exercise 3 · An exponential curve

1. [0.5 pt] f $(0) = 2$; limit 0 by comparing growth rates.

2. [1 pt] By the product rule,

$$f'(x) = e^{-x} - (x+2)e^{-x} = (-x-1)e^{-x}.$$

The two factors are respectively strictly negative and strictly positive on $[0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	2	$\searrow$	0

3. [1 pt] Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-1)e^{-x} = xe^{-x}.$$

On the domain, $x + 0 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity on the interior of the domain, so no inflection point.

4. [1.5 pt] Continuity, strict decrease, $f(0) = 2 > 1$ and limit $0 < 1$: exactly one solution.

5. [1 pt] The tangent at 0 has equation $y = f(0) + f'(0)x = 2 - x$. A convex function is above each of its tangents, so $(x+2)e^{-x} \geq 2 - x$ for all $x \geq 0$.

Exercise 4 · Forming a group

1. [1 pt] Order does not distinguish the groups: we must choose a subset of size 4 from 20 people, i.e. $\binom{20}{4} = 4845$ groups.

2. [1 pt] Two teachers are chosen from 9 and two pupils from 11. Each pair of choices gives a single group: $\binom{9}{2}\binom{11}{2} = 1980$.

3. [1 pt] The excluded groups are those made up only of teachers or only of pupils; these two categories are disjoint. The required number is $\binom{20}{4} - \binom{9}{4} - \binom{11}{4} = 4389$.

4. [1 pt] Each group produces 2 $\times$ 2 arrangements: 7920.

5. [1 pt] Each group of four distinct people corresponds to exactly 4 ! ordered lists. This multiplicity is constant, so dividing removes the overcounting.

DS3 · A

PAPER RESERVED FOR ASSESSMENT

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits and asymptotes — 5 points

Consider the function f defined by $f(x) = \frac{3x+8}{x-3}$. Let $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the domain of f. Find two reals A and B such that, for all $x \neq 3$, $f(x) = A + \frac{B}{x-3}$. (1 pt)
2. Determine the limits of f at $-\infty$ and at $+\infty$. (1 pt)
3. Determine the limits of f at 3^- and at 3^+. (1 pt)
4. Give the equations of the asymptotes to $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. (1 pt)
5. Independently of the previous questions, determine $\lim_{x \to +\infty} x^3 e^{-3x}$ with justification. (1 pt)

Exercise 2 · Distinguishing existence from uniqueness — 4 points

On $\mathbb{R}$, $f(x) = x^3 - 3x$.

1. Draw up the table of variations of f. (1 pt)
2. Show that $f(x) = 1$ has exactly three real solutions, giving three disjoint intervals that contain them. (2 pts)
3. Identify the inflection point of the curve, justifying the change of convexity. (1 pt)

DS3 · A · continued

Exercise 3 · An exponential curve — 5 points

Consider the function f defined on $[0;+\infty[$ by $f(x)=(x+5)e^{-x}$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Calculate $f''(x)$, then study the convexity of f. (1 pt)
4. Prove that the equation $f(x)=\frac{5}{2}$ has a unique solution α in $[0;+\infty[$. (1.5 pt)
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (1 pt)

Exercise 4 · Two counts of one group — 6 points

A club has 6 Terminale students and 4 Première students, all distinct. A group of three is chosen, with no roles.

1. Count all groups. (1 pt)
2. Count those with exactly two Terminale students. (1 pt)
3. Count those with at least one pupil from each level, in two ways. (2 pts)
4. Among these mixed groups, how many teams can be formed with a leader from Terminale chosen within the group? Explain why multiplying uniformly by 3 would be wrong. (2 pts)

Solutions · DS3 · A

Exercise 1 · Limits and asymptotes

1. [1 pt] The denominator is zero only at 3: $D_f = \mathbb{R} \setminus \{3\}$. Since 3 $x + 8 = 3(x-3) + 17$, we get $f(x) = 3 + \dfrac{17}{x-3}$.

2. [1 pt] Both limits are 3.

3. [1 pt] When $x \to 3^-$, $x - 3 \to 0$ through negative values; since 17 > 0, $\frac{17}{x-3} \to -\infty$, therefore $f(x) \to -\infty$. On the right, the denominator tends to 0 through positive values: the quotient, then $f(x)$, tend to $+\infty$.

4. [1 pt] x $= 3$ and $y = 3$; curve below for $x < 3$ and above for $x > 3$.

5. [1 pt] Let $t = 3x$. Then $t \to +\infty$ & $x^3 e^{-3x} = \frac{1}{27}\frac{t^3}{e^t}$. Comparing growth rates gives $\frac{t^3}{e^t} \to 0$; the required limit is therefore 0.

Exercise 2 · Distinguish existence and uniqueness

1. $f' = 3(x-1)(x+1)$; increasing on $]-\infty;-1]$, decreasing on $[-1;1]$, increasing on $[1;+\infty[$. $f(-1) = 2$, $f(1) = -2$.

2. Polynomial continuity and strict monotonicity on each branch. Values: $f(-2) = -2$, $f(-1) = 2$, $f(0) = 0$, $f(1) = -2$, $f(2) = 2$. The intermediate value theorem gives a unique solution in each of $]-2;-1[$, $]-1;0[$, $]1;2[$. The three branches cover $\mathbb{R}$ and each have only one solution. Existence 1 point, uniqueness and completeness 1 point.

3. $f'' = 6x$ changes sign from negative to positive at 0. The curve goes from concave to convex at $(0;0)$.

Exercise 3 · An exponential curve

1. [0.5 pt] f $(0) = 5$; limit 0 by comparing growth rates.

2. [1 pt] By the product rule,

$$f'(x) = e^{-x} - (x+5)e^{-x} = (-x-4)e^{-x}.$$

The two factors are respectively strictly negative and strictly positive on $[0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	5	$\searrow$	0

3. [1 pt] Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-4)e^{-x} = (x+3)e^{-x}.$$

On the domain, $x + 3 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity on the interior of the domain, so no inflection point.

4. [1.5 pt] Continuity, strict decrease, $f(0) = 5 > \frac{5}{2}$ and limit $0 < \frac{5}{2}$: exactly one solution.

5. [1 pt] The tangent at 0 has equation $y = f(0) + f'(0)x = 5 - 4x$. A convex function is above each of its tangents, so $(x+5)e^{-x} \geq 5 - 4x$ for all $x \geq 0$.

Exercise 4 · Two counts of one group

1. $\binom{10}{3} = 120$.
2. $\binom{6}{2}\binom{4}{1} = 60$.
3. Complement: 120 $- \binom{6}{3} - \binom{4}{3} = 96$. Disjoint cases: $\binom{6}{2}\binom{4}{1} + \binom{6}{1}\binom{4}{2} = 60 + 36 = 96$. One point per method.
4. With two Terminale students: 60 $\times 2 = 120$; with one Terminale: 36 $\times 1 = 36$, total 156. The leader must be in Terminale: the number of possibilities depends on the composition. Counting 1 point, explanation 1 point.

DS3 · B

PAPER RESERVED FOR ASSESSMENT

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits and asymptotes — 5 points

Consider the function f defined by $f(x) = \frac{2x+5}{x-1}$. Let $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the domain of f. Find two reals A and B such that, for all $x \neq 1$, $f(x) = A + \frac{B}{x-1}$. (1 pt)
2. Determine the limits of f at $-\infty$ and at $+\infty$. (1 pt)
3. Determine the limits of f at 1^- and at 1^+. (1 pt)
4. Give the equations of the asymptotes to $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. (1 pt)
5. Independently of the previous questions, determine $\lim_{x \to +\infty} x^2 e^{-1x}$ with justification. (1 pt)

Exercise 2 · Read a derivative before calculating — 4 points

We are given $F(x) = x + 2e^{-x}$ on $\mathbb{R}$ and $F'(x) = 1 - 2e^{-x}$.

1. Without logarithms, prove that F' is zero at a unique real number $a \in]0;1[$. You may use $e > 2$. (2 pts)
2. Deduce the variations and the convexity of F. (1 pt)
3. Can a horizontal line meet this curve at three distinct points? Justify. (1 pt)

DS3 · B · continued

Exercise 3 · An exponential curve — 5 points

Consider the function f defined on $[0;+\infty[$ by $f(x)=(x+4)e^{-x}$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Calculate $f''(x)$, then study the convexity of f. (1 pt)
4. Prove that the equation $f(x)=2$ has a unique solution α in $[0;+\infty[$. (1.5 pt)
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (1 pt)

Exercise 4 · A selection with an incompatibility — 6 points

From 8 distinct pupils, including Lina and Noé, a group of 3 is chosen.

1. Count all groups. (1 pt)
2. Count those that contain Lina and Noé. (1 pt)
3. Count those that contain exactly one of the two. (2 pts)
4. Only groups containing both of them together are excluded. Count the allowed groups by the complement, then by separating "none" and "exactly one". (2 pts)

Solutions · DS3 · B

Exercise 1 · Limits and asymptotes

1. [1 pt] The denominator is zero only at 1: $D_f = \mathbb{R} \setminus \{1\}$. Since $x + 5 = 2(x-1) + 7$, we get $f(x) = 2 + \dfrac{7}{x-1}$.

2. [1 pt] The two limits are 2.

3. [1 pt] When $x \to 1^-$, $x - 1 \to 0$ through negative values; since $7 > 0$, $\frac{7}{x-1} \to -\infty$, therefore $f(x) \to -\infty$. On the right, the denominator tends to 0 through positive values: the quotient, then $f(x)$, tend to $+\infty$.

4. [1 pt] x $= 1$ and $y = 2$; curve below for $x < 1$ and above for $x > 1$.

5. [1 pt] Comparing the growth of powers and the exponential directly gives $x^2 e^{-x} = \frac{x^2}{e^x} \to 0$ when $x \to +\infty$.

Exercise 2 · Read a derivative before calculating

1. F' is continuous, $(F')' = 2e^{-x} > 0$ therefore strictly increasing. $F'(0) = -1 < 0$ and $F'(1) = 1 - 2/e > 0$. The intermediate value theorem gives a root in $]0;1[$, strict monotonicity gives uniqueness on $\mathbb{R}$. Assumptions and signs 1 point, existence and uniqueness 1 point.

2. $F' < 0$ before a, $F' > 0$ after: F decreases then increases. $F'' > 0$ everywhere, so F is convex.

3. No: each strictly monotonic branch meets a horizontal line at most once, so at most two points in total.

Exercise 3 · An exponential curve

1. [0.5 pt] f $(0) = 4$; limit 0 by comparing growth rates.

2. [1 pt] By the product rule,

$$f'(x) = e^{-x} - (x+4)e^{-x} = (-x-3)e^{-x}.$$

The two factors are respectively strictly negative and strictly positive on $[0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	4	$\searrow$	0

3. [1 pt] Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-3)e^{-x} = (x+2)e^{-x}.$$

On the domain, $x + 2 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity on the interior of the domain, so no inflection point.

4. [1.5 pt] Continuity, strict decrease, $f(0) = 4 > 2$ and limit $0 < 2$: exactly one solution.

5. [1 pt] The tangent at 0 has equation $y = f(0) + f'(0)x = 4 - 3x$. A convex function is above each of its tangents, so $(x+4)e^{-x} \geq 4 - 3x$ for all $x \geq 0$.

Exercise 4 · A selection with an incompatibility

1. $\binom{8}{3} = 56$.
2. Choose the third from the other six: 6.
3. Choose which of the two is selected, then two among the other six: 2 $\binom{6}{2} = 30$. One point for each choice.
4. Complement: 56 $- 6 = 50$. Another method: $\binom{6}{3} + 30 = 20 + 30 = 50$. Both cases are disjoint and exhaustive. One point per method.

Mock exam 1 · Zero 1

SPECIMEN PAPER · PRACTICE

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Induction and bounds — 5 points

Consider the sequence (v_n) defined by $v_0 = 0$ and, for every natural number n, $v_{n+1} = \sqrt{42 + v_n}$.

1. Calculate v_1 and v_2. Give exact values. (0.25 pt)

2. Prove by induction that all terms are defined and that, for every $n \in \mathbb{N}$, $0 \leq v_n \leq 7$. (0.75 pt)

3. Prove by induction that, for every $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (0.75 pt)

4. Prove that, for every $n \in \mathbb{N}$, $v_n < 7$. (0.5 pt)

5. Justify that (v_n) converges, then determine its limit. (0.75 pt)

6. Prove that, for every $n \in \mathbb{N}$, $0 \leq 7 - v_{n+1} \leq \frac{7 - v_n}{7}$. (0.5 pt)

7. Deduce by induction an upper bound for $7 - v_n$ as a function of n. Then determine, by comparison of integer powers, an index from which this upper bound guarantees $7 - v_n \leq 10^{-3}$. (1 pt)

8. Write a Python function with no argument that returns the first integer n such that $7 - v_n \leq 10^{-3}$. Run the function and compare the index obtained with the sufficient index from the previous question. Explain any difference. (0.5 pt)

Exercise 2 · An exponential curve — 5 points

Consider the function f defined on $[0; +\infty[$ by $f(x) = (x + 5)e^{-x}$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Calculate $f''(x)$, then study the convexity of f. (0.5 pt)

4. Prove that the equation $f(x) = \frac{5}{2}$ has a unique solution α in $[0; +\infty[$. (1 pt)

5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (0.5 pt)

6. Show that $0 < \alpha < 2$. Write a Python function that applies bisection on $[0; 2]$ and returns an interval containing α of width at most 10^{-3}. (0.5 pt)

7. Determine the minimum number of bisection steps guaranteeing this width. (0.5 pt)

8. A student claims: "Since $f(x)$ tends to 0, the curve crosses the x-axis." Refute this statement by studying the sign of $f(x)$. (1 pt)

Mock exam 1 · Zero 1 · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=4\vec{\imath}+6\vec{\jmath}+10\vec{k}, \quad \vec{z}=4\vec{\imath}+6\vec{\jmath}+11\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (0.5 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (0.5 pt)

4. Express the vector $6\vec{u}-4\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (0.5 pt)

6. For questions 6 to 8, assume the basis is orthonormal. Calculate the distances AB, AC and BC. (0.5 pt)

7. Let I the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$, then determine whether the points A, I and D are collinear. (0.5 pt)

8. Define E by $\overrightarrow{AE}=\vec{z}$. Is the line (DE) contained in the plane (ABC) ? Justify. (1 pt)

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 19 times. The trials are independent and the probability of success is $p=\frac{3}{10}$ at each trial. Let X be the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. (0.5 pt)

2. Calculate the probability of exactly two successes. (0.5 pt)

3. Using complementary events, express and then calculate the probability of obtaining at least two successes. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same trial $N\geq 1$ times, still independently. Determine the smallest integer N for which the probability of getting at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. (0.5 pt)

6. For an integer $j\in\{0,\dots,19\}$, prove the formula for $P(X=j)$ by counting the possible positions of successes. (0.5 pt)

7. A series of 19 trials is accepted if its number of successes is between 1 and 3, endpoints included. Calculate the probability that a series is accepted. (0.5 pt)

8. Write a Python function with no argument that returns the smallest integer j such that $P(X\leq j)\geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Solutions · Mock exam 1 · Specimen 1

Exercise 1 · Induction and bounds

1. [0.25 pt] $v_1 = \sqrt{42+0} = \sqrt{42}$ and $v_2 = \sqrt{42+\sqrt{42}}$.

2. [0.75 pt] Let $P(n)$: "v_n is defined and $0 \le v_n \le 7$" .

   Base case. $v_0 = 0$ is defined and belongs to $[0,7]$; $P(0)$ is true.

   Inductive step. Fix $n \in \mathbb{N}$ and suppose $P(n)$ true. Then

   $$0 \le 42 \le 42 + v_n \le 42 + 7 = 49.$$

   The square root is therefore defined and, since this function is increasing on $[0,+\infty[$,

   $$0 \le v_{n+1} = \sqrt{42+v_n} \le \sqrt{49} = 7.$$

   Thus $P(n+1)$ is true.

   Conclusion. By induction, all terms are defined and belong to $[0,7]$.

3. [0.75 pt] For $n \in \mathbb{N}$, let $R(n)$: "$v_{n+1} \ge v_n$" .

   Base case. $v_1 = \sqrt{42} > 0 = v_0$; $R(0)$ is true.

   Inductive step. Fix $n \ge 0$ and suppose $v_{n+1} \ge v_n$. Adding 42 and then using the fact that the square root is increasing,

   $$v_{n+2} = \sqrt{42+v_{n+1}} \ge \sqrt{42+v_n} = v_{n+1}.$$

   This is $R(n+1)$.

   Conclusion. By induction, $v_{n+1} \ge v_n$ for all $n \ge 0$: (v_n) is increasing.

4. [0.5 pt] Let $S(n)$: "$v_n < 7$". At index 0, $v_0 = 0 < 7$. Fix $n \ge 0$ and suppose $S(n)$ true. Since the radicands are positive and the square root is strictly increasing, we get

   $$v_{n+1} = \sqrt{42+v_n} < \sqrt{42+7} = \sqrt{49} = 7.$$

   Hence $S(n+1)$ is true. The base case and the inductive step establish by induction that $v_n < 7$ for every $n \in \mathbb{N}$.

5. [0.75 pt] The sequence is increasing and bounded above by 7: the monotone convergence theorem gives a finite limit $\ell \in [0,7]$. The function $x \mapsto \sqrt{42+x}$ is continuous on this interval. Taking the limit in the relation, $\ell = \sqrt{42+\ell}$. Since $\ell \ge 0$, this is equivalent to

   $$\ell^2 - \ell - 42 = 0 \quad\Longleftrightarrow\quad (\ell-7)(\ell+6) = 0.$$

   The root -6 is excluded by $\ell \ge 0$; therefore $\lim v_n = 7$.

6. [0.5 pt] The previous result ensures $7 - v_{n+1} \ge 0$. Multiplying by the conjugate,

   $$\begin{aligned} 7 - v_{n+1} &= \frac{49 - (42+v_n)}{7+\sqrt{42+v_n}} \\ &= \frac{7-v_n}{7+\sqrt{42+v_n}} \le \frac{7-v_n}{7}. \end{aligned}$$

   The last inequality uses a positive numerator and a denominator at least equal to $7 > 0$.

7. [1 pt] Let us prove by induction $T(n)$: "$0 \leq 7 - v_n \leq 7^{1-n}$ ". At index 0, $7 - v_0 = 7 = 7^1$. Fix $n \geq 0$ and suppose $T(n)$ true. The previous question gives

$$0 \leq 7 - v_{n+1} \leq \frac{7 - v_n}{7} \leq \frac{7^{1-n}}{7} = 7^{1-(n+1)}.$$

This is $T(n+1)$. By induction, the bounds hold at every index. To have $7^{1-n} \leq 10^{-3}$, it suffices if $7^{n-1} \geq 1000$. Now $7^3 = 343 < 1000$ and $7^4 = 2401 \geq 1000$. Index 5 is therefore the first guaranteed by this upper bound, without using logarithms.

8. [0.5 pt]

```
from math import sqrt

def premier_rang():
    n, v = 0, 0.0
    while 7-v > 1e-3:
        v = sqrt(42+v)
        n += 1
    return n
```

Convergence to 7 guarantees termination. The loop examines the indices in order: it returns 4, the first index satisfying the criterion for the computed values. Here, the gaps at indices 3 and 4 are about 0 ,00265684 and 0,00018978. They lie clearly on either side of 10^{-3}. Guaranteed index 5 may be larger; it comes from an upper bound, not from the exact error.

Exercise 2 · An exponential curve

1. [0.5 pt] f $(0) = 5$; limit 0 by comparing growth rates.

2. [0.5 pt] By the product rule,

$$f'(x) = e^{-x} - (x+5)e^{-x} = (-x-4)e^{-x}.$$

The two factors are respectively strictly negative and strictly positive on $[0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	5	$\searrow$	0

3. [0.5 pt] Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-4)e^{-x} = (x+3)e^{-x}.$$

On the domain, $x + 3 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity on the interior of the domain, so no inflection point.

4. [1 pt] Continuity, strict decrease, $f(0) = 5 > \frac{5}{2}$ and limit $0 < \frac{5}{2}$: exactly one solution.

5. [0.5 pt] The tangent at 0 has equation $y = f(0) + f'(0)x = 5 - 4x$. A convex function is above each of its tangents, so $(x+5)e^{-x} \geq 5 - 4x$ for all $x \geq 0$.

6. [0.5 pt] We know $f(0) = 5 > \frac{5}{2}$. In addition, $f(2) = 7e^{-2} \simeq 0{,}9473 < \frac{5}{2}$. The zero of $f - \frac{5}{2}$ is therefore in $]0,2[$. A program is:

```
from math import exp

def encadrement():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+5)*exp(-m) > 5/2:
            a = m
        else :
            b = m
    return a, b
```

As f is decreasing, the bracketing is preserved at each step. The result has length at most 10^{-3} ; its midpoint has an error of at most 5×10^{-4}.

7. [0.5 pt] $\frac{2}{2^n} \leq 10^{-3}$: $n = 11$ steps suffice.

8. [1 pt] For $x \geq 0$, $x + 5 > 0$ and $e^{-x} > 0$, therefore $f(x) > 0$. A limit is not necessarily a attained value.

Exercise 3 · Vector calculation in a basis

1. [0.5 pt] In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, contradicting the uniqueness of the coordinates in a basis.

2. [0.5 pt] Expanding, 4 $\vec{u} + 6\vec{v} = 4(\vec{\imath} + \vec{k}) + 6(\vec{\jmath} + \vec{k}) = 4\vec{\imath} + 6\vec{\jmath} + 10\vec{k} = \vec{w}$.

3. [0.5 pt] Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis forces

$$\lambda = 4, \qquad \mu = 6, \qquad \lambda + \mu = 11.$$

The first two equalities give $\lambda + \mu = 10$, contradicting the third. The three vectors are therefore not coplanar.

4. [1 pt] $6\vec{u} - 4\vec{v} = 6(\vec{\imath} + \vec{k}) - 4(\vec{\jmath} + \vec{k}) = 6\vec{\imath} - 4\vec{\jmath} + 2\vec{k}$.

5. [0.5 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

6. [0.5 pt] AB $= AC = \sqrt{2}$; $\overrightarrow{BC} = \vec{v} - \vec{u} = (-1, 1, 0)$, therefore $BC = \sqrt{2}$.

7. [0.5 pt] The midpoint formula gives $\overrightarrow{AI} = \frac{1}{2}(\vec{u} + \vec{v})$. The vectors $\vec{u}, \vec{v}$ being non-collinear,

$$\overrightarrow{AD} = \lambda\overrightarrow{AI} \iff 4 = \frac{\lambda}{2} \text{ and } 6 = \frac{\lambda}{2}.$$

Here $4 \neq 6$: there is no such λ, therefore A, I, D are not collinear.

8. [1 pt] No: D lies in the plane but E does not, by the non-coplanarity proved above. The intersection is exactly D.

Exercise 4 · Repetitions and estimation

1. [0.5 pt] Repetition of a fixed number 19 of identical independent trials with two outcomes; $X \sim \mathcal{B}(19, \frac{3}{10})$.

2. [0.5 pt] P $(X = 2) = \binom{19}{2}(\frac{3}{10})^2(\frac{7}{10})^{17} \simeq 0{,}035802$.

3. [0.5 pt] $1 - P(X = 0) - P(X = 1) = 1 - (\frac{7}{10})^{19} - 19(\frac{3}{10})(\frac{7}{10})^{18} \simeq 0{,}989578$.

4. [1 pt] E $(X) = \frac{57}{10}$, $V(X) = \frac{399}{100}$.

5. [0.5 pt] For N independent trials, the probability of no success is $(1-p)^N$. The required prob. is therefore $1 - (\frac{7}{10})^N$. It is strictly increasing with N, because $0 < \frac{7}{10} < 1$. The calculations give

$$P_8 \simeq 0{,}942352 < 0{,}95, \qquad P_9 \simeq 0{,}959646 \geq 0{,}95.$$

Thus 9 is the smallest suitable number of trials.

6. [0.5 pt] Let $n = 19$. For $j \in \{0, \dots, n\}$, an ordered outcome containing j successes & $n - j$ failures has, by independence, the probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices of positions of successes. These outcomes are disjoint, therefore

$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{3}{10}.$$

7. [0.5 pt] The events $X = 1$, $X = 2$, $X = 3$ are disjoint, so

$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{19}{j} (\frac{3}{10})^j (\frac{7}{10})^{19-j} \simeq 0{,}132031.$$

8. [1 pt]

```
from math import comb

def quantile():
    n, p = 19, 3/10
    j, cumul = 0, (1-p)**n
    while cumul < 0.95:
        j += 1
        cumul += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The added probabilities are positive and the final cumulative total is 1, so the loop stops. It returns 9. The cumulative probabilities at 8 and 9 are about 0.916085 and 0.967447: this confirms minimality.

Mock exam 1 · Zero 2

SPECIMEN PAPER · PRACTICE

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Induction and bounds — 5 points

Consider the sequence (v_n) defined by $v_0 = 0$ and, for every natural number n, $v_{n+1} = \sqrt{110 + v_n}$.

1. Calculate v_1 and v_2. Give exact values. (0.25 pt)

2. Prove by induction that all terms are defined and that, for every $n \in \mathbb{N}$, $0 \leq v_n \leq 11$. (0.75 pt)

3. Prove by induction that, for every $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (0.75 pt)

4. Prove that, for every $n \in \mathbb{N}$, $v_n < 11$. (0.5 pt)

5. Justify that (v_n) converges, then determine its limit. (0.75 pt)

6. Prove that, for every $n \in \mathbb{N}$, $0 \leq 11 - v_{n+1} \leq \frac{11 - v_n}{11}$. (0.5 pt)

7. Deduce by induction an upper bound for $11 - v_n$ as a function of n. Then determine, by comparison of integer powers, an index from which this upper bound guarantees $11 - v_n \leq 10^{-3}$. (1 pt)

8. Write a Python function with no argument that returns the first integer n such that $11 - v_n \leq 10^{-3}$. Run the function and compare the index obtained with the sufficient index from the previous question. Explain any difference. (0.5 pt)

Exercise 2 · An exponential curve — 5 points

Consider the function f defined on $[0; +\infty[$ by $f(x) = (x + 4)e^{-x}$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Calculate $f''(x)$, then study the convexity of f. (0.5 pt)

4. Prove that the equation $f(x) = 2$ has a unique solution α in $[0; +\infty[$. (1 pt)

5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (0.5 pt)

6. Show that $0 < \alpha < 2$. Write a Python function that applies bisection on $[0; 2]$ and returns an interval containing α of width at most 10^{-3}. (0.5 pt)

7. Determine the minimum number of bisection steps guaranteeing this width. (0.5 pt)

8. A student claims: "Since $f(x)$ tends to 0, the curve crosses the x-axis." Refute this statement by studying the sign of $f(x)$. (1 pt)

Mock exam 1 · Zero 2 · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=3\vec{\imath}+8\vec{\jmath}+11\vec{k}, \quad \vec{z}=3\vec{\imath}+8\vec{\jmath}+12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (0.5 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (0.5 pt)

4. Express the vector 8 $\vec{u}-3\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (0.5 pt)

6. For questions 6 to 8, assume the basis is orthonormal. Calculate the distances AB, AC and BC. (0.5 pt)

7. Let I the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$, then determine whether the points A, I and D are collinear. (0.5 pt)

8. Define E by $\overrightarrow{AE}=\vec{z}$. Is the line (DE) contained in the plane (ABC) ? Justify. (1 pt)

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 16 times. The trials are independent dent and the probability of success is $p=\frac{1}{10}$ at each trial. Let X be the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. (0.5 pt)

2. Calculate the probability of exactly two successes. (0.5 pt)

3. Using complementary events, express and then calculate the probability of obtaining at least two successes. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same trial $N\geq 1$ times, still independently. Determine the smallest integer N for which the probability of getting at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. (0.5 pt)

6. For an integer $j\in\{0,\dots,16\}$, prove the formula for $P(X=j)$ by counting the possible positions of successes. (0.5 pt)

7. A series of 16 trials is accepted if its number of successes is between 1 and 3, endpoints included. Calculate the probability that a series is accepted. (0.5 pt)

8. Write a Python function with no argument that returns the smallest integer j such that $P(X\leq j)\geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Solutions · Mock exam 1 · Specimen 2

Exercise 1 · Induction and bounds

1. [0.25 pt] $v_1 = \sqrt{110+0} = \sqrt{110}$ and $v_2 = \sqrt{110+\sqrt{110}}$.

2. [0.75 pt] Let $P(n)$: "v_n is defined and $0 \le v_n \le 11$" .

   Base case. $v_0 = 0$ is defined and belongs to $[0,11]$; $P(0)$ is true.

   Inductive step. Fix $n \in \mathbb{N}$ and suppose $P(n)$ true. Then

   $$0 \le 110 \le 110 + v_n \le 110 + 11 = 121.$$

   The square root is therefore defined and, since this function is increasing on $[0,+\infty[$,

   $$0 \le v_{n+1} = \sqrt{110+v_n} \le \sqrt{121} = 11.$$

   Thus $P(n+1)$ is true.

   Conclusion. By induction, all terms are defined and belong to $[0,11]$.

3. [0.75 pt] For $n \in \mathbb{N}$, let $R(n)$: "$v_{n+1} \ge v_n$" .

   Base case. $v_1 = \sqrt{110} > 0 = v_0$; $R(0)$ is true.

   Inductive step. Fix $n \ge 0$ and suppose $v_{n+1} \ge v_n$. Adding 110 and then using the fact that the square root is increasing,

   $$v_{n+2} = \sqrt{110+v_{n+1}} \ge \sqrt{110+v_n} = v_{n+1}.$$

   This is $R(n+1)$.

   Conclusion. By induction, $v_{n+1} \ge v_n$ for all $n \ge 0$: (v_n) is increasing.

4. [0.5 pt] Let $S(n)$: "$v_n < 11$". At index 0, $v_0 = 0 < 11$. Fix $n \ge 0$ and suppose $S(n)$ true. Since the radicands are positive and the square root is strictly increasing, we get

   $$v_{n+1} = \sqrt{110+v_n} < \sqrt{110+11} = \sqrt{121} = 11.$$

   Hence $S(n+1)$ is true. The base case and the inductive step establish by induction that $v_n < 11$ for every $n \in \mathbb{N}$.

5. [0.75 pt] The sequence is increasing and bounded above by 11: the monotone convergence theorem gives a finite limit $\ell \in [0,11]$. The function $x \mapsto \sqrt{110+x}$ is continuous on this interval. Taking the limit in the relation, $\ell = \sqrt{110+\ell}$. Since $\ell \ge 0$, this is equivalent to

   $$\ell^2 - \ell - 110 = 0 \iff (\ell-11)(\ell+10) = 0.$$

   The root -10 is excluded by $\ell \ge 0$; therefore $\lim v_n = 11$.

6. [0.5 pt] The previous result ensures $11 - v_{n+1} \ge 0$. Multiplying by the conjugate,

   $$\begin{aligned} 11 - v_{n+1} &= \frac{121 - (110+v_n)}{11+\sqrt{110+v_n}} \\ &= \frac{11-v_n}{11+\sqrt{110+v_n}} \le \frac{11-v_n}{11}. \end{aligned}$$

   The last inequality uses a positive numerator and a denominator at least equal to $11 > 0$.

7. [1 pt] Let us prove by induction $T(n)$: "$0 \leq 11 - v_n \leq 11^{1-n}$ ". At index 0, $11 - v_0 = 11 = 11^1$. Fix $n \geq 0$ and suppose $T(n)$ true. The previous question gives

$$0 \leq 11 - v_{n+1} \leq \frac{11 - v_n}{11} \leq \frac{11^{1-n}}{11} = 11^{1-(n+1)}.$$

This is $T(n+1)$. By induction, the bounds hold at every index. To have $11^{1-n} \leq 10^{-3}$, it suffices if $11^{n-1} \geq 1000$. Now $11^2 = 121 < 1000$ and $11^3 = 1331 \geq 1000$. Index 4 is therefore the first guaranteed by this upper bound, without using logarithms.

8. [0.5 pt]

```
from math import sqrt

def premier_rang():
    n, v = 0, 0.0
    while 11-v > 1e-3:
        v = sqrt(110+v)
        n += 1
    return n
```

Convergence to 11 guarantees termination. The loop examines the indices in order: it returns 4, the first index satisfying the criterion for the computed values. Here, the gaps at indices 3 and 4 are about 0,00105884 and 0,00004813. They lie clearly on either side of 10^{-3}. Guaranteed index 4 may be larger; it comes from an upper bound, not from the exact error.

Exercise 2 · An exponential curve

1. [0.5 pt] $f(0) = 4$; limit 0 by comparing growth rates.

2. [0.5 pt] By the product rule,

$$f'(x) = e^{-x} - (x+4)e^{-x} = (-x-3)e^{-x}.$$

The two factors are respectively strictly negative and strictly positive on $[0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	4	$\searrow$	0

3. [0.5 pt] Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-3)e^{-x} = (x+2)e^{-x}.$$

On the domain, $x + 2 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity on the interior of the domain, so no inflection point.

4. [1 pt] Continuity, strict decrease, $f(0) = 4 > 2$ and limit $0 < 2$: exactly one solution.

5. [0.5 pt] The tangent at 0 has equation $y = f(0) + f'(0)x = 4 - 3x$. A convex function is above each of its tangents, so $(x+4)e^{-x} \geq 4 - 3x$ for all $x \geq 0$.

6. [0.5 pt] We know $f(0) = 4 > 2$. Moreover, $f(2) = 6e^{-2} \simeq 0{,}8120 < 2$. The zero of $f - 2$ is therefore in $]0, 2[$. A program is:

```
from math import exp

def encadrement():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+4)*exp(-m) > 4/2:
            a = m
        else :
            b = m
    return a, b
```

As f is decreasing, the bracketing is preserved at each step. The result has length at most 10^{-3} ; its midpoint has an error of at most 5×10^{-4}.

7. [0.5 pt] $\frac{2}{2^n} \leq 10^{-3}$: $n = 11$ steps suffice.

8. [1 pt] For $x \geq 0$, $x + 4 > 0$ and $e^{-x} > 0$, therefore $f(x) > 0$. A limit is not necessarily a attained value.

Exercise 3 · Vector calculation in a basis

1. [0.5 pt] In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, contradicting the uniqueness of the coordinates in a basis.

2. [0.5 pt] Expanding, 3 $\vec{u} + 8\vec{v} = 3(\vec{\imath} + \vec{k}) + 8(\vec{\jmath} + \vec{k}) = 3\vec{\imath} + 8\vec{\jmath} + 11\vec{k} = \vec{w}$.

3. [0.5 pt] Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis forces

$$\lambda = 3, \qquad \mu = 8, \qquad \lambda + \mu = 12.$$

The first two equalities give $\lambda + \mu = 11$, contradicting the third. The three vectors are therefore not coplanar.

4. [1 pt] $8\vec{u} - 3\vec{v} = 8(\vec{\imath} + \vec{k}) - 3(\vec{\jmath} + \vec{k}) = 8\vec{\imath} - 3\vec{\jmath} + 5\vec{k}$.

5. [0.5 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

6. [0.5 pt] AB $= AC = \sqrt{2}$; $\overrightarrow{BC} = \vec{v} - \vec{u} = (-1, 1, 0)$, therefore $BC = \sqrt{2}$.

7. [0.5 pt] The midpoint formula gives $\overrightarrow{AI} = \frac{1}{2}(\vec{u} + \vec{v})$. The vectors $\vec{u}, \vec{v}$ being non-collinear,

$$\overrightarrow{AD} = \lambda\overrightarrow{AI} \iff 3 = \frac{\lambda}{2} \text{ and } 8 = \frac{\lambda}{2}.$$

Here $3 \neq 8$: there is no such λ, therefore A, I, D are not collinear.

8. [1 pt] No: D lies in the plane but E does not, by the non-coplanarity proved above. The intersection is exactly D.

Exercise 4 · Repetitions and estimation

1. [0.5 pt] Repetition of a fixed number 16 of identical independent trials with two outcomes; $X \sim \mathcal{B}(16, \frac{1}{10})$.

2. [0.5 pt] P $(X = 2) = \binom{16}{2}(\frac{1}{10})^2(\frac{9}{10})^{14} \simeq 0{,}274522$.

3. [0.5 pt] $1 - P(X = 0) - P(X = 1) = 1 - (\frac{9}{10})^{16} - 16(\frac{1}{10})(\frac{9}{10})^{15} \simeq 0{,}485272$.

4. [1 pt] E $(X) = \frac{8}{5}$, $V(X) = \frac{36}{25}$.

5. [0.5 pt] For N independent trials, the probability of no success is $(1-p)^N$. The required prob. is therefore $1 - (\frac{9}{10})^N$. It is strictly increasing with N, because $0 < \frac{9}{10} < 1$. The calculations give

$$P_{28} \simeq 0{,}947665 < 0{,}95, \qquad P_{29} \simeq 0{,}952899 \geq 0{,}95.$$

Thus 29 is the smallest suitable number of trials.

6. [0.5 pt] Let $n = 16$. For $j \in \{0, \dots, n\}$, an ordered outcome containing j successes & $n - j$ failures has, by independence, the probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices of positions of successes. These outcomes are disjoint, therefore

$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{1}{10}.$$

7. [0.5 pt] The events $X = 1$, $X = 2$, $X = 3$ are disjoint, so

$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{16}{j} (\frac{1}{10})^j (\frac{9}{10})^{16-j} \simeq 0{,}746292.$$

8. [1 pt]

```
from math import comb

def quantile():
    n, p = 16, 1/10
    j, cumul = 0, (1-p)**n
    while cumul < 0.95:
        j += 1
        cumul += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The added probabilities are positive and the final cumulative total is 1, so the loop stops. It returns 4. The cumulative probabilities at 3 and 4 are about 0.931594 and 0.982996: this confirms minimality.

Mock exam 1 · A

PAPER RESERVED FOR ASSESSMENT

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Two arguments about a sequence — 4 points

We define $u_0 = 1$, $u_{n+1} = \sqrt{2 + u_n}$.

1. Prove by induction that $1 \leq u_n < 2$. (1 pt)

2. Show that $u_{n+1} \geq u_n$, then that the sequence converges. (1 pt)

3. Determine its limit, stating the assumption used to pass to the limit. (1 pt)

4. Show $0 \leq 2 - u_{n+1} \leq \frac{1}{3}(2 - u_n)$; explain the advantage of this information over a mere conjecture. (1 pt)

Exercise 2 · An exponential curve — 5 points

Consider the function f defined on $[0; +\infty[$ by $f(x) = (x + 3)e^{-x}$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Calculate $f''(x)$, then study the convexity of f. (0.5 pt)

4. Prove that the equation $f(x) = \frac{3}{2}$ has a unique solution α in $[0; +\infty[$. (1 pt)

5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (0.5 pt)

6. Show that $0 < \alpha < 2$. Write a Python function that applies bisection on $[0; 2]$ and returns an interval containing α of width at most 10^{-3}. (0.5 pt)

7. Determine the minimum number of bisection steps guaranteeing this width. (0.5 pt)

8. A student claims: "Since $f(x)$ tends to 0, the curve crosses the x-axis." Refute this statement by studying the sign of $f(x)$. (1 pt)

Mock exam 1 · A · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=2\vec{\imath}+5\vec{\jmath}+7\vec{k}, \quad \vec{z}=2\vec{\imath}+5\vec{\jmath}+8\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (0.5 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (0.5 pt)

4. Express the vector 5 $\vec{u}-2\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (0.5 pt)

6. For questions 6 to 8, assume the basis is orthonormal. Calculate the distances AB, AC and BC. (0.5 pt)

7. Let I the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$, then determine whether the points A, I and D are collinear. (0.5 pt)

8. Define E by $\overrightarrow{AE}=\vec{z}$. Is the line (DE) contained in the plane (ABC) ? Justify. (1 pt)

Exercise 4 · Two workshops and one order — 6 points

A component comes from A with probability 0 ,6, otherwise from B. The defect rates are respectively 0,02 and 0,05. The components of an order are assumed to be independent and to come from this same production.

1. Construct the tree and determine the probability p of a defect. (1 pt)

2. A component is defective. Calculate the probability that it comes from B. (1 pt)

3. For 20 components, define a binomial variable and calculate the probability of at least one defect. (1 pt)

4. Determine the expected number of defects. Interpret. (1 pt)

5. Write a Python function giving the smallest order size for which the probability of at least one defect exceeds 0.5, without logarithms. Justify stopping. (2 pts)

Solutions · Mock exam 1 · A

Exercise 1 · Two arguments about a sequence

1. $u_0 = 1 \in [1;2[$. If $1 \leq u_n < 2$, as the square root is increasing we get $\sqrt{3} \leq u_{n+1} < 2$, hence the invariant at every index.

2. $u_{n+1}^2 - u_n^2 = 2 + u_n - u_n^2 = (2 - u_n)(u_n + 1) > 0$. Both terms are positive, so $u_{n+1} > u_n$; being increasing and bounded above implies convergence.

3. By continuity of $x \mapsto \sqrt{2+x}$ on $[1;2]$, $\ell = \sqrt{2+\ell}$. Hence $(\ell - 2)(\ell + 1) = 0$; $\ell \geq 1$ requires $\ell = 2$.

4. Conjugate: $2 - u_{n+1} = (2 - u_n)/(2 + \sqrt{2+u_n})$. The denominator is at least 3, hence the bound. It controls the error at every index, not only for the first ones calculated.

Exercise 2 · An exponential curve

1. [0.5 pt] f $(0) = 3$; limit 0 by comparing growth rates.

2. [0.5 pt] By the product rule,

$$f'(x) = e^{-x} - (x+3)e^{-x} = (-x-2)e^{-x}.$$

The two factors are respectively strictly negative and strictly positive on $[0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	3	$\searrow$	0

3. [0.5 pt] Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-2)e^{-x} = (x+1)e^{-x}.$$

On the domain, $x + 1 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity on the interior of the domain, so no inflection point.

4. [1 pt] Continuity, strict decrease, $f(0) = 3 > \frac{3}{2}$ and limit $0 < \frac{3}{2}$: exactly one solution.

5. [0.5 pt] The tangent at 0 has equation $y = f(0) + f'(0)x = 3 - 2x$. A convex function is above each of its tangents, so $(x+3)e^{-x} \geq 3 - 2x$ for all $x \geq 0$.

6. [0.5 pt] We know $f(0) = 3 > \frac{3}{2}$. In addition, $f(2) = 5e^{-2} \simeq 0{,}6767 < \frac{3}{2}$. The zero of $f - \frac{3}{2}$ is therefore in $]0,2[$. A program is:

```
from math import exp

def encadrement():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+3)*exp(-m) > 3/2:
            a = m
        else :
            b = m
    return a, b
```

As f is decreasing, the bracketing is preserved at each step. The result has length at most 10^{-3} ; its midpoint has an error of at most 5×10^{-4}.

7. [0.5 pt] $\frac{2}{2^n} \leq 10^{-3}$: $n = 11$ steps suffice.

8. [1 pt] For $x \geq 0$, $x + 3 > 0$ and $e^{-x} > 0$, therefore $f(x) > 0$. A limit is not necessarily a attained value.

Exercise 3 · Vector calculation in a basis

1. [0.5 pt] In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, contradicting the uniqueness of the coordinates in a basis.

2. [0.5 pt] Expanding, 2 $\vec{u} + 5\vec{v} = 2(\vec{\imath} + \vec{k}) + 5(\vec{j} + \vec{k}) = 2\vec{\imath} + 5\vec{j} + 7\vec{k} = \vec{w}$.

3. [0.5 pt] Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis forces

$$\lambda = 2, \qquad \mu = 5, \qquad \lambda + \mu = 8.$$

The first two equalities give $\lambda + \mu = 7$, contradicting the third. The three vectors are therefore not coplanar.

4. [1 pt] $5\vec{u} - 2\vec{v} = 5(\vec{\imath} + \vec{k}) - 2(\vec{j} + \vec{k}) = 5\vec{\imath} - 2\vec{j} + 3\vec{k}$.

5. [0.5 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

6. [0.5 pt] AB $= AC = \sqrt{2}$; $\overrightarrow{BC} = \vec{v} - \vec{u} = (-1, 1, 0)$, therefore $BC = \sqrt{2}$.

7. [0.5 pt] The midpoint formula gives $\overrightarrow{AI} = \frac{1}{2}(\vec{u} + \vec{v})$. The vectors $\vec{u}, \vec{v}$ being non-collinear,

$$\overrightarrow{AD} = \lambda\overrightarrow{AI} \iff 2 = \frac{\lambda}{2} \text{ and } 5 = \frac{\lambda}{2}.$$

Here $2 \neq 5$: there is no such λ, therefore A, I, D are not collinear.

8. [1 pt] No: D lies in the plane but E does not, by the non-coplanarity proved above. The intersection is exactly D.

Exercise 4 · Two workshops and one order

1. Tree: A/0 ,6 then defective/0 ,02 ; B/0,4 then defective/0 ,05, with the complements. $p = 0{,}6(0{,}02) + 0{,}4(0{,}05) = 0{,}032$.

2. $P_D(B) = 0{,}4(0{,}05)/0{,}032 = 0{,}625$.

3. X counts the defects: $X \sim \mathcal{B}(20; 0{,}032)$. $P(X \geq 1) = 1 - 0{,}968^{20}$.

4. $E(X) = 20(0{,}032) = 0.64$ defects per order on average over many orders; this is not a number guaranteed in each one.

5. Program (1 pt):

```
def taille():
    n, aucun = 0, 1.0
    while 1 - aucun <= 0.5:
        n += 1
        aucun *= 0.968
    return n
```

$0{,}968^n \to 0$, so the threshold is crossed; as it is decreasing, the first index is guaranteed (1 pt). Result: 22; check indices 21 and 22.

Mock exam 1 · B

PAPER RESERVED FOR ASSESSMENT

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A calculation that never reaches its target — 4 points

For $n \in \mathbb{N}$, we define $v_0 = 0$, $v_{n+1} = (v_n + 3)/2$.

1. Show by induction that $v_n = 3 - 3(1/2)^n$. (1 pt)

2. Determine the monotonicity and the limit. (1 pt)

3. A program repeats the update as long as $v < 3$. Why does it never terminate in exact ar.? (1 pt)

4. Suggest a stopping condition that guarantees $|v_n - 3| < 10^{-3}$ and determine the first index that works. (1 pt)

Exercise 2 · An exponential curve — 5 points

Consider the function f defined on $[0;+\infty[$ by $f(x) = (x+2)e^{-x}$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Calculate $f''(x)$, then study the convexity of f. (0.5 pt)

4. Prove that the equation $f(x) = 1$ has a unique solution α in $[0;+\infty[$. (1 pt)

5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (0.5 pt)

6. Show that $0 < \alpha < 2$. Write a Python function that applies bisection on $[0;2]$ and returns an interval containing α of width at most 10^{-3}. (0.5 pt)

7. Determine the minimum number of bisection steps guaranteeing this width. (0.5 pt)

8. A student claims: "Since $f(x)$ tends to 0, the curve crosses the x-axis." Refute this statement by studying the sign of $f(x)$. (1 pt)

Mock exam 1 · B · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=5\vec{\imath}+6\vec{\jmath}+11\vec{k}, \quad \vec{z}=5\vec{\imath}+6\vec{\jmath}+12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (0.5 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (0.5 pt)

4. Express the vector 6 $\vec{u}-5\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (0.5 pt)

6. For questions 6 to 8, assume the basis is orthonormal. Calculate the distances AB, AC and BC. (0.5 pt)

7. Let I the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$, then determine whether the points A, I and D are collinear. (0.5 pt)

8. Define E by $\overrightarrow{AE}=\vec{z}$. Is the line (DE) contained in the plane (ABC) ? Justify. (1 pt)

Exercise 4 · Recovering an unknown probability — 6 points

A two-outcome experiment is repeated independently three times, with probability of success $p\in[0;1]$ constant. Let X be the number of successes and $q(p)=P(X\geq 2)$.

1. Justify the model and establish $q(p)=3p^2-2p^3$. (1 pt)

2. Study q on $[0;1]$. (1 pt)

3. Prove that $q(p)=0{,}8$ has a unique solution and bound it between two consecutive hundredths. (2 pts)

4. If $p=0.8$, calculate $E(X)$ and $P(X=0)$. (1 pt)

5. Another protocol stops at the first success. Does the number of successes obtained still follow $\mathcal{B}(3;p)$? Justify for $0<p<1$. (1 pt)

Solutions · Mock exam 1 · B

Exercise 1 · A calculation that never reaches its target

1. Base case: $3 - 3 = 0$. If the formula holds at index n fixed, $(v_n + 3)/2 = 3 - 3(1/2)^{n+1}$. By induction it holds for every index.

2. $v_{n+1} - v_n = \frac{3}{2}(1/2)^n > 0$; $v_n \to 3$ since $(1/2)^n \to 0$.

3. The formula gives $v_n < 3$ for every integer: the threshold is never reached. Rounding by a computer may change this behaviour, without contradicting the proof.

4. Continue while $3 - v \geq 10^{-3}$. The error is $3/2^n$; $2^{11} = 2048 < 3000 < 4096 = 2^{12}$, hence the first index 12.

Exercise 2 · An exponential curve

1. [0.5 pt] f $(0) = 2$; limit 0 by comparing growth rates.

2. [0.5 pt] By the product rule,

$$f'(x) = e^{-x} - (x+2)e^{-x} = (-x-1)e^{-x}.$$

The two factors are respectively strictly negative and strictly positive on $[0, +\infty[$; therefore $f' < 0$.

x	0		$+\infty$
$f'(x)$		$-$	
$f(x)$	2	$\searrow$	0

3. [0.5 pt] Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-1)e^{-x} = xe^{-x}.$$

On the domain, $x + 0 \geq 0$ and $e^{-x} > 0$: f is convex. There is no change of convexity on the interior of the domain, so no inflection point.

4. [1 pt] Continuity, strict decrease, $f(0) = 2 > 1$ and limit $0 < 1$: exactly one solution.

5. [0.5 pt] The tangent at 0 has equation $y = f(0) + f'(0)x = 2 - x$. A convex function is above each of its tangents, so $(x+2)e^{-x} \geq 2 - x$ for all $x \geq 0$.

6. [0.5 pt] We know $f(0) = 2 > 1$. Moreover, $f(2) = 4e^{-2} \simeq 0{,}5413 < 1$. The zero of $f - 1$ is therefore in $]0,2[$. A program is:

```
from math import exp

def encadrement():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+2)*exp(-m) > 2/2:
            a = m
        else :
            b = m
    return a, b
```

As f is decreasing, the bracketing is preserved at each step. The result has length at most 10^{-3} ; its midpoint has an error of at most 5×10^{-4}.

7. [0.5 pt] $\frac{2}{2^n} \leq 10^{-3}$: $n = 11$ steps suffice.

8. [1 pt] For $x \geq 0$, $x + 2 > 0$ and $e^{-x} > 0$, therefore $f(x) > 0$. A limit is not necessarily a attained value.

Exercise 3 · Vector calculation in a basis

1. [0.5 pt] In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, contradicting the uniqueness of the coordinates in a basis.

2. [0.5 pt] Expanding, $5\vec{u} + 6\vec{v} = 5(\vec{\imath} + \vec{k}) + 6(\vec{\jmath} + \vec{k}) = 5\vec{\imath} + 6\vec{\jmath} + 11\vec{k} = \vec{w}$.

3. [0.5 pt] Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis forces

$$\lambda = 5, \qquad \mu = 6, \qquad \lambda + \mu = 12.$$

The first two equalities give $\lambda + \mu = 11$, contradicting the third. The three vectors are therefore not coplanar.

4. [1 pt] $6\vec{u} - 5\vec{v} = 6(\vec{\imath} + \vec{k}) - 5(\vec{\jmath} + \vec{k}) = 6\vec{\imath} - 5\vec{\jmath} + \vec{k}$.

5. [0.5 pt] $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

6. [0.5 pt] AB $= AC = \sqrt{2}$; $\overrightarrow{BC} = \vec{v} - \vec{u} = (-1, 1, 0)$, therefore $BC = \sqrt{2}$.

7. [0.5 pt] The midpoint formula gives $\overrightarrow{AI} = \frac{1}{2}(\vec{u} + \vec{v})$. The vectors $\vec{u}, \vec{v}$ being non-collinear,

$$\overrightarrow{AD} = \lambda\overrightarrow{AI} \iff 5 = \frac{\lambda}{2} \text{ and } 6 = \frac{\lambda}{2}.$$

Here $5 \neq 6$: there is no such λ, therefore A, I, D are not collinear.

8. [1 pt] No: D lies in the plane but E does not, by the non-coplanarity proved above. The intersection is exactly D.

Exercise 4 · Reconstitute an unknown probability

1. $X \sim \mathcal{B}(3; p)$ by independence, same probability and fixed number of three trials. $q = \binom{3}{2}p^2(1 - p) + p^3 = 3p^2 - 2p^3$.

2. $q' = 6p(1-p) > 0$ on $]0;1[$; strictly increasing from 0 to 1.

3. Polynomial continuity, strict increase and $0 < 0{,}8 < 1$ give existence and uniqueness (1 pt).
   $q(0{,}71) = 0{,}796478 < 0{,}8 < q(0{,}72) = 0.808704$, therefore solution in $]0{,}71;0{,}72[$ (1 pt).

4. $E(X) = 2{,}4$; $P(X=0) = 0{,}2^3 = 0{,}008$.

5. No: at most one success is observed, as we stop at the first; the binomial distribution would give a strictly positive probability for two and three successes.

DS4 · Specimen 1

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x)=\frac{3+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (1 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)

3. State the maximum value of f, then solve $f(x)=0$. (1 pt)

4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)

5. Prove that the equation $f(x)=\frac{1}{2}e^2$ with justification. $]0;+\infty[$. (1 pt)

Exercise 2 · Antiderivatives and an imposed condition — 5 points

The first four questions are independent. For each, give the expression of the required antiderivatives and check the result by differentiation.

1. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto 2x^2-6x+2$. (1 pt)

2. Determine the antiderivative F of $x\mapsto e^{2x}$ on $\mathbb{R}$ such that $F(0)=6$. (1 pt)

3. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto\frac{2x}{x^2+6}$. (1 pt)

4. Determine an antiderivative on $\mathbb{R}$ of $x\mapsto 2x(x^2+2)^2$. (1 pt)

5. Let F and G two antiderivatives of the same function over an interval I. Prove that there exists a real constant C such that $F(x)-G(x)=C$ for all $x\in I$. (1 pt)

DS4 · Specimen 1 · continued

Exercise 3 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{5}y + \frac{8}{5}$ on $[0;+\infty[$. In questions 2 to 5, we call y the solution that satisfies $y(0) = 13$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution satisfying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ as t tends to $+\infty$. (1 pt)

4. Study the direction of variation of y, then find the smallest real number $t \geq 0$ such that $y(t) \leq 9$. (1 pt)

5. Euler's method is applied with a step $h > 0$ and let Y_k call the approx. of $y(kh)$. Give the value of Y_0 and express Y_{k+1} as a function of Y_k and of h. (1 pt)

Exercise 4 · Projection and volume — 5 points

In an orthonormal coordinate system with origin O, we consider $A(11;0;0)$, $B(0;11;0)$ and $C(0;0;11)$. Let $\vec{n} = (1;1;1)$. Lengths, areas and volumes are expressed in units of length, of area and volume.

1. Prove that $\vec{n}$ is normal to the plane (ABC). (1 pt)

2. Let H the orthogonal projection of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Prove that OH is the distance from O to the plane (ABC) : for every $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the area of the triangle ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Solutions · DS4 · Specimen 1

Exercise 1 · A logarithmic function

1. [1 pt] At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+: limit $-\infty$; at $+\infty$, limit 0 by comparing growth rates.

2. [1 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (3+\ln x)}{x^2} = \frac{-2-\ln x}{x^2}.$$

The denominator is positive. The numerator is zero at $s = e^{-2}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f strictly increases on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [1 pt] The maximum is reached only at $s = e^{-2}$ and is

$$f(s) = \frac{3-2}{e^{-2}} = e^2.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $+\ln x = 0$, i.e. $x = e^{-3}$.

4. [1 pt] We differentiate $(-2-\ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-2-\ln x)x^{-3} = \frac{2\ln x + 3}{x^3}.$$

The numerator is zero at $t = e^{-\frac{3}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at t and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. [1 pt] Let $m = \frac{e^2}{2}$ & $s = e^{-2}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and with value 2 m at s: the corollary of the intermediate value theorem gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from 2 m to 0 $< m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

Exercise 2 · Antiderivatives and an imposed condition

1. [1 pt] By linearity, the antiderivatives are $F(x) = \frac{2}{3}x^3 - 3x^2 + 2x + C$, $C \in \mathbb{R}$. Indeed, $F'(x) = 2x^2 - 6x + 2$. The derivative does not depend on C, and all antiderivatives differ by a constant on $\mathbb{R}$.

2. [1 pt] The antiderivatives are $F(x) = \frac{e^{2x}}{2} + C$. The condition $F(0) = 6$ requires $\frac{1}{2} + C = 6$, hence $C = \frac{11}{2}$. The derivative of $\frac{e^{2x}}{2}$ is e^{2x}: the function obtained does satisfy both conditions.

3. [1 pt] Let $u(x) = x^2 + 6 > 0$ on $\mathbb{R}$; $u'(x) = 2x$. The antiderivatives are therefore $\ln(x^2+6) + C$, since their derivative is $\frac{u'}{u} = \frac{2x}{x^2+6}$.

4. [1 pt] An antiderivative is $F(x) = \frac{(x^2+2)^3}{3}$. Indeed, by the chain rule,

$$F'(x) = \frac{3}{3}\,2x(x^2+2)^2 = 2x(x^2+2)^2.$$

5. [1 pt] The derivative of their difference is zero on the interval; the difference is therefore constant. The fact that it is an interval matters in this result.

Exercise 3 · Differential equation and evolution

1. [1 pt] $0 = -\frac{1}{5}y + \frac{8}{5}$ gives $y = 8$.

2. [1 pt] The constant solution is 8. Let $z = y - 8$; the equation becomes $z' = -\frac{z}{5}$. Its solutions are $z(t) = Ce^{-\frac{t}{5}}$, so all solutions of the original equation are $y(t) = 8 + Ce^{-\frac{t}{5}}$. The condition $y(0) = 13$ gives $C = 5$. For the function obtained,

$$y'(t) = -e^{-\frac{t}{5}} = -\frac{y(t)}{5} + \frac{8}{5}, \qquad y(0) = 13.$$

The solution of the initial problem is therefore $y(t) = 8 + 5e^{-\frac{t}{5}}$.

3. [1 pt] y $' = -e^{-\frac{t}{5}} < 0$; limit 8.

4. [1 pt] Since $e^{-\frac{t}{5}} > 0$, the derivative $y'(t) = -e^{-\frac{t}{5}}$ is negative: y is strictly decreasing.
Let us solve

$$\begin{aligned} y(t) \leq 9 &\iff 5e^{-\frac{t}{5}} \leq 1 \\ &\iff -\frac{t}{5} \leq -\ln 5 \\ &\iff t \geq 5\ln 5. \end{aligned}$$

The last step reverses the inequality because we multiply by $-5 < 0$. The first time is 5 $\ln 5 \simeq$ 8,0472.

5. [1 pt] Y $= 13$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{5} + \frac{8}{5})$, $t_j = jh$.

Exercise 4 · Projection and volume

1. [1 pt] $\overrightarrow{AB} = (-11, 11, 0)$ and $\overrightarrow{AC} = (-11, 0, 11)$ are not collinear and their dot products with $\vec{n}$ are zero.

2. [1 pt] The line normal to the plane through O has direction vector $\vec{n} = (1, 1, 1)$. Its point H is thus as (t, t, t). For H to lie in (ABC), it is necessary and sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t - 11) + t + t = 3t - 11 = 0.$$

Thus $t = \frac{11}{3}$ & $H = (\frac{11}{3}, \frac{11}{3}, \frac{11}{3})$. This point is in the plane and $\overrightarrow{OH}$ is collinear with the normal vector; it is indeed the required projection.

3. [1 pt] $OH = \frac{11}{\sqrt{3}}$; for every P of the plane, $OP^2 = OH^2 + HP^2 \geq OH^2$, with equality only for $P = H$.

4. [1 pt] Differences in coordinates give $AB = AC = BC = 11\sqrt{2}$. The triangle is equilateral. Its altitude cuts one side into two equal parts and equals

$$\sqrt{(11\sqrt{2})^2 - (\frac{11\sqrt{2}}{2})^2} = \frac{11\sqrt{6}}{2}.$$

Its area is therefore $\frac{1}{2} \times 11\sqrt{2} \times \frac{11\sqrt{6}}{2} = \frac{121}{2}\sqrt{3}$ units of area.

5. [1 pt] Base ABC : volume $\frac{(\frac{121}{2}\sqrt{3})(\frac{11}{\sqrt{3}})}{3} = \frac{1331}{6}$. Base OAB : area $\frac{121}{2}$ and height 11, same value.

DS4 · Specimen 2

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x)=\frac{2+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (1 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)

3. State the maximum value of f, then solve $f(x)=0$. (1 pt)

4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)

5. Prove that the equation $f(x)=\frac{1}{2}e^1$ with justification. $]0;+\infty[$. (1 pt)

Exercise 2 · Antiderivatives and an imposed condition — 5 points

The first four questions are independent. For each, give the expression of the required antiderivatives and check the result by differentiation.

1. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto 5x^2-7x+3$. (1 pt)

2. Determine the antiderivative F of $x\mapsto e^{5x}$ on $\mathbb{R}$ such that $F(0)=7$. (1 pt)

3. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto\frac{2x}{x^2+7}$. (1 pt)

4. Determine an antiderivative on $\mathbb{R}$ of $x\mapsto 2x(x^2+3)^5$. (1 pt)

5. Let F and G two antiderivatives of the same function over an interval I. Prove that there exists a real constant C such that $F(x)-G(x)=C$ for all $x\in I$. (1 pt)

DS4 · Specimen 2 · continued

Exercise 3 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{4}y + \frac{5}{2}$ on $[0;+\infty[$. In questions 2 to 5, we call y the solution that satisfies $y(0) = 16$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution satisfying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ as t tends to $+\infty$. (1 pt)

4. Study the direction of variation of y, then find the smallest real number $t \geq 0$ such that $y(t) \leq 11$. (1 pt)

5. Euler's method is applied with a step $h > 0$ and let Y_k call the approx. of $y(kh)$. Give the value of Y_0 and express Y_{k+1} as a function of Y_k and of h. (1 pt)

Exercise 4 · Projection and volume — 5 points

In an orthonormal coordinate system with origin O, we consider $A(7;0;0)$, $B(0;7;0)$ and $C(0;0;7)$. Let $\vec{n} = (1;1;1)$. Lengths, areas and volumes are expressed in units of length, of area and volume.

1. Prove that $\vec{n}$ is normal to the plane (ABC). (1 pt)

2. Let H the orthogonal projection of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Prove that OH is the distance from O to the plane (ABC) : for every $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the area of the triangle ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Solutions · DS4 · Specimen 2

Exercise 1 · A logarithmic function

1. [1 pt] At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+: limit $-\infty$; at $+\infty$, limit 0 by comparing growth rates.

2. [1 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (2+\ln x)}{x^2} = \frac{-1-\ln x}{x^2}.$$

The denominator is positive. The numerator is zero at $s = e^{-1}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f strictly increases on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [1 pt] The maximum is reached only at $s = e^{-1}$ and is

$$f(s) = \frac{2-1}{e^{-1}} = e^1.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $2 + \ln x = 0$, i.e. $x = e^{-2}$.

4. [1 pt] We differentiate $(-1-\ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-1-\ln x)x^{-3} = \frac{2\ln x + 1}{x^3}.$$

The numerator is zero at $t = e^{-\frac{1}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at t and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. [1 pt] Let $m = \frac{e^1}{2}$ & $s = e^{-1}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and with value 2 m at s: the corollary of the intermediate value theorem gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from 2 m to 0 $< m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

Exercise 2 · Antiderivatives and an imposed condition

1. [1 pt] By linearity, the antiderivatives are $F(x) = \frac{5}{3}x^3 - \frac{7}{2}x^2 + 3x + C$, $C \in \mathbb{R}$. Indeed, $F'(x) = 5x^2 - 7x + 3$. The derivative does not depend on C, and all antiderivatives differ by a constant on $\mathbb{R}$.

2. [1 pt] The antiderivatives are $F(x) = \frac{e^{5x}}{} + C$. The condition $F(0) = 7$ requires $^1 + C = 7$, hence $C = \frac{34}{5}$. The derivative of $\frac{e^{5x}}{5}$ is e^{5x}: the function obtained does satisfy both conditions.

3. [1 pt] Let $u(x) = x^2 + 7 > 0$ on $\mathbb{R}$; $u'(x) = 2x$. The antiderivatives are therefore $\ln(x^2+7) + C$, since their derivative is $\frac{u'}{u} = \frac{2x}{x^2+7}$.

4. [1 pt] An antiderivative is $F(x) = \frac{(x^2+3)^6}{6}$. Indeed, by the chain rule,

$$F'(x) = \frac{6}{6}\,2x(x^2+3)^5 = 2x(x^2+3)^5.$$

5. [1 pt] The derivative of their difference is zero on the interval; the difference is therefore constant. The fact that it is an interval matters in this result.

Exercise 3 · Differential equation and evolution

1. [1 pt] $0 = -\frac{1}{4}y + \frac{5}{2}$ gives $y = 10$.

2. [1 pt] The constant solution is 10. Let $z = y - 10$; the equation becomes $z' = -\frac{z}{4}$. Its solutions are $z(t) = Ce^{-\frac{t}{4}}$, so all solutions of the original equation are $y(t) = 10 + Ce^{-\frac{t}{4}}$. The condition $y(0) = 16$ gives $C = 6$. For the function obtained,

$$y'(t) = -\frac{3}{2}e^{-\frac{t}{4}} = -\frac{y(t)}{4} + \frac{5}{2}, \qquad y(0) = 16.$$

The solution of the initial problem is therefore $y(t) = 10 + 6e^{-\frac{t}{4}}$.

3. [1 pt] y $' = -\frac{3}{2}e^{-\frac{t}{4}} < 0$; limit 10.

4. [1 pt] Since $e^{-\frac{t}{4}} > 0$, the derivative $y'(t) = -\frac{3}{2}e^{-\frac{t}{4}}$ is negative: y is strictly decreasing.
Let us solve

$$\begin{aligned} y(t) \leq 11 &\iff 6e^{-\frac{t}{4}} \leq 1 \\ &\iff -\frac{t}{4} \leq -\ln 6 \\ &\iff t \geq 4\ln 6. \end{aligned}$$

The last step reverses the inequality because we multiply by $-4 < 0$. The first time is 4 $\ln 6 \simeq$ 7,1670.

5. [1 pt] Y $= 16$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{4} + \frac{5}{2})$, $t_j = jh$.

Exercise 4 · Projection and volume

1. [1 pt] $\overrightarrow{AB} = (-7,7,0)$ and $\overrightarrow{AC} = (-7,0,7)$ are not collinear and their dot products with $\vec{n}$ are zero.

2. [1 pt] The line normal to the plane through O has direction vector $\vec{n} = (1,1,1)$. Its point H is thus as (t,t,t). For H to lie in (ABC), it is necessary and sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t-7) + t + t = 3t - 7 = 0.$$

Thus $t = \frac{7}{3}$ & $H = (\frac{7}{3}, \frac{7}{3}, \frac{7}{3})$. This point is in the plane and $\overrightarrow{OH}$ is collinear with the normal vector; it is indeed the required projection.

3. [1 pt] $OH = \frac{7}{\sqrt{3}}$; for every P of the plane, $OP^2 = OH^2 + HP^2 \geq OH^2$, with equality only for $P = H$.

4. [1 pt] Differences in coordinates give $AB = AC = BC = 7\sqrt{2}$. The triangle is equilateral. Its altitude cuts one side into two equal parts and equals

$$\sqrt{(7\sqrt{2})^2 - (\frac{7\sqrt{2}}{2})^2} = \frac{7\sqrt{6}}{2}.$$

Its area is therefore $\frac{1}{2} \times 7\sqrt{2} \times \frac{7\sqrt{6}}{2} = \frac{49}{2}\sqrt{3}$ units of area.

5. [1 pt] Base ABC : volume $\frac{(\frac{49}{2}\sqrt{3})(\frac{7}{\sqrt{3}})}{3} = \frac{343}{6}$. Base OAB : area $\frac{49}{2}$ and height 7, same value.

DS4 · A

PAPER RESERVED FOR ASSESSMENT

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x)=\frac{5+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (1 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. State the maximum value of f, then solve $f(x)=0$. (1 pt)
4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)
5. Prove that the equation $f(x)=\frac{1}{2}e^4$ with justification. $]0;+\infty[$. (1 pt)

Exercise 2 · Antiderivatives and an imposed condition — 5 points

The first four questions are independent. For each, give the expression of the required antiderivatives and check the result by differentiation.

1. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto 4x^2-4x+1$. (1 pt)
2. Determine the antiderivative F of $x\mapsto e^{4x}$ on $\mathbb{R}$ such that $F(0)=4$. (1 pt)
3. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto\frac{2x}{x^2+4}$. (1 pt)
4. Determine an antiderivative on $\mathbb{R}$ of $x\mapsto 2x(x^2+1)^4$. (1 pt)
5. Let F and G two antiderivatives of the same function over an interval I. Prove that there exists a real constant C such that $F(x)-G(x)=C$ for all $x\in I$. (1 pt)

DS4 · A · continued

Exercise 3 · Identifying and solving a cooling model — 4 points

The temperature $T(t)$ in degrees Celsius, t in minutes, satisfies $T' = -k(T - 20)$, for $k > 0$. We measure $T(0) = 80$ and $T(10) = 50$.

1. Express T as a function of k. (1 pt)

2. Determine k and specify its unit. (1 pt)

3. Determine the first time at which $T(t) \leq 35$ and interpret. (2 pts)

Exercise 4 · The closest point on a line — 6 points

In an orthonormal coordinate system, $A(1;0;0)$, $M(0;2;3)$ and of d is the line passing through A w/ direction $u = (1;1;0)$. A point P of d satisfies $\overrightarrow{AP} = tu$.

1. Express the coordinates of P. (1 pt)

2. Determine the point H on d such that $\overrightarrow{MH} \cdot u = 0$. (2 pts)

3. Expand MP^2 and write it in the form 2 $(t - \frac{1}{2})^2 + c$. (2 pts)

4. Find the minimum distance again and compare the two methods. (1 pt)

Solutions · DS4 · A

Exercise 1 · A logarithmic function

1. [1 pt] At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+: limit $-\infty$; at $+\infty$, limit 0 by comparing growth rates.

2. [1 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (5 + \ln x)}{x^2} = \frac{-4 - \ln x}{x^2}.$$

The denominator is positive. The numerator is zero at $s = e^{-4}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f strictly increases on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [1 pt] The maximum is reached only at $s = e^{-4}$ and is

$$f(s) = \frac{5 - 4}{e^{-4}} = e^4.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $5 + \ln x = 0$, i.e. $x = e^{-5}$.

4. [1 pt] We differentiate $(-4 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-4 - \ln x)x^{-3} = \frac{2\ln x + 7}{x^3}.$$

The numerator is zero at $t = e^{-\frac{7}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at t and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. [1 pt] Let $m = \frac{e^4}{2}$ & $s = e^{-4}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and with value 2 m at s: the corollary of the intermediate value theorem gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from 2 m to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

Exercise 2 · Antiderivatives and an imposed condition

1. [1 pt] By linearity, the antiderivatives are $F(x) = \frac{4}{3}x^3 - 2x^2 + x + C$, $C \in \mathbb{R}$. Indeed, $F'(x) = 4x^2 - 4x + 1$. The derivative does not depend on C, and all antiderivatives differ by a constant on $\mathbb{R}$.

2. [1 pt] The antiderivatives are $F(x) = \frac{e^{4x}}{} + C$. The condition $F(0) = 4$ requires $^1 + C = 4$, hence $C = \frac{15}{4}$. The derivative of $\frac{e^{4x}}{4}$ is e^{4x}: the function obtained does satisfy both conditions.

3. [1 pt] Let $u(x) = x^2 + 4 > 0$ on $\mathbb{R}$; $u'(x) = 2x$. The antiderivatives are therefore $\ln(x^2 + 4) + C$, since their derivative is $\frac{u'}{u} = \frac{2x}{x^2+4}$.

4. [1 pt] An antiderivative is $F(x) = \frac{(x^2+1)^5}{5}$. Indeed, by the chain rule,

$$F'(x) = \frac{5}{5}2x(x^2 + 1)^4 = 2x(x^2 + 1)^4.$$

5. [1 pt] The derivative of their difference is zero on the interval; the difference is therefore constant. The fact that it is an interval matters in this result.

Exercise 3 · Identifying and solving a cooling model

1. Set $z = T - 20$, then $z' = -kz$ and $z(0) = 60$. Therefore, $T(t) = 20 + 60e^{-kt}$.
2. $50 = 20 + 60e^{-10k}$, therefore $e^{-10k} = 1/2$ and $k = \ln 2/10$ in min $^{-1}$.
3. $T \leq 35 \iff e^{-kt} \leq 1/4 \iff t \geq \ln 4/k = 20$. The temperature is strictly decreasing, since $T' = -60ke^{-kt} < 0$; the first time is 20 min. Solving 1 point, first time and interpretation 1 point.

Exercise 4 · The closest point of a line

1. $P(1+t;t;0)$.
2. $MH = (1+t;t-2;-3)$; $MH \cdot u = 2t - 1$. Hence $t = 1/2$ and $H(3/2;1/2;0)$. Equation 1 point, solving 1 point.
3. $MP^2 = (1+t)^2 + (t-2)^2 + 9 = 2t^2 - 2t + 14 = 2(t-1/2)^2 + 27/2$. Expansion 1 point, square 1 point.
4. The minimum is $\sqrt{27/2} = 3\sqrt{6}/2$, attained only at $t = 1/2$. The algebraic minimisation gives the orthogonal projection again.

DS4 · B

PAPER RESERVED FOR ASSESSMENT

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x)=\frac{4+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (1 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. State the maximum value of f, then solve $f(x)=0$. (1 pt)
4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)
5. Prove that the equation $f(x)=\frac{1}{2}e^3$ with justification. $]0;+\infty[$. (1 pt)

Exercise 2 · Antiderivatives and an imposed condition — 5 points

The first four questions are independent. For each, give the expression of the required antiderivatives and check the result by differentiation.

1. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto 3x^2-6x+2$. (1 pt)
2. Determine the antiderivative F of $x\mapsto e^{3x}$ on $\mathbb{R}$ such that $F(0)=6$. (1 pt)
3. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto\frac{2x}{x^2+6}$. (1 pt)
4. Determine an antiderivative on $\mathbb{R}$ of $x\mapsto 2x(x^2+2)^3$. (1 pt)
5. Let F and G two antiderivatives of the same function over an interval I. Prove that there exists a real constant C such that $F(x)-G(x)=C$ for all $x\in I$. (1 pt)

DS4 · B · continued

Exercise 3 · A variable input — 4 points

We look for a differentiable function y on $[0;+\infty[$ such that $y' = -y + 2t + 1$ and $y(0) = 3$.

1. Determine a particular affine solution $y_p(t) = at + b$. (1 pt)

2. Show that $z = y - y_p$ satisfies $z' = -z$, then determine y. (2 pts)

3. Check the initial condition and explain why y does not tend to a constant equilibrium value. (1 pt)

Exercise 4 · An altitude calculated in two ways — 6 points

In an orthonormal coordinate system, $O(0;0;0)$, $A(6;0;0)$, $B(0;4;0)$, $C(0;0;3)$. Coordinates are in metres.

1. Calculate the volume of $OABC$ with the base OAB. (1 pt)

2. Verify that $n = (2;3;4)$ is normal to the plane (ABC). (2 pts)

3. The point H defined by $\overrightarrow{OH} = tn$ lies in the plane if $\overrightarrow{AH}$ is orthogonal to n. Determine t then OH. (2 pts)

4. Deduce the area of ABC. (1 pt)

Solutions · DS4 · B

Exercise 1 · A logarithmic function

1. [1 pt] At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+: limit $-\infty$; at $+\infty$, limit 0 by comparing growth rates.

2. [1 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (4 + \ln x)}{x^2} = \frac{-3 - \ln x}{x^2}.$$

The denominator is positive. The numerator is zero at $s = e^{-3}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f strictly increases on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [1 pt] The maximum is reached only at $s = e^{-3}$ and is

$$f(s) = \frac{4 - 3}{e^{-3}} = e^3.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $4 + \ln x = 0$, i.e. $x = e^{-4}$.

4. [1 pt] We differentiate $(-3 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-3 - \ln x)x^{-3} = \frac{2\ln x + 5}{x^3}.$$

The numerator is zero at $t = e^{-\frac{5}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at t and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. [1 pt] Let $m = \frac{e^3}{2}$ & $s = e^{-3}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and with value 2 m at s: the corollary of the intermediate value theorem gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from 2 m to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

Exercise 2 · Antiderivatives and an imposed condition

1. [1 pt] By linearity, the antiderivatives are $F(x) = x^3 - 3x^2 + 2x + C$, $C \in \mathbb{R}$. Indeed, $F'(x) = 3x^2 - 6x + 2$. The derivative does not depend on C, and all antiderivatives differ by a constant on $\mathbb{R}$.

2. [1 pt] The antiderivatives are $F(x) = \frac{e^{3x}}{} + C$. The condition $F(0) = 6$ requires $\frac{1}{} + C = 6$, hence $C = \frac{17}{3}$. The derivative of $\frac{e^{3x}}{3}$ is e^{3x}: the function obtained does satisfy both conditions.

3. [1 pt] Let $u(x) = x^2 + 6 > 0$ on $\mathbb{R}$; $u'(x) = 2x$. The antiderivatives are therefore $\ln(x^2 + 6) + C$, since their derivative is $\frac{u'}{u} = \frac{2x}{x^2+6}$.

4. [1 pt] An antiderivative is $F(x) = \frac{(x^2+2)^4}{4}$. Indeed, by the chain rule,

$$F'(x) = \frac{4}{4}\,2x(x^2+2)^3 = 2x(x^2+2)^3.$$

5. [1 pt] The derivative of their difference is zero on the interval; the difference is therefore constant. The fact that it is an interval matters in this result.

Exercise 3 · A variable contribution

1. $a = -at - b + 2t + 1$ at every time requires $a = 2$, $b = -1$; $y_p = 2t - 1$.
2. By subtraction, $z' = -z$ so $z = Ce^{-t}$. $y(0) = 3$ gives $-1 + C = 3$, i.e. $C = 4$. Thus $y = 2t - 1 + 4e^{-t}$. Transformation 1 point, constant 1 point.
3. $y(0) = -1 + 4 = 3$; $y(t) \to +\infty$ since the affine term diverges and the exponential tends to 0. The right-hand side includes a variable input of t.

Exercise 4 · An altitude calculated in two ways

1. $\mathcal{A}(OAB) = 6 \times 4/2 = 12$ m^2 ; height $OC = 3$ m, so volume 12 m ^{3}.
2. $AB = (-6;4;0)$, $AC = (-6;0;3)$ are not collinear. $n \cdot AB = -12+12 = 0$, $n \cdot AC = -12+12 = 0$. Tests 1 point, non-collinearity and conclusion 1 point.
3. $AH = (2t - 6; 3t; 4t)$, therefore $AH \cdot n = 29t - 12 = 0$. Thus $t = 12/29$ and $OH = (12/29)\sqrt{29} = 12/\sqrt{29}$ m. Condition 1 point, distance 1 point.
4. $12 = \mathcal{A}(ABC)\, OH/3$, so $\mathcal{A}(ABC) = 3\sqrt{29}$ m^2.

DS5 · Specimen 1

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Integral and sequence — 5 points

For every natural number n, set $I_n = \int_0^1 x^n e^{4x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{4x}$.

1. Calculate I_0. (1 pt)

2. Using integration by parts, calculate I_1. (1 pt)

3. Prove, by comparing the integrands, that (I_n) is positive and decreasing. (1 pt)

4. Show that $I_n \leq \frac{e^4}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of (I_n). (1 pt)

5. Using integration by parts, express I_{n+1} as a function of I_n and of n. (1 pt)

Exercise 2 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{3}y + \frac{8}{3}$ on $[0;+\infty[$. In questions 2 to 5, we call y the solution that satisfies $y(0) = 12$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution satisfying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ as t tends to $+\infty$. (1 pt)

4. Study the direction of variation of y, then find the smallest real number $t \geq 0$ such that $y(t) \leq 9$. (1 pt)

5. Euler's method is applied with a step $h > 0$ and let Y_k call the approx. of $y(kh)$. Give the value of Y_0 and express Y_{k+1} as a function of Y_k and of h. (1 pt)

DS5 · Specimen 1 · continued

Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system with origin O, we consider $A(9;0;0)$, $B(0;9;0)$ and $C(0;0;9)$. Let $\vec{n} = (1;1;1)$. Lengths, areas and volumes are expressed in units of length, of area and volume.

1. Prove that $\vec{n}$ is normal to the plane (ABC). (1 pt)

2. Let H the orthogonal projection of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Prove that OH is the distance from O to the plane (ABC) : for every $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the area of the triangle ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Exercise 4 · Sums of random variables — 5 points

Let $n = 17$. Consider n independent random variables $X_1, \ldots, X_{17}$, which all follow the Bernoulli distribution with parameter $p = \frac{1}{5}$. Set $S = \sum_{i=1}^{17} X_i$ and $G = 9S - 17$.

1. Justify the distribution of S and calculate $E(S)$. (1 pt)

2. From the variance of a Bernoulli variable and independence, calculate $V(S)$. (1 pt)

3. Calculate $E(G)$ and $V(G)$. (1 pt)

4. Set $M = \frac{S}{17}$. Calculate $E(M)$ and $V(M)$. (1 pt)

5. For this question only, the 17 variables are replaced by 17 variables all equal to X_1. Then calculate the variance of their mean and compare it with that of question 4. Explain why the results differ. (1 pt)

Solutions · DS5 · Specimen 1

Exercise 1 · Integral and sequence

1. [1 pt] The function $x \mapsto \frac{e^{4x}}{4}$ is an antiderivative of e^{4x}. Thus $I_0 = [\frac{e^{4x}}{4}]_0^1 = \frac{e^4-1}{4}$.

2. [1 pt] Let us choose $u(x) = x$ and $v(x) = \frac{e^{4x}}{4}$, two differentiable functions with continuous derivatives on $[0,1]$. Then $u' = 1$ and $v' = e^{4x}$. Integration by parts gives

$$\begin{aligned} I_1 &= \left[\frac{xe^{4x}}{4}\right]_0^1 - \frac{1}{4}\int_0^1 e^{4x}\,\mathrm{d}x \\ &= \frac{e^4}{4} - \frac{e^4-1}{16} = \frac{3e^4+1}{16}. \end{aligned}$$

3. [1 pt] For $x \in [0,1]$, $0 \le x^{n+1} \le x^n$. Since $e^{4x} > 0$, integrating with bounds in increasing order gives $0 \le I_{n+1} \le I_n$. The sequence is therefore positive and non-increasing. The inequalities are even strict: the integrands of I_n and of $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. [1 pt] e $^{4x} \le e^4$; therefore $0 \le I_n \le e^4 \int_0^1 x^n\,\mathrm{d}x = \frac{e^4}{n+1} \to 0$.

5. [1 pt] Apply integration by parts to $u(x) = x^{n+1}$ & $v(x) = \frac{e^{4x}}{4}$. Since $n+1 \ge 1$, the term at $x = 0$ is zero. Hence

$$\begin{aligned} I_{n+1} &= \left[\frac{x^{n+1}e^{4x}}{4}\right]_0^1 - \frac{n+1}{4}\int_0^1 x^n e^{4x}\,\mathrm{d}x \\ &= \frac{e^4}{4} - \frac{n+1}{4}I_n. \end{aligned}$$

Exercise 2 · Differential equation and evolution

1. [1 pt] $0 = -\frac{1}{3}y + \frac{8}{3}$ gives $y = 8$.

2. [1 pt] The constant solution is 8. Let $z = y - 8$; the equation becomes $z' = -\frac{z}{3}$. Its solutions are $z(t) = Ce^{-\frac{t}{3}}$, so all solutions of the original equation are $y(t) = 8 + Ce^{-\frac{t}{3}}$. The condition $y(0) = 12$ gives $C = 4$. For the function obtained,

$$y'(t) = -\frac{4}{3}e^{-\frac{t}{3}} = -\frac{y(t)}{3} + \frac{8}{3}, \qquad y(0) = 12.$$

The solution of the initial problem is therefore $y(t) = 8 + 4e^{-\frac{t}{3}}$.

3. [1 pt] y $' = -\frac{4}{3}e^{-\frac{t}{3}} < 0$; limit 8.

4. [1 pt] Since $e^{-\frac{t}{3}} > 0$, the derivative $y'(t) = -\frac{4}{3}e^{-\frac{t}{3}}$ is negative: y is strictly decreasing. Let us solve

$$\begin{aligned} y(t) \le 9 &\iff 4e^{-\frac{t}{3}} \le 1 \\ &\iff -\frac{t}{3} \le -\ln 4 \\ &\iff t \ge 3\ln 4. \end{aligned}$$

The last step reverses the inequality because we multiply by $-3 < 0$. The first time is 3 $\ln 4 \simeq$ 4,1589.

5. [1 pt] Y $= 12$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{3} + \frac{8}{3})$, $t_j = jh$.

Exercise 3 · Projection and volume

1. [1 pt] $\overrightarrow{AB} = (-9,9,0)$ and $\overrightarrow{AC} = (-9,0,9)$ are not collinear and their dot products with $\vec{n}$ are zero.

2. [1 pt] The line normal to the plane through O has direction vector $\vec{n} = (1,1,1)$. Its point H is thus as (t,t,t). For H to lie in (ABC), it is necessary and sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t-9) + t + t = 3t - 9 = 0.$$

Thus $t = 3$ and $H = (3,3,3)$. This point is in the plane and $\overrightarrow{OH}$ is collinear with the normal vector; it is indeed the required projection.

3. [1 pt] $OH = \frac{9}{\sqrt{3}}$; for every P of the plane, $OP^2 = OH^2 + HP^2 \geq OH^2$, with equality only for $P = H$.

4. [1 pt] Differences in coordinates give $AB = AC = BC = 9\sqrt{2}$. The triangle is equilateral. Its altitude cuts one side into two equal parts and equals

$$\sqrt{(9\sqrt{2})^2 - (\frac{9\sqrt{2}}{2})^2} = \frac{9\sqrt{6}}{2}.$$

Its area is therefore $\frac{1}{2} \times 9\sqrt{2} \times \frac{9\sqrt{6}}{2} = \frac{81}{2}\sqrt{3}$ units of area.

5. [1 pt] Base ABC : volume $\frac{(\frac{81}{2}\sqrt{3})(\frac{9}{\sqrt{3}})}{3} = \frac{243}{2}$. Base OAB : area $\frac{81}{2}$ and height 9, same value.

Exercise 4 · Sums of random variables

1. [1 pt] S $\sim \mathcal{B}(17, \frac{1}{5})$; $E(S) = \frac{17}{5}$.

2. [1 pt] Each X_i is 0 or 1, so $X_i^2 = X_i$. Thus $E(X_i) = p = \frac{1}{5}$ and

$$V(X_i) = E(X_i^2) - E(X_i)^2 = p - p^2 = p(1-p) = \frac{4}{25}.$$

The variables are independent; the additivity of the variance gives

$$V(S) = \sum_{i=1}^{17} V(X_i) = 17p(1-p) = \frac{68}{25}.$$

3. [1 pt] By linearity of expectation and by $V(\lambda S + \mu) = \lambda^2 V(S)$,

$$E(G) = 9E(S) - 17 = \frac{68}{5}, \qquad V(G) = 81V(S) = \frac{5508}{25}.$$

The constant -17 does not change the variance.

4. [1 pt] E $(M) = \frac{1}{5}$ & $V(M) = \frac{4}{425}$; the variance factor is $\frac{1}{n^2}$.

5. [1 pt] No. The new mean is X_1, its variance is $\frac{4}{25}$. The additivity of variance used previously requires independence.

DS5 · Specimen 2

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Integral and sequence — 5 points

For every natural number n, set $I_n = \int_0^1 x^n e^{3x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{3x}$.

1. Calculate I_0. (1 pt)

2. Using integration by parts, calculate I_1. (1 pt)

3. Prove, by comparing the integrands, that (I_n) is positive and decreasing. (1 pt)

4. Show that $I_n \leq \frac{e^3}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of (I_n). (1 pt)

5. Using integration by parts, express I_{n+1} as a function of I_n and of n. (1 pt)

Exercise 2 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{2}y + 5$ on $[0;+\infty[$. In questions 2 to 5, we call y the solution that satisfies $y(0) = 15$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution satisfying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ as t tends to $+\infty$. (1 pt)

4. Study the direction of variation of y, then find the smallest real number $t \geq 0$ such that $y(t) \leq 11$. (1 pt)

5. Euler's method is applied with a step $h > 0$ and let Y_k call the approx. of $y(kh)$. Give the value of Y_0 and express Y_{k+1} as a function of Y_k and of h. (1 pt)

DS5 · Specimen 2 · continued

Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system with origin O, we consider $A(13;0;0)$, $B(0;13;0)$ and $C(0;0;13)$. Let $\vec{n} = (1;1;1)$. Lengths, areas and volumes are expressed in units of length, of area and volume.

1. Prove that $\vec{n}$ is normal to the plane (ABC). (1 pt)

2. Let H the orthogonal projection of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Prove that OH is the distance from O to the plane (ABC) : for every $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the area of the triangle ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Exercise 4 · Sums of random variables — 5 points

Let $n = 14$. Consider n independent random variables $X_1, \ldots, X_{14}$, which all follow the Bernoulli distribution with parameter $p = \frac{1}{4}$. Set $S = \sum_{i=1}^{14} X_i$ and $G = 6S - 14$.

1. Justify the distribution of S and calculate $E(S)$. (1 pt)

2. From the variance of a Bernoulli variable and independence, calculate $V(S)$. (1 pt)

3. Calculate $E(G)$ and $V(G)$. (1 pt)

4. Set $M = \frac{S}{14}$. Calculate $E(M)$ and $V(M)$. (1 pt)

5. For this question only, the 14 variables are replaced by 14 variables all equal to X_1. Then calculate the variance of their mean and compare it with that of question 4. Explain why the results differ. (1 pt)

Solutions · DS5 · Specimen 2

Exercise 1 · Integral and sequence

1. [1 pt] The function $x \mapsto \frac{e^{3x}}{3}$ is an antiderivative of e^{3x}. Thus $I_0 = [\frac{e^{3x}}{3}]_0^1 = \frac{e^3-1}{3}$.

2. [1 pt] Let us choose $u(x) = x$ and $v(x) = \frac{e^{3x}}{3}$, two differentiable functions with continuous derivatives on $[0,1]$. Then $u' = 1$ and $v' = e^{3x}$. Integration by parts gives

$$\begin{aligned} I_1 &= \left[\frac{xe^{3x}}{3}\right]_0^1 - \frac{1}{3}\int_0^1 e^{3x}\,\mathrm{d}x \\ &= \frac{e^3}{3} - \frac{e^3-1}{9} = \frac{2e^3+1}{9}. \end{aligned}$$

3. [1 pt] For $x \in [0,1]$, $0 \leq x^{n+1} \leq x^n$. Since $e^{3x} > 0$, integrating with bounds in increasing order gives $0 \leq I_{n+1} \leq I_n$. The sequence is therefore positive and non-increasing. The inequalities are even strict: the integrands of I_n and of $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. [1 pt] e $^{3x} \leq e^3$; therefore $0 \leq I_n \leq e^3 \int_0^1 x^n\,\mathrm{d}x = \frac{e^3}{n+1} \to 0$.

5. [1 pt] Apply integration by parts to $u(x) = x^{n+1}$ & $v(x) = \frac{e^{3x}}{3}$. Since $n+1 \geq 1$, the term at $x = 0$ is zero. Hence

$$\begin{aligned} I_{n+1} &= \left[\frac{x^{n+1}e^{3x}}{3}\right]_0^1 - \frac{n+1}{3}\int_0^1 x^n e^{3x}\,\mathrm{d}x \\ &= \frac{e^3}{3} - \frac{n+1}{3}I_n. \end{aligned}$$

Exercise 2 · Differential equation and evolution

1. [1 pt] $0 = -\frac{1}{2}y + 5$ gives $y = 10$.

2. [1 pt] The constant solution is 10. Let $z = y - 10$; the equation becomes $z' = -\frac{z}{2}$. Its solutions are $z(t) = Ce^{-\frac{t}{2}}$, so all solutions of the original equation are $y(t) = 10 + Ce^{-\frac{t}{2}}$. The condition $y(0) = 15$ gives $C = 5$. For the function obtained,

$$y'(t) = -\frac{5}{2}e^{-\frac{t}{2}} = -\frac{y(t)}{2} + 5, \qquad y(0) = 15.$$

The solution of the initial problem is therefore $y(t) = 10 + 5e^{-\frac{t}{2}}$.

3. [1 pt] y $' = -\frac{5}{2}e^{-\frac{t}{2}} < 0$; limit 10.

4. [1 pt] Since $e^{-\frac{t}{2}} > 0$, the derivative $y'(t) = -\frac{5}{2}e^{-\frac{t}{2}}$ is negative: y is strictly decreasing. Let us solve

$$\begin{aligned} y(t) \leq 11 &\iff 5e^{-\frac{t}{2}} \leq 1 \\ &\iff -\frac{t}{2} \leq -\ln 5 \\ &\iff t \geq 2\ln 5. \end{aligned}$$

The last step reverses the inequality because we multiply by $-2 < 0$. The first time is 2 $\ln 5 \simeq$ 3,2189.

5. [1 pt] Y $= 15$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{2} + 5)$, $t_j = jh$.

Exercise 3 · Projection and volume

1. [1 pt] $\overrightarrow{AB} = (-13, 13, 0)$ and $\overrightarrow{AC} = (-13, 0, 13)$ are not collinear and their dot products with $\vec{n}$ are zero.

2. [1 pt] The line normal to the plane through O has direction vector $\vec{n} = (1, 1, 1)$. Its point H is thus as (t, t, t). For H to lie in (ABC), it is necessary and sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t - 13) + t + t = 3t - 13 = 0.$$

Thus $t = \frac{13}{3}$ & $H = (\frac{13}{3}, \frac{13}{3}, \frac{13}{3})$. This point is in the plane and $\overrightarrow{OH}$ is collinear with the normal vector; it is indeed the required projection.

3. [1 pt] $OH = \frac{13}{\sqrt{3}}$; for every P of the plane, $OP^2 = OH^2 + HP^2 \geq OH^2$, with equality only for $P = H$.

4. [1 pt] Differences in coordinates give $AB = AC = BC = 13\sqrt{2}$. The triangle is equilateral. Its altitude cuts one side into two equal parts and equals

$$\sqrt{(13\sqrt{2})^2 - (\frac{13\sqrt{2}}{2})^2} = \frac{13\sqrt{6}}{2}.$$

Its area is therefore $\frac{1}{2} \times 13\sqrt{2} \times \frac{13\sqrt{6}}{2} = \frac{169}{2}\sqrt{3}$ units of area.

5. [1 pt] Base ABC : volume $\frac{(\frac{169}{2}\sqrt{3})(\frac{13}{\sqrt{3}})}{3} = \frac{2197}{6}$. Base OAB : area $\frac{169}{2}$ and height 13, same value.

Exercise 4 · Sums of random variables

1. [1 pt] S $\sim \mathcal{B}(14, \frac{1}{4})$; $E(S) = \frac{7}{2}$.

2. [1 pt] Each X_i is 0 or 1, so $X_i^2 = X_i$. Thus $E(X_i) = p = \frac{1}{4}$ and

$$V(X_i) = E(X_i^2) - E(X_i)^2 = p - p^2 = p(1 - p) = \frac{3}{16}.$$

The variables are independent; the additivity of the variance gives

$$V(S) = \sum_{i=1}^{14} V(X_i) = 14p(1 - p) = \frac{21}{8}.$$

3. [1 pt] By linearity of expectation and by $V(\lambda S + \mu) = \lambda^2 V(S)$,

$$E(G) = 6E(S) - 14 = 7, \qquad V(G) = 36V(S) = \frac{189}{2}.$$

The constant -14 does not change the variance.

4. [1 pt] E $(M) = \frac{1}{4}$ & $V(M) = \frac{3}{224}$; the variance factor is $\frac{1}{n^2}$.

5. [1 pt] No. The new mean is X_1, its variance is $\frac{3}{16}$. The additivity of variance used previously requires independence.

DS5 · A

PAPER RESERVED FOR ASSESSMENT

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Choosing two integrations by parts — 4 points

Let $I = \int_0^1 x^2 e^x \,\mathrm{d}x$.

1. Justify the choice of functions in a first integration by parts and reduce I to $J = \int_0^1 xe^x \,\mathrm{d}x$. (1 pt)

2. Calculate J then I. (2 pts)

3. Check the bounds $\frac{1}{3} \leq I \leq \frac{e}{3}$ without calculating an antiderivative. (1 pt)

Exercise 2 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{5}y + \frac{6}{5}$ on $[0;+\infty[$. In questions 2 to 5, we call y the solution that satisfies $y(0) = 12$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution satisfying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ as t tends to $+\infty$. (1 pt)

4. Study the direction of variation of y, then find the smallest real number $t \geq 0$ such that $y(t) \leq 7$. (1 pt)

5. Euler's method is applied with a step $h > 0$ and let Y_k call the approx. of $y(kh)$. Give the value of Y_0 and express Y_{k+1} as a function of Y_k and of h. (1 pt)

DS5 · A · continued

Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system with origin O, we consider $A(10;0;0)$, $B(0;10;0)$ and $C(0;0;10)$. Let $\vec{n} = (1;1;1)$. Lengths, areas and volumes are expressed in units of length, of area and volume.

1. Prove that $\vec{n}$ is normal to the plane (ABC). (1 pt)

2. Let H the orthogonal projection of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Prove that OH is the distance from O to the plane (ABC) : for every $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the area of the triangle ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Exercise 4 · A cost that is not binomial — 6 points

A random cost X is 0, 2 or 5 euros with respective probabilities $0{,}5$, $0{,}3$, $0{,}2$. The charge is $Y = 0{,}8X + 1$. For n independent operations with the same distribution, let S_n be the sum of the invoices & $M_n = S_n / n$.

1. Calculate $E(X)$ and $E(X^2)$. (1 pt)

2. Deduce $V(X)$, $E(Y)$ and $V(Y)$. (2 pts)

3. Calculate $E(M_n)$ and $V(M_n)$ indicating where independence is used. (2 pts)

4. What happens to the standard deviation of M_n if we multiply n by 4? (1 pt)

Solutions · DS5 · A

Exercise 1 · Choosing two integrations by parts

1. Take $u = x^2$, $v = e^x$, of class $\mathcal{C}^1$ on $[0;1]$. Then $I = [x^2e^x]_0^1 - 2\int_0^1 xe^x\,\mathrm{d}x = e - 2J$.

2. By parts again with $u = x$, $v = e^x$: $J = [xe^x]_0^1 - \int_0^1 e^x\,\mathrm{d}x = e - (e-1) = 1$. Hence $I = e - 2$. One point for each justified result.

3. $1 \leq e^x \leq e$ and $x^2 \geq 0$, therefore $x^2 \leq x^2e^x \leq ex^2$. Integrating on $[0;1]$ gives the bounds.

Exercise 2 · Differential equation and evolution

1. [1 pt] $0 = -\frac{1}{5}y + \frac{6}{5}$ gives $y = 6$.

2. [1 pt] The constant solution is 6. Let $z = y - 6$; the equation becomes $z' = -\frac{z}{5}$. Its solutions are $z(t) = Ce^{-\frac{t}{5}}$, so all solutions of the original equation are $y(t) = 6 + Ce^{-\frac{t}{5}}$. The condition $y(0) = 12$ gives $C = 6$. For the function obtained,

$$y'(t) = -\frac{6}{5}e^{-\frac{t}{5}} = -\frac{y(t)}{5} + \frac{6}{5}, \qquad y(0) = 12.$$

The solution of the initial problem is therefore $y(t) = 6 + 6e^{-\frac{t}{5}}$.

3. [1 pt] y $' = -\frac{6}{5}e^{-\frac{t}{5}} < 0$; limit 6.

4. [1 pt] Since $e^{-\frac{t}{5}} > 0$, the derivative $y'(t) = -\frac{6}{5}e^{-\frac{t}{5}}$ is negative: y is strictly decreasing.
Let us solve

$$\begin{aligned} y(t) \leq 7 &\iff 6e^{-\frac{t}{5}} \leq 1 \\ &\iff -\frac{t}{5} \leq -\ln 6 \\ &\iff t \geq 5\ln 6. \end{aligned}$$

The last step reverses the inequality because we multiply by $-5 < 0$. The first time is 5 $\ln 6 \simeq$ 8,9588.

5. [1 pt] Y $= 12$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{5} + \frac{6}{5})$, $t_j = jh$.

Exercise 3 · Projection and volume

1. [1 pt] $\overrightarrow{AB} = (-10, 10, 0)$ and $\overrightarrow{AC} = (-10, 0, 10)$ are not collinear and their dot products with $\vec{n}$ are zero.

2. [1 pt] The line normal to the plane through O has direction vector $\vec{n} = (1,1,1)$. Its point H is thus as (t,t,t). For H to lie in (ABC), it is necessary and sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t-10) + t + t = 3t - 10 = 0.$$

Thus $t = \frac{10}{3}$ & $H = (\frac{10}{3}, \frac{10}{3}, \frac{10}{3})$. This point is in the plane and $\overrightarrow{OH}$ is collinear with the normal vector; it is indeed the required projection.

3. [1 pt] $OH = \frac{10}{\sqrt{3}}$; for every P of the plane, $OP^2 = OH^2 + HP^2 \geq OH^2$, with equality only for $P = H$.

4. [1 pt] Differences in coordinates give $AB = AC = BC = 10\sqrt{2}$. The triangle is equilateral. Its altitude cuts one side into two equal parts and equals

$$\sqrt{(10\sqrt{2})^2 - (\frac{10\sqrt{2}}{2})^2} = \frac{10\sqrt{6}}{2}.$$

Its area is therefore $\frac{1}{2} \times 10\sqrt{2} \times \frac{10\sqrt{6}}{2} = 50\sqrt{3}$ units of area.

5. [1 pt] Base ABC : volume $\frac{(50\sqrt{3})(\frac{10}{\sqrt{3}})}{3} = \frac{500}{3}$. Base OAB : area 50 and height 10, same value.

Exercise 4 · A cost that is not binomial

1. $E(X) = 0{,}6 + 1 = 1{,}6$; $E(X^2) = 1{,}2 + 5 = 6.2$ (0.5 each).

2. $V(X) = 6{,}2 - 1{,}6^2 = 3{,}64$; $E(Y) = 0{,}8(1{,}6) + 1 = 2{,}28$; $V(Y) = 0{,}8^2(3{,}64) = 2{,}3296$. Variance of X 1 point, transformations 1 point.

3. Linearity: $E(S_n) = 2{,}28n$, therefore $E(M_n) = 2{,}28$ (1 pt). Independence: $V(S_n) = 2{,}3296n$, therefore $V(M_n) = 2{,}3296/n$ (1 pt).

4. It is divided by 2 because $\sigma(M_n) = \sqrt{2{,}3296/n}$.

DS5 · B

PAPER RESERVED FOR ASSESSMENT

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A family of integrals with no antiderivative required — 4 points

For $n \in \mathbb{N}$, $I_n = \int_0^1 \frac{x^n}{1+x}\,\mathrm{d}x$.

1. Calculate I_0 and establish $I_n + I_{n+1} = \frac{1}{n+1}$. (1 pt)

2. Show that (I_n) is decreasing and positive. (1 pt)

3. Give bounds for I_n which makes it possible to determine its limit. (2 pts)

Exercise 2 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{4}y + 2$ on $[0;+\infty[$. In questions 2 to 5, we call y the solution that satisfies $y(0) = 12$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution satisfying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ as t tends to $+\infty$. (1 pt)

4. Study the direction of variation of y, then find the smallest real number $t \geq 0$ such that $y(t) \leq 9$. (1 pt)

5. Euler's method is applied with a step $h > 0$ and let Y_k call the approx. of $y(kh)$. Give the value of Y_0 and express Y_{k+1} as a function of Y_k and of h. (1 pt)

DS5 · B · continued

Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system with origin O, we consider $A(6;0;0)$, $B(0;6;0)$ and $C(0;0;6)$. Let $\vec{n} = (1;1;1)$. Lengths, areas and volumes are expressed in units of length, of area and volume.

1. Prove that $\vec{n}$ is normal to the plane (ABC). (1 pt)

2. Let H the orthogonal projection of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Prove that OH is the distance from O to the plane (ABC) : for every $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the area of the triangle ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Exercise 4 · Detecting a hidden dependence — 6 points

X and Y are two independent Bernoulli random variables with parameter $1/2$. Set $S = X + Y$, $Z = 2X$, $U = S/2$ and $W = Z/2$.

1. Give the distribution of S. (1 pt)

2. Give the distribution of Z and calculate $E(S), E(Z)$. (1 pt)

3. Calculate $V(S)$ and $V(Z)$. Explain why the two results differ. (2 pts)

4. Compare the variances of U and W. Which one corresponds to the mean of two independent observations? (2 pts)

Solutions · DS5 · B

Exercise 1 · A family of integrals with no antiderivative required

1. $I_0 = [\ln(1+x)]_0^1 = \ln 2$. The sum of the integrands is x^n, therefore $I_n + I_{n+1} = \int_0^1 x^n \, dx = 1/(n+1)$.
2. For $x \in [0;1]$, $0 \leq x^{n+1} \leq x^n$; dividing by $1 + x > 0$, then integrating gives $0 \leq I_{n+1} \leq I_n$.
3. $1/2 \leq 1/(1+x) \leq 1$, hence $1/[2(n+1)] \leq I_n \leq 1/(n+1)$ (1 pt). The bounds tend towards 0; the squeeze theorem gives $I_n \to 0$ (1 pt).

Exercise 2 · Differential equation and evolution

1. [1 pt] $0 = -\frac{1}{4}y + 2$ gives $y = 8$.
2. [1 pt] The constant solution is 8. Let $z = y - 8$; the equation becomes $z' = -\frac{z}{4}$. Its solutions are $z(t) = Ce^{-\frac{t}{4}}$, so all solutions of the original equation are $y(t) = 8 + Ce^{-\frac{t}{4}}$. The condition $y(0) = 12$ gives $C = 4$. For the function obtained,

$$y'(t) = -e^{-\frac{t}{4}} = -\frac{y(t)}{4} + 2, \qquad y(0) = 12.$$

The solution of the initial problem is therefore $y(t) = 8 + 4e^{-\frac{t}{4}}$.

3. [1 pt] $y' = -e^{-\frac{t}{4}} < 0$; limit 8.
4. [1 pt] Since $e^{-\frac{t}{4}} > 0$, the derivative $y'(t) = -e^{-\frac{t}{4}}$ is negative: y is strictly decreasing.
Let us solve

$$\begin{aligned} y(t) \leq 9 &\iff 4e^{-\frac{t}{4}} \leq 1 \\ &\iff -\frac{t}{4} \leq -\ln 4 \\ &\iff t \geq 4\ln 4. \end{aligned}$$

The last step reverses the inequality because we multiply by $-4 < 0$. The first time is $4\ln 4 \simeq 5{,}5452$.

5. [1 pt] $Y_0 = 12$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{4} + 2)$, $t_j = jh$.

Exercise 3 · Projection and volume

1. [1 pt] $\overrightarrow{AB} = (-6,6,0)$ and $\overrightarrow{AC} = (-6,0,6)$ are not collinear and their dot products with $\vec{n}$ are zero.
2. [1 pt] The line normal to the plane through O has direction vector $\vec{n} = (1,1,1)$. Its point H is thus as (t,t,t). For H to lie in (ABC), it is necessary and sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t-6) + t + t = 3t - 6 = 0.$$

Thus $t = 2$ and $H = (2,2,2)$. This point is in the plane and $\overrightarrow{OH}$ is collinear with the normal vector; it is indeed the required projection.

3. [1 pt] $OH = \frac{6}{\sqrt{3}}$; for every P of the plane, $OP^2 = OH^2 + HP^2 \geq OH^2$, with equality only for $P = H$.
4. [1 pt] Differences in coordinates give $AB = AC = BC = 6\sqrt{2}$. The triangle is equilateral. Its altitude cuts one side into two equal parts and equals

$$\sqrt{(6\sqrt{2})^2 - (\frac{6\sqrt{2}}{2})^2} = \frac{6\sqrt{6}}{2}.$$

Its area is therefore $\frac{1}{2} \times 6\sqrt{2} \times \frac{6\sqrt{6}}{2} = 18\sqrt{3}$ units of area.

5. [1 pt] Base ABC : volume $\frac{(18\sqrt{3})(\frac{6}{\sqrt{3}})}{3} = 36$. Base OAB : area 18 and height 6, same value.

Exercise 4 · Detecting a hidden dependence

1. S is 0, 1, 2 with probabilities 1/4, 1/2, 1/4: distribution $\mathcal{B}(2; 1/2)$.
2. Z is 0 or 2 with probabilities 1/2 each. $E(S) = E(Z) = 1$.
3. $V(S) = V(X) + V(Y) = 1/2$ by independence. $V(Z) = 4V(X) = 1$. Adding two copies of the same variable is not the same as adding independent variables. Calculations 1 point, explanation 1 point.
4. $V(U) = V(S)/4 = 1/8$ and $V(W) = V(Z)/4 = 1/4$. $U = (X + Y)/2$ is the mean of independent variables; $W = X$ does not reduce the spread. Calculations 1 point, interpretation 1 point.

Mock exam 2 · Zero 1

SPECIMEN PAPER · PRACTICE

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Integral and sequence — 5 points

For every natural number n, set $I_n = \int_0^1 x^n e^{5x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{5x}$.

1. Calculate I_0. (0.5 pt)

2. Using integration by parts, calculate I_1. (0.5 pt)

3. Prove, by comparing the integrands, that (I_n) is positive and decreasing. (0.5 pt)

4. Show that $I_n \leq \frac{e^5}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of (I_n). (1 pt)

5. Using integration by parts, express I_{n+1} as a function of I_n and of n. (0.5 pt)

6. Show that $I_n \geq \frac{1}{n+1}$ for all $n \in \mathbb{N}$. (0.5 pt)

7. We divide $[0;1]$ in N intervals of the same length, with $N \geq 1$. Write the sums L_N and U_N of areas of the rectangles built with the values of e^{5x} at the left and right ends. Justify $L_N \leq I_0 \leq U_N$, then calculate $U_N - L_N$. (0.5 pt)

8. Determine the smallest integer N guaranteeing that this bracketing has a width at most equal to 10^{-2}. Write a Python function that returns N, L_N and U_N. (1 pt)

Exercise 2 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x) = \frac{4+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. State the maximum value of f, then solve $f(x) = 0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)

5. Prove that the equation $f(x) = \frac{1}{2}e^3$ has exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine an antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the mean value of f on $[1;e]$. (0.5 pt)

8. Independently of the previous questions, solve $\ln(x-3) + \ln(x+3) = \ln(27)$, after specifying the set on which this equation is defined. (1 pt)

Mock exam 2 · Zero 1 · continued

Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 10$ and the point $M(3;0;0)$. The line d passes through M and has direction vector $\vec{n} = (1;2;2)$.

1. Give a parametric representation of d. (0.5 pt)

2. Prove that d meets $\mathcal{P}$ at a point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal projection of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deduce the distance from M to the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ through M. (0.5 pt)

7. Determine the angle between d and the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define the point N by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from N to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. (1 pt)

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 20 times. The trials are independent and the probability of success is $p = \frac{3}{10}$ at each trial. Let X be the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. (0.5 pt)

2. Calculate the probability of exactly two successes. (0.5 pt)

3. Using complementary events, express and then calculate the probability of obtaining at least two successes. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same trial $N \geq 1$ times, still independently. Determine the smallest integer N for which the probability of getting at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 20\}$, prove the formula for $P(X = j)$ by counting the possible positions of successes. (0.5 pt)

7. A series of 20 trials is accepted if its number of successes is between 1 and 3, endpoints included. Calculate the probability that a series is accepted. (0.5 pt)

8. Write a Python function with no argument that returns the smallest integer j such that $P(X \leq j) \geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Solutions · Mock exam 2 · Specimen 1

Exercise 1 · Integral and sequence

1. [0.5 pt] The function $x \mapsto \frac{e^{5x}}{5}$ is an antiderivative of e^{5x}. Thus $I_0 = [\frac{e^{5x}}{5}]_0^1 = \frac{e^5-1}{5}$.

2. [0.5 pt] Let us choose $u(x) = x$ and $v(x) = \frac{e^{5x}}{5}$, two differentiable functions with continuous derivatives on $[0,1]$. Then $u' = 1$ and $v' = e^{5x}$. Integration by parts gives

$$\begin{aligned} I_1 &= \left[\frac{xe^{5x}}{5}\right]_0^1 - \frac{1}{5}\int_0^1 e^{5x}\,\mathrm{d}x \\ &= \frac{e^5}{5} - \frac{e^5-1}{25} = \frac{4e^5+1}{25}. \end{aligned}$$

3. [0.5 pt] For $x \in [0,1]$, $0 \leq x^{n+1} \leq x^n$. Since $e^{5x} > 0$, integrating with bounds in increasing order gives $0 \leq I_{n+1} \leq I_n$. The sequence is therefore positive and non-increasing. The inequalities are even strict: the integrands of I_n and of $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. [1 pt] e $^{5x} \leq e^5$; therefore $0 \leq I_n \leq e^5 \int_0^1 x^n\,\mathrm{d}x = \frac{e^5}{n+1} \to 0$.

5. [0.5 pt] Apply integration by parts to $u(x) = x^{n+1}$ & $v(x) = \frac{e^{5x}}{5}$. Since $n+1 \geq 1$, the term at $x = 0$ is zero. Hence

$$\begin{aligned} I_{n+1} &= \left[\frac{x^{n+1}e^{5x}}{5}\right]_0^1 - \frac{n+1}{5}\int_0^1 x^n e^{5x}\,\mathrm{d}x \\ &= \frac{e^5}{5} - \frac{n+1}{5}I_n. \end{aligned}$$

6. [0.5 pt] For $0 \leq x \leq 1$, $e^{5x} \geq 1$ and $x^n \geq 0$. Hence $x^n \leq x^n e^{5x}$; by integration,

$$I_n \geq \int_0^1 x^n\,\mathrm{d}x = \left[\frac{x^{n+1}}{n+1}\right]_0^1 = \frac{1}{n+1}.$$

7. [0.5 pt] The function $x \mapsto e^{5x}$ is increasing. On each interval $[\frac{j}{N}, \frac{j+1}{N}]$, the values at the endpoints bound the function. By integrating then summing,

$$L_N = \frac{1}{N}\sum_{j=0}^{N-1} e^{\frac{5j}{N}} \leq I_0 \leq U_N = \frac{1}{N}\sum_{j=1}^{N} e^{\frac{5j}{N}}.$$

The intermediate terms cancel in the difference, so $U_N - L_N = \frac{e^5-1}{N}$.

8. [1 pt] The width is at most 10^{-2} if $N \geq 100(e^5 - 1)$. The first integer satisfying this bound is $N = 14742$. Here is the calculation of the bounds:

```
from math import exp, ceil

def bornes():
    N = ceil(100*(exp(5)-1))
    bas = sum (exp(5*j/N) for j in range (N))/N
    haut = bas+(exp(5)-1)/N
    return N, bas, haut
```

Exercise 2 · A logarithmic function

1. [0.5 pt] At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+: limit $-\infty$; at $+\infty$, limit 0 by comparing growth rates.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (4 + \ln x)}{x^2} = \frac{-3 - \ln x}{x^2}.$$

The denominator is positive. The numerator is zero at $s = e^{-3}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f strictly increases on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [0.5 pt] The maximum is reached only at $s = e^{-3}$ and is

$$f(s) = \frac{4 - 3}{e^{-3}} = e^3.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $4 + \ln x = 0$, i.e. $x = e^{-4}$.

4. [1 pt] We differentiate $(-3 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-3 - \ln x)x^{-3} = \frac{2\ln x + 5}{x^3}.$$

The numerator is zero at $t = e^{-\frac{5}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at t and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Let $m = \frac{e^3}{2}$ & $s = e^{-3}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and with value 2 m at s: the corollary of the intermediate value theorem gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from 2 m to 0 $< m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. [0.5 pt] Let $F(x) = 4\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{4}{x} + \frac{\ln x}{x} = f(x)$; F is an antiderivative on $]0, +\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{9}{2}$ & $F(1) = 0$. The integral therefore equals $\frac{9}{2}$, and the average value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{9}{2(e-1)}.$$

8. [1 pt] Domain $x > 3$; $x^2 - 9 = 27$, so $x = 6$, the negative root being excluded.

Exercise 3 · Line, plane and projection

1. [0.5 pt] $(x, y, z) = (3 + t, 2t, 2t)$, $t \in \mathbb{R}$.

2. [0.5 pt] $3 + 9t = 10$ gives $t = \frac{7}{9}$; $H = (\frac{34}{9}, \frac{14}{9}, \frac{14}{9})$.

3. [0.5 pt] The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Since $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{7}{9}\vec{n}$, H is the orthogonal projection of M onto the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1 + 4 + 4} = 3$. Hence

$$d(M, \mathcal{P}) = MH = |\frac{7}{9}| 3 = \frac{7}{3}.$$

4. [1 pt] Let $y = z = s$; $x = 10 - 4s$. Hence $(x, y, z) = (10, 0, 0) + s(-4, 1, 1)$.

5. [0.5 pt] Yes, its last two coordinates are equal. H is on the line and $\overrightarrow{MH}$ is orthogonal to every direction in the plane, hence to this line. Distance $\frac{7}{3}$.

6. [0.5 pt] x $+ 2y + 2z = 3$; same normal vector and passing through M.

7. [0.5 pt] $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. [1 pt] $N = M + \vec{n}$ is on the same normal line as M ; its projection onto $\mathcal{P}$ is therefore also H. Now $H \in \mathcal{P} \cap \mathcal{Q}$. For every P of this line, $NP^2 = NH^2 + HP^2$, therefore its minimum distance to N is NH. Finally,

$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{7}{9} - 1| \, \|\vec{n}\| = \frac{2}{3}.$$

Exercise 4 · Repetitions and estimation

1. [0.5 pt] Repetition of a fixed number 20 of identical independent trials with two outcomes; $X \sim \mathcal{B}(20, \frac{3}{10})$.

2. [0.5 pt] P $(X = 2) = \binom{20}{2}(\frac{3}{10})^2(\frac{7}{10})^{18} \simeq 0{,}027846$.

3. [0.5 pt] $1 - P(X = 0) - P(X = 1) = 1 - (\frac{7}{10})^{20} - 20(\frac{3}{10})(\frac{7}{10})^{19} \simeq 0{,}992363$.

4. [1 pt] E $(X) = 6$, $V(X) = \frac{21}{5}$.

5. [0.5 pt] For N independent trials, the probability of no success is $(1 - p)^N$. The required prob. is therefore $1 - (\frac{7}{10})^N$. It is strictly increasing with N, because $0 < \frac{7}{10} < 1$. The calculations give

$$P_8 \simeq 0{,}942352 < 0{,}95, \qquad P_9 \simeq 0{,}959646 \geq 0{,}95.$$

Thus 9 is the smallest suitable number of trials.

6. [0.5 pt] Let $n = 20$. For $j \in \{0, \ldots, n\}$, an ordered outcome containing j successes & $n - j$ failures has, by independence, the probability $p^j(1 - p)^{n-j}$. There are $\binom{n}{j}$ choices of positions of successes. These outcomes are disjoint, therefore

$$P(X = j) = \binom{n}{j} p^j (1 - p)^{n-j}, \qquad p = \frac{3}{10}.$$

7. [0.5 pt] The events $X = 1$, $X = 2$, $X = 3$ are disjoint, so

$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{20}{j} (\frac{3}{10})^j (\frac{7}{10})^{20-j} \simeq 0{,}106289.$$

8. [1 pt]

```
from math import comb

def quantile():
    n, p = 20, 3/10
    j, cumul = 0, (1-p)**n
    while cumul < 0.95:
        j += 1
        cumul += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The added probabilities are positive and the final cumulative total is 1, so the loop stops. It returns 9. The cumulative probabilities at 8 and 9 are about 0.886669 and 0.952038: this confirms minimality.

Mock exam 2 · Zero 2

SPECIMEN PAPER · PRACTICE

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Integral and sequence — 5 points

For every natural number n, set $I_n = \int_0^1 x^n e^{4x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{4x}$.

1. Calculate I_0. (0.5 pt)

2. Using integration by parts, calculate I_1. (0.5 pt)

3. Prove, by comparing the integrands, that (I_n) is positive and decreasing. (0.5 pt)

4. Show that $I_n \leq \frac{e^4}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of (I_n). (1 pt)

5. Using integration by parts, express I_{n+1} as a function of I_n and of n. (0.5 pt)

6. Show that $I_n \geq \frac{1}{n+1}$ for all $n \in \mathbb{N}$. (0.5 pt)

7. We divide $[0;1]$ in N intervals of the same length, with $N \geq 1$. Write the sums L_N and U_N of areas of the rectangles built with the values of e^{4x} at the left and right ends. Justify $L_N \leq I_0 \leq U_N$, then calculate $U_N - L_N$. (0.5 pt)

8. Determine the smallest integer N guaranteeing that this bracketing has a width at most equal to 10^{-2}. Write a Python function that returns N, L_N and U_N. (1 pt)

Exercise 2 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x) = \frac{3+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. State the maximum value of f, then solve $f(x) = 0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)

5. Prove that the equation $f(x) = \frac{1}{2}e^2$ has exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine an antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the mean value of f on $[1;e]$. (0.5 pt)

8. Independently of the previous questions, solve $\ln(x-1) + \ln(x+1) = \ln(63)$, after specifying the set on which this equation is defined. (1 pt)

Mock exam 2 · Zero 2 · continued

Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 6$ and the point $M(1;0;0)$. The line d passes through M and has direction vector $\vec{n} = (1;2;2)$.

1. Give a parametric representation of d. (0.5 pt)

2. Prove that d meets $\mathcal{P}$ at a point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal projection of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deduce the distance from M to the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ through M. (0.5 pt)

7. Determine the angle between d and the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define the point N by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from N to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. (1 pt)

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 17 times. The trials are independent and the probability of success is $p = \frac{3}{10}$ at each trial. Let X be the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. (0.5 pt)

2. Calculate the probability of exactly two successes. (0.5 pt)

3. Using complementary events, express and then calculate the probability of obtaining at least two successes. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same trial $N \geq 1$ times, still independently. Determine the smallest integer N for which the probability of getting at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 17\}$, prove the formula for $P(X = j)$ by counting the possible positions of successes. (0.5 pt)

7. A series of 17 trials is accepted if its number of successes is between 1 and 3, endpoints included. Calculate the probability that a series is accepted. (0.5 pt)

8. Write a Python function with no argument that returns the smallest integer j such that $P(X \leq j) \geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Solutions · Mock exam 2 · Specimen 2

Exercise 1 · Integral and sequence

1. [0.5 pt] The function $x \mapsto \frac{e^{4x}}{4}$ is an antiderivative of e^{4x}. Thus $I_0 = [\frac{e^{4x}}{4}]_0^1 = \frac{e^4-1}{4}$.

2. [0.5 pt] Let us choose $u(x) = x$ and $v(x) = \frac{e^{4x}}{4}$, two differentiable functions with continuous derivatives on $[0,1]$. Then $u' = 1$ and $v' = e^{4x}$. Integration by parts gives

$$I_1 = \left[\frac{xe^{4x}}{4}\right]_0^1 - \frac{1}{4}\int_0^1 e^{4x}\,\mathrm{d}x$$
$$= \frac{e^4}{4} - \frac{e^4-1}{16} = \frac{3e^4+1}{16}.$$

3. [0.5 pt] For $x \in [0,1]$, $0 \leq x^{n+1} \leq x^n$. Since $e^{4x} > 0$, integrating with bounds in increasing order gives $0 \leq I_{n+1} \leq I_n$. The sequence is therefore positive and non-increasing. The inequalities are even strict: the integrands of I_n and of $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. [1 pt] e $^{4x} \leq e^4$; therefore $0 \leq I_n \leq e^4 \int_0^1 x^n\,\mathrm{d}x = \frac{e^4}{n+1} \to 0$.

5. [0.5 pt] Apply integration by parts to $u(x) = x^{n+1}$ & $v(x) = \frac{e^{4x}}{4}$. Since $n+1 \geq 1$, the term at $x = 0$ is zero. Hence

$$I_{n+1} = \left[\frac{x^{n+1}e^{4x}}{4}\right]_0^1 - \frac{n+1}{4}\int_0^1 x^n e^{4x}\,\mathrm{d}x$$
$$= \frac{e^4}{4} - \frac{n+1}{4}I_n.$$

6. [0.5 pt] For $0 \leq x \leq 1$, $e^{4x} \geq 1$ and $x^n \geq 0$. Hence $x^n \leq x^n e^{4x}$; by integration,

$$I_n \geq \int_0^1 x^n\,\mathrm{d}x = \left[\frac{x^{n+1}}{n+1}\right]_0^1 = \frac{1}{n+1}.$$

7. [0.5 pt] The function $x \mapsto e^{4x}$ is increasing. On each interval $[\frac{j}{N}, \frac{j+1}{N}]$, the values at the endpoints bound the function. By integrating then summing,

$$L_N = \frac{1}{N}\sum_{j=0}^{N-1} e^{\frac{4j}{N}} \leq I_0 \leq U_N = \frac{1}{N}\sum_{j=1}^{N} e^{\frac{4j}{N}}.$$

The intermediate terms cancel in the difference, so $U_N - L_N = \frac{e^4-1}{N}$.

8. [1 pt] The width is at most 10^{-2} if $N \geq 100(e^4 - 1)$. The first integer satisfying this bound is $N = 5360$. Here is the calculation of the bounds:

```
from math import exp, ceil

def bornes():
    N = ceil(100*(exp(4)-1))
    bas = sum (exp(4*j/N) for j in range (N))/N
    haut = bas+(exp(4)-1)/N
    return N, bas, haut
```

Exercise 2 · A logarithmic function

1. [0.5 pt] At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+: limit $-\infty$; at $+\infty$, limit 0 by comparing growth rates.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (3 + \ln x)}{x^2} = \frac{-2 - \ln x}{x^2}.$$

The denominator is positive. The numerator is zero at $s = e^{-2}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f strictly increases on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [0.5 pt] The maximum is reached only at $s = e^{-2}$ and is

$$f(s) = \frac{3 - 2}{e^{-2}} = e^2.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $3 + \ln x = 0$, i.e. $x = e^{-3}$.

4. [1 pt] We differentiate $(-2 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-2 - \ln x)x^{-3} = \frac{2\ln x + 3}{x^3}.$$

The numerator is zero at $t = e^{-\frac{3}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at t and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Let $m = \frac{e^2}{2}$ & $s = e^{-2}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and with value 2 m at s: the corollary of the intermediate value theorem gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from 2 m to 0 $< m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. [0.5 pt] Let $F(x) = 3\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{3}{x} + \frac{\ln x}{x} = f(x)$; F is an antiderivative on $]0, +\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{7}{2}$ & $F(1) = 0$. The integral therefore equals $\frac{7}{2}$, and the average value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{7}{2(e-1)}.$$

8. [1 pt] Domain $x > 1$; $x^2 - 1 = 63$, therefore $x = 8$, the negative root being excluded.

Exercise 3 · Line, plane and projection

1. [0.5 pt] $(x, y, z) = (1 + t, 2t, 2t)$, $t \in \mathbb{R}$.

2. [0.5 pt] $1 + 9t = 6$ gives $t = \frac{5}{9}$; $H = (\frac{14}{9}, \frac{10}{9}, \frac{10}{9})$.

3. [0.5 pt] The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Since $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{5}{9}\vec{n}$, H is the orthogonal projection of M onto the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1 + 4 + 4} = 3$. Hence

$$d(M, \mathcal{P}) = MH = |\frac{5}{9}| 3 = \frac{5}{3}.$$

4. [1 pt] Let $y = z = s$; $x = 6 - 4s$. Hence $(x, y, z) = (6, 0, 0) + s(-4, 1, 1)$.

5. [0.5 pt] Yes, its last two coordinates are equal. H is on the line and $\overrightarrow{MH}$ is orthogonal to every direction in the plane, hence to this line. Distance $\frac{5}{3}$.

6. [0.5 pt] x $+ 2y + 2z = 1$; same normal vector and passing through M.

7. [0.5 pt] $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. [1 pt] $N = M + \vec{n}$ is on the same normal line as M ; its projection onto $\mathcal{P}$ is therefore also H. Now $H \in \mathcal{P} \cap \mathcal{Q}$. For every P of this line, $NP^2 = NH^2 + HP^2$, therefore its minimum distance to N is NH. Finally,

$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{5}{9} - 1| \, \|\vec{n}\| = \frac{4}{3}.$$

Exercise 4 · Repetitions and estimation

1. [0.5 pt] Repetition of a fixed number 17 of identical independent trials with two outcomes; $X \sim \mathcal{B}(17, \frac{3}{10})$.

2. [0.5 pt] P $(X = 2) = \binom{17}{2}(\frac{3}{10})^2(\frac{7}{10})^{15} \simeq 0{,}058110$.

3. [0.5 pt] $1 - P(X = 0) - P(X = 1) = 1 - (\frac{7}{10})^{17} - 17(\frac{3}{10})(\frac{7}{10})^{16} \simeq 0{,}980725$.

4. [1 pt] E $(X) = \frac{51}{10}$, $V(X) = \frac{357}{100}$.

5. [0.5 pt] For N independent trials, the probability of no success is $(1 - p)^N$. The required prob. is therefore $1 - (\frac{7}{10})^N$. It is strictly increasing with N, because $0 < \frac{7}{10} < 1$. The calculations give

$$P_8 \simeq 0{,}942352 < 0{,}95, \qquad P_9 \simeq 0{,}959646 \geq 0{,}95.$$

Thus 9 is the smallest suitable number of trials.

6. [0.5 pt] Let $n = 17$. For $j \in \{0, \ldots, n\}$, an ordered outcome containing j successes & $n - j$ failures has, by independence, the probability $p^j(1 - p)^{n-j}$. There are $\binom{n}{j}$ choices of positions of successes. These outcomes are disjoint, therefore

$$P(X = j) = \binom{n}{j} p^j (1 - p)^{n-j}, \qquad p = \frac{3}{10}.$$

7. [0.5 pt] The events $X = 1$, $X = 2$, $X = 3$ are disjoint, so

$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{17}{j} (\frac{3}{10})^j (\frac{7}{10})^{17-j} \simeq 0{,}199581.$$

8. [1 pt]

```
from math import comb

def quantile():
    n, p = 17, 3/10
    j, cumul = 0, (1-p)**n
    while cumul < 0.95:
        j += 1
        cumul += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The added probabilities are positive and the final cumulative total is 1, so the loop stops. It returns 8. The cumulative probabilities at 7 and 8 are about 0.895360 and 0.959723: this confirms minimality.

Mock exam 2 · A

PAPER RESERVED FOR ASSESSMENT

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · An area to split — 4 points

On $[0;1]$, we consider $f(x) = x^2$. Set $L_n = \frac{1}{n}\sum_{k=0}^{n-1}\left(\frac{k}{n}\right)^2$ and $U_n = \frac{1}{n}\sum_{k=1}^{n}\left(\frac{k}{n}\right)^2$ for $n \geq 1$.

1. Draw the corresponding rectangles for $n = 4$ and justify $L_n \leq \int_0^1 x^2\,\mathrm{d}x \leq U_n$. (1 pt)

2. Calculate $U_n - L_n$ then determine how it varies. n which guarantees a width of at most 10^{-3}. (1 pt)

3. Calculate the integral exactly and compare it with the bounds. (1 pt)

4. Determine the x-coordinate $a \in [0;1]$ which divides this area into two equal areas. (1 pt)

Exercise 2 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x) = \frac{2+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. State the maximum value of f, then solve $f(x) = 0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)

5. Prove that the equation $f(x) = \frac{1}{2}e^1$ has exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine an antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the mean value of f on $[1;e]$. (0.5 pt)

8. Independently of the previous questions, solve $\ln(x-2) + \ln(x+2) = \ln(21)$, after specifying the set on which this equation is defined. (1 pt)

Mock exam 2 · A · continued

Exercise 3 · Two planes and a projection — 6 points

In an orthonormal coordinate system, $\mathcal{P} : x + y + z = 3$, $\mathcal{Q} : x - y = 0$ and $M(3;0;0)$.

1. Justify that the two planes intersect. (1 pt)

2. Give a parametric representation of their intersection d. (1 pt)

3. Determine the projection H of M on d. (2 pts)

4. Calculate the distance from M to d. (1 pt)

5. Is this projection also that of M on $\mathcal{P}$? Justify. (1 pt)

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 18 times. The trials are independent and the probability of success is $p = \frac{1}{5}$ at each trial. Let X be the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. (0.5 pt)

2. Calculate the probability of exactly two successes. (0.5 pt)

3. Using complementary events, express and then calculate the probability of obtaining at least two successes. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same trial $N \geq 1$ times, still independently. Determine the smallest integer N for which the probability of getting at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 18\}$, prove the formula for $P(X = j)$ by counting the possible positions of successes. (0.5 pt)

7. A series of 18 trials is accepted if its number of successes is between 1 and 3, endpoints included. Calculate the probability that a series is accepted. (0.5 pt)

8. Write a Python function with no argument that returns the smallest integer j such that $P(X \leq j) \geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Solutions · Mock exam 2 · A

Exercise 1 · An area to split

1. On each interval of width 1 /n, x^2 lies between its values at the endpoints. The rectangles on the left give lower bounds and those on the right give upper bounds. For $n = 4$, left heights 0, 1 /16, 1/4, 9/16, right heights 1/16, 1/4, 9/16, 1.

2. The interior terms cancel: $U_n - L_n = 1/n$. Take $n \geq 1000$.

3. $\int_0^1 x^2\,\mathrm{d}x = 1/3$. For $n = 4$, $L_4 = 7/32$ and $U_4 = 15/32$ do bound 1/3; the width is 1/4.

4. $\int_0^a x^2\,\mathrm{d}x = a^3/3 = 1/6$ gives $a = \sqrt[3]{1/2}$. The strict monotonicity of $a \mapsto a^3$ guarantees uniqueness.

Exercise 2 · A logarithmic function

1. [0.5 pt] At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+: limit $-\infty$; at $+\infty$, limit 0 by comparing growth rates.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (2 + \ln x)}{x^2} = \frac{-1 - \ln x}{x^2}.$$

The denominator is positive. The numerator is zero at $s = e^{-1}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f strictly increases on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [0.5 pt] The maximum is reached only at $s = e^{-1}$ and is

$$f(s) = \frac{2 - 1}{e^{-1}} = e^1.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $2 + \ln x = 0$, i.e. $x = e^{-2}$.

4. [1 pt] We differentiate $(-1 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-1 - \ln x)x^{-3} = \frac{2\ln x + 1}{x^3}.$$

The numerator is zero at $t = e^{-\frac{1}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at t and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Let $m = \frac{e^1}{2}$ & $s = e^{-1}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and with value 2 m at s: the corollary of the intermediate value theorem gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from 2 m to 0 $< m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. [0.5 pt] Let $F(x) = 2\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{2}{x} + \frac{\ln x}{x} = f(x)$; F is an antiderivative on $]0, +\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{5}{2}$ & $F(1) = 0$. The integral therefore equals $\frac{5}{2}$, and the average value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{5}{2(e-1)}.$$

8. [1 pt] Domain $x > 2$; $x^2 - 4 = 21$, therefore $x = 5$, the negative root being excluded.

Exercise 3 · Two planes and a projection

1. Normal vectors $(1;1;1)$ and $(1;-1;0)$ non-collinear, so the planes intersect.

2. $x = y = t$, $z = 3 - 2t$. Thus $d : (t;t;3-2t)$, dir. vector $u = (1;1;-2)$.

3. $H = (t;t;3-2t)$; $MH = (t-3;t;3-2t)$. The equation $MH \cdot u = 6t - 9 = 0$ gives $t = 3/2$, i.e. $H(3/2;3/2;0)$. One coordinate and the system: 1 point; solving and checking membership: 1 point.

4. $MH = \sqrt{(3/2)^2 + (3/2)^2} = 3/\sqrt{2}$.

5. No: M already belongs to $\mathcal{P}$ and its projection onto this plane is M, while $H \neq M$.

Exercise 4 · Repetitions and estimation

1. [0.5 pt] Repetition of a fixed number 18 of identical independent trials with two outcomes; $X \sim \mathcal{B}(18, \frac{1}{5})$.

2. [0.5 pt] P $(X = 2) = \binom{18}{2}(\frac{1}{5})^2(\frac{4}{5})^{16} \simeq 0{,}172263$.

3. [0.5 pt] $1 - P(X = 0) - P(X = 1) = 1 - (\frac{4}{5})^{18} - 18(\frac{1}{5})(\frac{4}{5})^{17} \simeq 0{,}900921$.

4. [1 pt] E $(X) = \frac{18}{5}$, $V(X) = \frac{72}{25}$.

5. [0.5 pt] For N independent trials, the probability of no success is $(1-p)^N$. The required prob. is therefore $1 - (\frac{4}{5})^N$. It is strictly increasing with N, because $0 < \frac{4}{5} < 1$. The calculations give

$$P_{13} \simeq 0{,}945024 < 0{,}95, \qquad P_{14} \simeq 0{,}956020 \geq 0{,}95.$$

Thus 14 is the smallest suitable number of trials.

6. [0.5 pt] Let $n = 18$. For $j \in \{0, \ldots, n\}$, an ordered outcome containing j successes & $n - j$ failures has, by independence, the probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices of positions of successes. These outcomes are disjoint, therefore

$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{1}{5}.$$

7. [0.5 pt] The events $X = 1$, $X = 2$, $X = 3$ are disjoint, so

$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{18}{j} (\frac{1}{5})^j (\frac{4}{5})^{18-j} \simeq 0{,}483011.$$

8. [1 pt]

```
from math import comb

def quantile():
    n, p = 18, 2/10
    j, cumul = 0, (1-p)**n
    while cumul < 0.95:
        j += 1
        cumul += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The added probabilities are positive and the final cumulative total is 1, so the loop stops. It returns 7. The cumulative probabilities at 6 and 7 are about 0.948729 and 0.983720: this confirms minimality.

Mock exam 2 · B

PAPER RESERVED FOR ASSESSMENT

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Signed area, geometric area — 4 points

Consider $f(x) = x - 1$ on $[0;3]$.

1. Draw the curve and indicate the sign of f. (1 pt)
2. Calculate $\int_0^3 f(x)\,\mathrm{d}x$, then the geometric area between the curve and the x-axis. (1 pt)
3. Determine the mean value of f and construct the rectangle with the same signed area on $[0;3]$. (1 pt)
4. For $a \in [1;3]$, solve $\int_0^a f(x)\,\mathrm{d}x = 0$ and explain how the areas cancel. (1 pt)

Exercise 2 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x) = \frac{5+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)
3. State the maximum value of f, then solve $f(x) = 0$. (0.5 pt)
4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)
5. Prove that the equation $f(x) = \frac{1}{2}e^4$ has exactly two solutions in $]0;+\infty[$. (0.5 pt)
6. Determine an antiderivative of f on $]0;+\infty[$. (0.5 pt)
7. Calculate the mean value of f on $[1;e]$. (0.5 pt)
8. Independently of the previous questions, solve $\ln(x-3) + \ln(x+3) = \ln(27)$, after specifying the set on which this equation is defined. (1 pt)

Mock exam 2 · B · continued

Exercise 3 · A line whose position depends on a parameter — 6 points

In an orthonormal coordinate system, $\mathcal{P} : x + 2y - z = 1$. For $m \in \mathbb{R}$, d_m is defined by $(x;y;z) = (0;0;m) + t(1;m;1)$, $t \in \mathbb{R}$.

1. Show that each d_m is indeed a line. (1 pt)

2. Determine the intersection with $\mathcal{P}$ when $m \neq 0$. (2 pts)

3. Study separately $m = 0$. (1 pt)

4. For $m = 1$, calculate the point of intersection. (1 pt)

5. Is the line d_1 orthogonal to the plane? (1 pt)

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 20 times. The trials are independent and the probability of success is $p = \frac{1}{10}$ at each trial. Let X be the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. (0.5 pt)

2. Calculate the probability of exactly two successes. (0.5 pt)

3. Using complementary events, express and then calculate the probability of obtaining at least two successes. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same trial $N \geq 1$ times, still independently. Determine the smallest integer N for which the probability of getting at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 20\}$, prove the formula for $P(X = j)$ by counting the possible positions of successes. (0.5 pt)

7. A series of 20 trials is accepted if its number of successes is between 1 and 3, endpoints included. Calculate the probability that a series is accepted. (0.5 pt)

8. Write a Python function with no argument that returns the smallest integer j such that $P(X \leq j) \geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Solutions · Mock exam 2 · B

Exercise 1 · Signed area, geometric area

1. Line through $(0;-1)$ and $(1;0)$; negative sign up to 1, positive after.

2. $[x^2/2-x]_0^3 = 3/2$. Negative area in absolute value: 1 /2; positive area: 2, so the geometric area 5/2. Two results, 0.5 point each.

3. Mean $(3/2)/3 = 1/2$. Rectangle of width 3 and height 1/2, so area 3/2.

4. $a^2/2 - a = a(a-2)/2 = 0$; the only solution in $[1;3]$ is $a = 2$. The two triangles on $[0;1]$ and $[1;2]$ each have area 1/2 and opposite signs.

Exercise 2 · A logarithmic function

1. [0.5 pt] At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+ : limit $-\infty$; at $+\infty$, limit 0 by comparing growth rates.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (5+\ln x)}{x^2} = \frac{-4-\ln x}{x^2}.$$

The denominator is positive. The numerator is zero at $s = e^{-4}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f strictly increases on $]0,s]$, then decreases strictly on $[s,+\infty[$.

3. [0.5 pt] The maximum is reached only at $s = e^{-4}$ and is

$$f(s) = \frac{5-4}{e^{-4}} = e^4.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $5 + \ln x = 0$, i.e. $x = e^{-5}$.

4. [1 pt] We differentiate $(-4-\ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-4-\ln x)x^{-3} = \frac{2\ln x + 7}{x^3}.$$

The numerator is zero at $t = e^{-\frac{7}{2}}$ and changes from negative to positive. The function is concave on $]0,t]$, convex on $[t,+\infty[$. It changes convexity at t and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Let $m = \frac{e^4}{2}$ & $s = e^{-4}$. The function is continuous on $]0,+\infty[$. On $]0,s]$, it is strictly increasing, with limit $-\infty$ at 0 and with value 2 m at s : the corollary of the intermediate value theorem gives exactly one solution in $]0,s[$. On $[s,+\infty[$, it is strictly decreasing from 2 m to 0 $< m$: exactly one solution in $]s,+\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. [0.5 pt] Let $F(x) = 5\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{5}{x} + \frac{\ln x}{x} = f(x)$; F is an antiderivative on $]0,+\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{11}{2}$ & $F(1) = 0$. The integral therefore equals $\frac{11}{2}$, and the average value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{11}{2(e-1)}.$$

8. [1 pt] Domain $x > 3$; $x^2 - 9 = 27$, so $x = 6$, the negative root being excluded.

Exercise 3 · A line whose position depends on a parameter

1. $(1;m;1) \neq 0$ because its first coordinate is 1.

2. The substitution gives $t+2mt-(m+t)=1$, i.e. $2mt=1+m$. If $m \neq 0$, $t=(1+m)/(2m)$: unique point $((1+m)/(2m);(1+m)/2;m+(1+m)/(2m))$. Equation 1 point, solution and coordinates 1 point.

3. For $m=0$, the equality is $0=1$: empty intersection, line strictly parallel to the plane.

4. $t=1$ and point $(1;1;2)$, which satisfies $1+2-2=1$.

5. No: $(1;1;1)$ is not collinear with the normal $(1;2;-1)$. Intersecting does not mean being perpendicular.

Exercise 4 · Repetitions and estimation

1. [0.5 pt] Repetition of a fixed number 20 of identical independent trials with two outcomes; $X \sim \mathcal{B}(20, \frac{1}{10})$.

2. [0.5 pt] $P(X=2)=\binom{20}{2}(\frac{1}{10})^2(\frac{9}{10})^{18} \simeq 0{,}285180$.

3. [0.5 pt] $1-P(X=0)-P(X=1)=1-(\frac{9}{10})^{20}-20(\frac{1}{10})(\frac{9}{10})^{19} \simeq 0{,}608253$.

4. [1 pt] $E(X)=2$, $V(X)=\frac{9}{5}$.

5. [0.5 pt] For N independent trials, the probability of no success is $(1-p)^N$. The required prob. is therefore $1-(\frac{9}{10})^N$. It is strictly increasing with N, because $0<\frac{9}{10}<1$. The calculations give

$$P_{28} \simeq 0{,}947665 < 0{,}95, \qquad P_{29} \simeq 0{,}952899 \geq 0{,}95.$$

   Thus 29 is the smallest suitable number of trials.

6. [0.5 pt] Let $n=20$. For $j \in \{0,\dots,n\}$, an ordered outcome containing j successes & $n-j$ failures has, by independence, the probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices of positions of successes. These outcomes are disjoint, therefore

$$P(X=j)=\binom{n}{j}p^j(1-p)^{n-j}, \qquad p=\frac{1}{10}.$$

7. [0.5 pt] The events $X=1$, $X=2$, $X=3$ are disjoint, so

$$P(1 \leq X \leq 3)=\sum_{j=1}^{3}\binom{20}{j}(\frac{1}{10})^j(\frac{9}{10})^{20-j} \simeq 0{,}745470.$$

8. [1 pt]

```
from math import comb

def quantile():
    n, p = 20, 1/10
    j, cumul = 0, (1-p)**n
    while cumul < 0.95:
        j += 1
        cumul += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The added probabilities are positive and the final cumulative total is 1, so the loop stops. It returns 4. The cumulative probabilities at 3 and 4 are about 0.867047 and 0.956826: this confirms minimality.

Final mock exam · Zero 1

SPECIMEN PAPER · PRACTICE

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x)=\frac{2+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)
3. State the maximum value of f, then solve $f(x)=0$. (0.5 pt)
4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)
5. Prove that the equation $f(x)=\frac{1}{2}e^1$ has exactly two solutions in $]0;+\infty[$. (0.5 pt)
6. Determine an antiderivative of f on $]0;+\infty[$. (0.5 pt)
7. Calculate the mean value of f on $[1;e]$. (0.5 pt)
8. Independently of the previous questions, solve $\ln(x-2)+\ln(x+2)=\ln(32)$, after specifying the set on which this equation is defined. (1 pt)

Exercise 2 · Periodic phenomenon — 5 points

A point moves on a wheel turning at constant speed in a vertical plane. Its height above the ground is modelled by $h(t)=9-3\cos\left(\frac{\pi t}{5}\right)$, for t expressed in seconds and $h(t)$ in metres. We study one complete revolution, for $t\in[0;10]$.

1. Give a period of the model extended to $\mathbb{R}$, then the minimum and maximum heights on a turn. (0.5 pt)
2. Calculate $h'(t)$ and draw up the table of variations of h on $[0;10]$. (0.5 pt)
3. Determine all the times $t\in[0;10]$ such that $h(t)=\frac{21}{2}$. (0.5 pt)
4. Calculate the mean value of h on $[0;10]$. (1 pt)
5. Study the convexity of h on the interval studied and determine the coordinates of the points of inflection of its curve. (0.5 pt)
6. Determine the total time during which the height is at least the level of question 3. (0.5 pt)
7. The vertical speed is given by $h'(t)$. Determine its maximum and minimum values, in metres per second, as well as the times at which they are reached. (0.5 pt)
8. Calculate the vertical displacement $h(10)-h(0)$, then the sum of the height climbed on the way up and the height descended on the way down. Explain why these two results are different. (1 pt)

Final mock exam · Zero 1 · continued

Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x+2y+2z=11$ and point $M(2;0;0)$. The line d passes through M and has direction vector $\vec{n}=(1;2;2)$.

1. Give a parametric representation of d. (0.5 pt)

2. Prove that d meets $\mathcal{P}$ at a point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal projection of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. Consider the plane $\mathcal{Q}$ with equation $y=z$. Prove that $\mathcal{P}\cap\mathcal{Q}$ is a line and give a parametric representation of it. (1 pt)

5. Verify that $H\in\mathcal{Q}$. Deduce the distance from M to the line $\mathcal{P}\cap\mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ through M. (0.5 pt)

7. Determine the angle between d and the line $\mathcal{P}\cap\mathcal{Q}$. (0.5 pt)

8. Define the point N by $\overrightarrow{MN}=\vec{n}$. Calculate the distances from N to the plane $\mathcal{P}$ and to the line $\mathcal{P}\cap\mathcal{Q}$, then compare them. (1 pt)

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 20 times. The trials are independent dent and the probability of success is $p=\frac{1}{5}$ at each trial. Let X be the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. (0.5 pt)

2. Calculate the probability of exactly two successes. (0.5 pt)

3. Using complementary events, express and then calculate the probability of obtaining at least two successes. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same trial $N\geq 1$ times, still independently. Determine the smallest integer N for which the probability of getting at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. (0.5 pt)

6. For an integer $j\in\{0,\dots,20\}$, prove the formula for $P(X=j)$ by counting the possible positions of successes. (0.5 pt)

7. For N independent trials with the same parameter p, let F_N be the frequency of successes. Prove the expressions for $E(F_N)$ and $V(F_N)$ as a function of N and p. (0.5 pt)

8. Using Bienaymé-Chebyshev, determine a sample size N sufficient to ensure $P(|F_N-p|<0{,}02)\geq 0{,}95$. Is the size obtained necessarily the smallest possible for this probability ? Justify. (1 pt)

Solutions · Final mock exam · Specimen 1

Exercise 1 · A logarithmic function

1. [0.5 pt] At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+: limit $-\infty$; at $+\infty$, limit 0 by comparing growth rates.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (2+\ln x)}{x^2} = \frac{-1-\ln x}{x^2}.$$

The denominator is positive. The numerator is zero at $s = e^{-1}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f strictly increases on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [0.5 pt] The maximum is reached only at $s = e^{-1}$ and is

$$f(s) = \frac{2-1}{e^{-1}} = e^1.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $2 + \ln x = 0$, i.e. $x = e^{-2}$.

4. [1 pt] We differentiate $(-1-\ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-1-\ln x)x^{-3} = \frac{2\ln x + 1}{x^3}.$$

The numerator is zero at $t = e^{-\frac{1}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at t and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Let $m = \frac{e^1}{2}$ & $s = e^{-1}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and with value 2 m at s: the corollary of the intermediate value theorem gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from 2 m to 0 $< m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. [0.5 pt] Let $F(x) = 2\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{2}{x} + \frac{\ln x}{x} = f(x)$; F is an antiderivative on $]0, +\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{5}{2}$ & $F(1) = 0$. The integral therefore equals $\frac{5}{2}$, and the average value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{5}{2(e-1)}.$$

8. [1 pt] Domain $x > 2$; $x^2 - 4 = 32$, so $x = 6$, the negative root being excluded.

Exercise 2 · Periodic phenomenon

1. [0.5 pt] Period 10 s; minimum 6 m, maximum 12 m.

2. [0.5 pt] h $'(t) = (\frac{3}{5}\pi)\sin(\frac{\pi t}{5})$; increasing on $[0, 5]$, decreasing on $[5, 10]$.

3. [0.5 pt] The equation is equivalent to $\cos(\frac{\pi t}{5}) = -\frac{1}{2}$. Over one turn, $\frac{\pi t}{5} \in [0, 2\pi]$; the suitable angles are $\frac{2\pi}{3}$ & $\frac{4\pi}{3}$. Thus $t = \frac{10}{3}$ s or $t = \frac{20}{3}$ s.

4. [1 pt] An antiderivative is $F(t) = 9t - \frac{15}{\pi}\sin(\frac{\pi t}{5})$. Since $\sin 0 = \sin(2\pi) = 0$,

$$\overline{h} = \frac{1}{10}\int_0^{10} h(t)\,\mathrm{d}t = \frac{F(10) - F(0)}{10} = \frac{90}{10} = 9 \text{ m}.$$

5. [0.5 pt] The second derivative is $h''(t) = \frac{3\pi^2}{25}\cos(\frac{\pi t}{5})$. Its sign is that of the cosine: positive on $[0, \frac{5}{2}[$ & $]\frac{15}{2}, 10]$, negative between these two intervals. The function is convex on $[0, \frac{5}{2}]$ and $[\frac{15}{2}, 10]$, concave on $[\frac{5}{2}, \frac{15}{2}]$. The two changes of convexity give the inflection points $(\frac{5}{2}, 9)$ and $(\frac{15}{2}, 9)$.

6. [0.5 pt] Between $\frac{10}{3}$ & $\frac{20}{3}$; duration $\frac{10}{3}$ s.

7. [0.5 pt] h ′ reaches $\frac{3}{5}\pi$ m s^{-1} in $t = \frac{5}{2}$ and its opposite at $t = \frac{15}{2}$.

8. [1 pt] Displacement $h(10) - h(0) = 0$. Ascent of 6 m then descent of 6 m: cumulative distance 12 m, equal to the integral of $|h'|$.

Exercise 3 · Line, plane and projection

1. [0.5 pt] $(x, y, z) = (2 + t, 2t, 2t)$, $t \in \mathbb{R}$.

2. [0.5 pt] $2 + 9t = 11$ gives $t = 1$; $H = (3, 2, 2)$.

3. [0.5 pt] The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Since $H \in \mathcal{P}$ and $\overrightarrow{MH} = \vec{n}$, H is the orthogonal projection of M onto the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1 + 4 + 4} = 3$. Hence

$$d(M, \mathcal{P}) = MH = |1|\,3 = 3.$$

4. [1 pt] Let $y = z = s$; $x = 11 - 4s$. Hence $(x, y, z) = (11, 0, 0) + s(-4, 1, 1)$.

5. [0.5 pt] Yes, its last two coordinates are equal. H is on the line and $\overrightarrow{MH}$ is orthogonal to every direction in the plane, hence to this line. Distance 3.

6. [0.5 pt] x $+ 2y + 2z = 2$; same normal vector and passing through M.

7. [0.5 pt] $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. [1 pt] $N = M + \vec{n}$ is on the same normal line as M ; its projection onto $\mathcal{P}$ is therefore also H. Now $H \in \mathcal{P} \cap \mathcal{Q}$. For every P of this line, $NP^2 = NH^2 + HP^2$, therefore its minimum distance to N is NH. Finally,

$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |1 - 1|\,\|\vec{n}\| = 0.$$

Exercise 4 · Repetitions and estimation

1. [0.5 pt] Repetition of a fixed number 20 of identical independent trials with two outcomes; $X \sim \mathcal{B}(20, \frac{1}{5})$.

2. [0.5 pt] P $(X = 2) = \binom{20}{2}(\frac{1}{5})^2(\frac{4}{5})^{18} \simeq 0{,}136909$.

3. [0.5 pt] $1 - P(X = 0) - P(X = 1) = 1 - (\frac{4}{5})^{20} - 20(\frac{1}{5})(\frac{4}{5})^{19} \simeq 0{,}930825$.

4. [1 pt] E $(X) = 4$, $V(X) = \frac{16}{5}$.

5. [0.5 pt] For N independent trials, the probability of no success is $(1 - p)^N$. The required prob. is therefore 1 $- (\frac{4}{5})^N$. It is strictly increasing with N, because $0 < \frac{4}{5} < 1$. The calculations give

$$P_{13} \simeq 0{,}945024 < 0{,}95, \qquad P_{14} \simeq 0{,}956020 \geq 0{,}95.$$

Thus 14 is the smallest suitable number of trials.

6. [0.5 pt] Let $n = 20$. For $j \in \{0, \ldots, n\}$, an ordered outcome containing j successes & $n - j$ failures has, by independence, the probability $p^j(1 - p)^{n-j}$. There are $\binom{n}{j}$ choices of positions of successes. These outcomes are disjoint, therefore

$$P(X = j) = \binom{n}{j} p^j (1 - p)^{n-j}, \qquad p = \frac{1}{5}.$$

7. [0.5 pt] If $B_1, \dots, B_N$ are the independent indicator variables of the successes, $F_N = \frac{B_1+\cdots+B_N}{N}$. With $E(B_i) = p$ & $V(B_i) = p(1-p)$,

$$E(F_N) = \frac{Np}{N} = \frac{1}{5}, \qquad V(F_N) = \frac{Np(1-p)}{N^2} = \frac{\frac{4}{25}}{N}.$$

Independence is required for the addition of variances, not for the linearity of expectation.

8. [1 pt] Bienaymé-Chebyshev applied to F_N gives

$$P(|F_N - p| \geq 0{,}02) \leq \frac{p(1-p)}{N(0{,}02)^2}.$$

To ensure that this probability is at most 0.05, it is enough to

$$N \geq \frac{\frac{4}{25}}{0{,}05 \times (0{,}02)^2} = 8000.$$

By complement, we then have $P(|F_N - p| < 0{,}02) \geq 0{,}95$. The size 8000 is sufficient; the inequality being an upper bound, it does not necessarily give the minimum size for the exact prob.

Final mock exam · Zero 2

SPECIMEN PAPER · PRACTICE

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x)=\frac{5+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)
3. State the maximum value of f, then solve $f(x)=0$. (0.5 pt)
4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)
5. Prove that the equation $f(x)=\frac{1}{2}e^4$ has exactly two solutions in $]0;+\infty[$. (0.5 pt)
6. Determine an antiderivative of f on $]0;+\infty[$. (0.5 pt)
7. Calculate the mean value of f on $[1;e]$. (0.5 pt)
8. Independently of the previous questions, solve $\ln(x-3)+\ln(x+3)=\ln(40)$, after specifying the set on which this equation is defined. (1 pt)

Exercise 2 · Periodic phenomenon — 5 points

A point moves on a wheel turning at constant speed in a vertical plane. Its height above the ground is modelled by $h(t)=12-4\cos\left(\frac{\pi t}{4}\right)$, for t expressed in seconds and $h(t)$ in metres. We study one complete revolution, for $t\in[0;8]$.

1. Give a period of the model extended to $\mathbb{R}$, then the minimum and maximum heights on a turn. (0.5 pt)
2. Calculate $h'(t)$ and draw up the table of variations of h on $[0;8]$. (0.5 pt)
3. Determine all the times $t\in[0;8]$ such that $h(t)=14$. (0.5 pt)
4. Calculate the mean value of h on $[0;8]$. (1 pt)
5. Study the convexity of h on the interval studied and determine the coordinates of the points of inflection of its curve. (0.5 pt)
6. Determine the total time during which the height is at least the level of question 3. (0.5 pt)
7. The vertical speed is given by $h'(t)$. Determine its maximum and minimum values, in metres per second, as well as the times at which they are reached. (0.5 pt)
8. Calculate the vertical displacement $h(8)-h(0)$, then the sum of the height climbed on the way up and the height descended on the way down. Explain why these two results are different. (1 pt)

Final mock exam · Zero 2 · continued

Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 7$ and point $M(3;0;0)$. The line d passes through M and has direction vector $\vec{n} = (1;2;2)$.

1. Give a parametric representation of d. (0.5 pt)

2. Prove that d meets $\mathcal{P}$ at a point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal projection of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deduce the distance from M to the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ through M. (0.5 pt)

7. Determine the angle between d and the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define the point N by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from N to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. (1 pt)

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 17 times. The trials are independent and the probability of success is $p = \frac{3}{10}$ at each trial. Let X be the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. (0.5 pt)

2. Calculate the probability of exactly two successes. (0.5 pt)

3. Using complementary events, express and then calculate the probability of obtaining at least two successes. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same trial $N \geq 1$ times, still independently. Determine the smallest integer N for which the probability of getting at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 17\}$, prove the formula for $P(X = j)$ by counting the possible positions of successes. (0.5 pt)

7. For N independent trials with the same parameter p, let F_N be the frequency of successes. Prove the expressions for $E(F_N)$ and $V(F_N)$ as a function of N and p. (0.5 pt)

8. Using Bienaymé-Chebyshev, determine a sample size N sufficient to ensure $P(|F_N - p| < 0{,}02) \geq 0{,}95$. Is the size obtained necessarily the smallest possible for this probability ? Justify. (1 pt)

Solutions · Final mock exam · Specimen 2

Exercise 1 · A logarithmic function

1. [0.5 pt] At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+: limit $-\infty$; at $+\infty$, limit 0 by comparing growth rates.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (5+\ln x)}{x^2} = \frac{-4-\ln x}{x^2}.$$

The denominator is positive. The numerator is zero at $s = e^{-4}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f strictly increases on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [0.5 pt] The maximum is reached only at $s = e^{-4}$ and is

$$f(s) = \frac{5-4}{e^{-4}} = e^4.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $5 + \ln x = 0$, i.e. $x = e^{-5}$.

4. [1 pt] We differentiate $(-4-\ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-4-\ln x)x^{-3} = \frac{2\ln x + 7}{x^3}.$$

The numerator is zero at $t = e^{-\frac{7}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at t and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Let $m = \frac{e^4}{2}$ & $s = e^{-4}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and with value 2 m at s: the corollary of the intermediate value theorem gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from 2 m to 0 $< m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. [0.5 pt] Let $F(x) = 5\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{5}{x} + \frac{\ln x}{x} = f(x)$; F is an antiderivative on $]0, +\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{11}{2}$ & $F(1) = 0$. The integral therefore equals $\frac{11}{2}$, and the average value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{11}{2(e-1)}.$$

8. [1 pt] Domain $x > 3$; $x^2 - 9 = 40$, so $x = 7$, the negative root being excluded.

Exercise 2 · Periodic phenomenon

1. [0.5 pt] Period 8 s; minimum 8 m, maximum 16 m.

2. [0.5 pt] $h'(t) = \pi\sin(\frac{\pi t}{4})$; increasing on $[0, 4]$, decreasing on $[4, 8]$.

3. [0.5 pt] The equation is equivalent to $\cos(\frac{\pi t}{4}) = -\frac{1}{2}$. Over one turn, $\frac{\pi t}{4} \in [0, 2\pi]$; the suitable angles are $\frac{2\pi}{3}$ & $\frac{4\pi}{3}$. Thus $t = \frac{8}{3}$ s or $t = \frac{16}{3}$ s.

4. [1 pt] An antiderivative is $F(t) = 12t - \frac{16}{\pi}\sin(\frac{\pi t}{4})$. Since $\sin 0 = \sin(2\pi) = 0$,

$$\overline{h} = \frac{1}{8}\int_0^8 h(t)\,\mathrm{d}t = \frac{F(8)-F(0)}{8} = \frac{96}{8} = 12 \text{ m}.$$

5. [0.5 pt] The second derivative is $h''(t) = \frac{4\pi^2}{16}\cos(\frac{\pi t}{4})$. Its sign is that of the cosine: positive on $[0,2[$ & $]6,8]$, negative between these two intervals. The function is convex on $[0,2]$ and $[6,8]$, concave on $[2,6]$. The two changes of convexity give the inflection points $(2,12)$ and $(6,12)$.

6. [0.5 pt] Between $\frac{8}{3}$ & $\frac{16}{3}$; duration $\frac{8}{3}$ s.

7. [0.5 pt] h $'$ reaches 1π m s^{-1} at $t = 2$ and its opposite at $t = 6$.

8. [1 pt] Displacement $h(8) - h(0) = 0$. Ascent of 8 m then descent of 8 m: cumulative distance 16 m, equal to the integral of $|h'|$.

Exercise 3 · Line, plane and projection

1. [0.5 pt] $(x,y,z) = (3+t, 2t, 2t)$, $t \in \mathbb{R}$.

2. [0.5 pt] $3 + 9t = 7$ gives $t = \frac{4}{9}$; $H = (\frac{31}{9}, \frac{8}{9}, \frac{8}{9})$.

3. [0.5 pt] The vector $\vec{n} = (1,2,2)$ is normal to $\mathcal{P}$. Since $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{4}{9}\vec{n}$, H is the orthogonal projection of M onto the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1+4+4} = 3$. Hence

$$d(M,\mathcal{P}) = MH = |\frac{4}{9}|\,3 = \frac{4}{3}.$$

4. [1 pt] Let $y = z = s$; $x = 7 - 4s$. Hence $(x,y,z) = (7,0,0) + s(-4,1,1)$.

5. [0.5 pt] Yes, its last two coordinates are equal. H is on the line and $\overrightarrow{MH}$ is orthogonal to every direction in the plane, hence to this line. Distance $\frac{4}{3}$.

6. [0.5 pt] x $+ 2y + 2z = 3$; same normal vector and passing through M.

7. [0.5 pt] $\vec{n} \cdot (-4,1,1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. [1 pt] $N = M + \vec{n}$ is on the same normal line as M ; its projection onto $\mathcal{P}$ is therefore also H. Now $H \in \mathcal{P} \cap \mathcal{Q}$. For every P of this line, $NP^2 = NH^2 + HP^2$, therefore its minimum distance to N is NH. Finally,

$$d(N,\mathcal{P}) = d(N,\mathcal{P}\cap\mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{4}{9} - 1|\,\|\vec{n}\| = \frac{5}{3}.$$

Exercise 4 · Repetitions and estimation

1. [0.5 pt] Repetition of a fixed number 17 of identical independent trials with two outcomes; $X \sim \mathcal{B}(17, \frac{3}{10})$.

2. [0.5 pt] P $(X = 2) = \binom{17}{2}(\frac{3}{10})^2(\frac{7}{10})^{15} \simeq 0{,}058110$.

3. [0.5 pt] $1 - P(X=0) - P(X=1) = 1 - (\frac{7}{10})^{17} - 17(\frac{3}{10})(\frac{7}{10})^{16} \simeq 0{,}980725$.

4. [1 pt] E $(X) = \frac{51}{10}$, $V(X) = \frac{357}{100}$.

5. [0.5 pt] For N independent trials, the probability of no success is $(1-p)^N$. The required prob. is therefore $1 - (\frac{7}{10})^N$. It is strictly increasing with N, because $0 < \frac{7}{10} < 1$. The calculations give

$$P_8 \simeq 0{,}942352 < 0{,}95, \qquad P_9 \simeq 0{,}959646 \geq 0{,}95.$$

Thus 9 is the smallest suitable number of trials.

6. [0.5 pt] Let $n = 17$. For $j \in \{0,\dots,n\}$, an ordered outcome containing j successes & $n-j$ failures has, by independence, the probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices of positions of successes. These outcomes are disjoint, therefore

$$P(X=j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{3}{10}.$$

7. [0.5 pt] If $B_1, \ldots, B_N$ are the independent indicator variables of the successes, $F_N = \frac{B_1+\cdots+B_N}{N}$. With $E(B_i) = p$ & $V(B_i) = p(1-p)$,

$$E(F_N) = \frac{Np}{N} = \frac{3}{10}, \qquad V(F_N) = \frac{Np(1-p)}{N^2} = \frac{\frac{21}{100}}{N}.$$

Independence is required for the addition of variances, not for the linearity of expectation.

8. [1 pt] Bienaymé-Chebyshev applied to F_N gives

$$P(|F_N - p| \geq 0{,}02) \leq \frac{p(1-p)}{N(0{,}02)^2}.$$

To ensure that this probability is at most 0.05, it is enough to

$$N \geq \frac{\frac{21}{100}}{0{,}05 \times (0{,}02)^2} = 10500.$$

By complement, we then have $P(|F_N - p| < 0{,}02) \geq 0{,}95$. The size 10500 is sufficient; the inequality being an upper bound, it does not necessarily give the minimum size for the exact prob.

Final mock exam · A

PAPER RESERVED FOR ASSESSMENT

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x)=\frac{4+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)
3. State the maximum value of f, then solve $f(x)=0$. (0.5 pt)
4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)
5. Prove that the equation $f(x)=\frac{1}{2}e^3$ has exactly two solutions in $]0;+\infty[$. (0.5 pt)
6. Determine an antiderivative of f on $]0;+\infty[$. (0.5 pt)
7. Calculate the mean value of f on $[1;e]$. (0.5 pt)
8. Independently of the previous questions, solve $\ln(x-1)+\ln(x+1)=\ln(15)$, after specifying the set on which this equation is defined. (1 pt)

Exercise 2 · A variable trigonometric area — 4 points

For $a\in[0;\pi/2]$, set $A(a)=\int_0^a \sin x\cos x\,\mathrm{d}x$.

1. Justify that $A(a)$ is a geometric area. (1 pt)
2. Calculate $A(a)$ and $A(\pi/2)$. (1 pt)
3. Show that this total area is split in two by the vertical line x = $\pi/4$. (1 pt)
4. Determine the maximum of $x\mapsto \sin x\cos x$ on $[0;\pi/2]$ justifying. (1 pt)

Final mock exam · A · continued

Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 8$ & the point $M(1;0;0)$. The line d passes through M and has direction vector $\vec{n} = (1;2;2)$.

1. Give a parametric representation of d. (0.5 pt)

2. Prove that d meets $\mathcal{P}$ at a point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal projection of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deduce the distance from M to the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ through M. (0.5 pt)

7. Determine the angle between d and the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define the point N by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from N to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. (1 pt)

Exercise 4 · A guarantee and its real accuracy — 6 points

We toss n times independently, and F_n is the frequency of heads.

1. Calculate $E(F_n)$ and $V(F_n)$. (1 pt)

2. Establish a bound for $P(|F_n - \frac{1}{2}| \geq 0{,}1)$. (1 pt)

3. Determine the smallest size for which this bound guarantees a probability of at most 0 ,05. (1 pt)

4. For $n = 100$, write the exact probability of a deviation of at least 0 ,1 as a binomial sum. (1 pt)

5. A student concludes "500 trials are needed and each extra trial improves the frequency ". Discuss these two statements separately. (2 pts)

Solutions · Final mock exam · A

Exercise 1 · A logarithmic function

1. [0.5 pt] At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+: limit $-\infty$; at $+\infty$, limit 0 by comparing growth rates.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (4 + \ln x)}{x^2} = \frac{-3 - \ln x}{x^2}.$$

The denominator is positive. The numerator is zero at $s = e^{-3}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f strictly increases on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [0.5 pt] The maximum is reached only at $s = e^{-3}$ and is

$$f(s) = \frac{4 - 3}{e^{-3}} = e^3.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $4 + \ln x = 0$, i.e. $x = e^{-4}$.

4. [1 pt] We differentiate $(-3 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-3 - \ln x)x^{-3} = \frac{2\ln x + 5}{x^3}.$$

The numerator is zero at $t = e^{-\frac{5}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at t and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Let $m = \frac{e^3}{2}$ & $s = e^{-3}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and with value 2 m at s: the corollary of the intermediate value theorem gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from 2 m to 0 $< m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. [0.5 pt] Let $F(x) = 4\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{4}{x} + \frac{\ln x}{x} = f(x)$; F is an antiderivative on $]0, +\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{9}{2}$ & $F(1) = 0$. The integral therefore equals $\frac{9}{2}$, and the average value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{9}{2(e-1)}.$$

8. [1 pt] Domain $x > 1$; $x^2 - 1 = 15$, so $x = 4$, the negative root being excluded.

Exercise 2 · A variable trigonometric area

1. Both factors are positive or zero on the interval, so the integral measures the area under the curve.

2. An antiderivative is $\frac{1}{2}\sin^2 x$ by differentiation; $A(a) = \frac{1}{2}\sin^2 a$ and $A(\pi/2) = 1/2$.

3. $A(\pi/4) = \frac{1}{2}(\sqrt{2}/2)^2 = 1/4$, half of 1 $/2$. The strict monotonicity of sin over this interval ensures the uniqueness of the division.

4. The derivative is $\cos^2 x - \sin^2 x = \cos 2x$, positive before $\pi/4$, negative after. Maximum 1 $/2$ at $\pi/4$.

Exercise 3 · Line, plane and projection

1. [0.5 pt] $(x, y, z) = (1 + t, 2t, 2t)$, $t \in \mathbb{R}$.

2. [0.5 pt] $1 + 9t = 8$ gives $t = \frac{7}{9}$; $H = (\frac{16}{9}, \frac{14}{9}, \frac{14}{9})$.

3. [0.5 pt] The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Since $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{7}{9}\vec{n}$, H is the orthogonal projection of M onto the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1 + 4 + 4} = 3$. Hence

$$d(M, \mathcal{P}) = MH = |\frac{7}{9}| 3 = \frac{7}{3}.$$

4. [1 pt] Let $y = z = s$; $x = 8 - 4s$. Hence $(x, y, z) = (8, 0, 0) + s(-4, 1, 1)$.

5. [0.5 pt] Yes, its last two coordinates are equal. H is on the line and $\overrightarrow{MH}$ is orthogonal to every direction in the plane, hence to this line. Distance $\frac{7}{3}$.

6. [0.5 pt] x $+ 2y + 2z = 1$; same normal vector and passing through M.

7. [0.5 pt] $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. [1 pt] $N = M + \vec{n}$ is on the same normal line as M ; its projection onto $\mathcal{P}$ is therefore also H. Now $H \in \mathcal{P} \cap \mathcal{Q}$. For every P of this line, $NP^2 = NH^2 + HP^2$, therefore its minimum distance to N is NH. Finally,

$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{7}{9} - 1| \, \|\vec{n}\| = \frac{2}{3}.$$

Exercise 4 · A guarantee and its real accuracy

1. Mean of n independent Bernoullis: $E(F_n) = 1/2$, $V(F_n) = 1/(4n)$.

2. Chebyshev with $a = 0.1$: probability at most $1/(4n \times 0{,}01) = 25/n$ (and always at most 1).

3. $25/n \leq 0{,}05 \iff n \geq 500$. First integer for this guarantee: 500.

4. For $X = 100F_{100} \sim \mathcal{B}(100; 1/2)$, the event is $X \leq 40$ or $X \geq 60$. By symmetry, probability $2\sum_{k=0}^{40} \binom{100}{k} 2^{-100}$.

5. 500 is sufficient by an upper bound, not proved necessary (1 pt). A trial may move the frequency away from 1/2: after heads then tails it is 1/2, after another heads 2/3 (1 pt).

Final mock exam · B

PAPER RESERVED FOR ASSESSMENT

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x)=\frac{3+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. State the maximum value of f, then solve $f(x)=0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)

5. Prove that the equation $f(x)=\frac{1}{2}e^2$ has exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine an antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the mean value of f on $[1;e]$. (0.5 pt)

8. Independently of the previous questions, solve $\ln(x-2)+\ln(x+2)=\ln(32)$, after specifying the set on which this equation is defined. (1 pt)

Exercise 2 · Studying a period without guessing it — 4 points

Let $f(x)=2+\sin(2x)$ on $\mathbb{R}$.

1. Prove that π is a period and that $\pi/2$ is not one. (1 pt)

2. Give the variations of f on $[0;\pi]$. (1 pt)

3. Solve $f(x)\geq 5/2$ on $[0;\pi]$. (1 pt)

4. Calculate the mean value of f on $[0;\pi]$ and explain how the oscillation cancels out. (1 pt)

Final mock exam · B · continued

Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x+2y+2z=12$ and the point $M(2;0;0)$. The line d passes through M and has direction vector $\vec{n}=(1;2;2)$.

1. Give a parametric representation of d. (0.5 pt)

2. Prove that d meets $\mathcal{P}$ at a point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal projection of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. Consider the plane $\mathcal{Q}$ with equation $y=z$. Prove that $\mathcal{P}\cap\mathcal{Q}$ is a line and give a parametric representation of it. (1 pt)

5. Verify that $H\in\mathcal{Q}$. Deduce the distance from M to the line $\mathcal{P}\cap\mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ through M. (0.5 pt)

7. Determine the angle between d and the line $\mathcal{P}\cap\mathcal{Q}$. (0.5 pt)

8. Define the point N by $\overrightarrow{MN}=\vec{n}$. Calculate the distances from N to the plane $\mathcal{P}$ and to the line $\mathcal{P}\cap\mathcal{Q}$, then compare them. (1 pt)

Exercise 4 · Stable mean or dependent copies — 6 points

A variable X takes the values 1 and 3 with probabilities 1 /2. For $n\geq 1$, M_n is the mean of n independent copies of X.

1. Calculate $E(X)$ and $V(X)$. (1 pt)

2. Calculate $E(M_n)$, $V(M_n)$ and bound above $P(|M_n-2|\geq 0{,}2)$. (1 pt)

3. Determine the smallest sample size guaranteeing, by this bound, $P(|M_n-2|<0{,}2)\geq 0{,}99$. (1 pt)

4. The independent copies are replaced by $X_1=\cdots=X_n=X$. Calculate the variance of their mean and the probability of a deviation strictly less than 0.2. (1 pt)

5. In the independent model, express M_n by means of a binomial variable, then write a event that allows this probability to be calculated exactly. (2 pts)

Solutions · Final mock exam · B

Exercise 1 · A logarithmic function

1. [0.5 pt] At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+: limit $-\infty$; at $+\infty$, limit 0 by comparing growth rates.

2. [0.5 pt] By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (3+\ln x)}{x^2} = \frac{-2-\ln x}{x^2}.$$

The denominator is positive. The numerator is zero at $s = e^{-2}$, is positive for $0 < x < s$ and negative for $x > s$. Thus f strictly increases on $]0, s]$, then decreases strictly on $[s, +\infty[$.

3. [0.5 pt] The maximum is reached only at $s = e^{-2}$ and is

$$f(s) = \frac{3-2}{e^{-2}} = e^2.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $3 + \ln x = 0$, i.e. $x = e^{-3}$.

4. [1 pt] We differentiate $(-2-\ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-2-\ln x)x^{-3} = \frac{2\ln x + 3}{x^3}.$$

The numerator is zero at $t = e^{-\frac{3}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at t and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. [0.5 pt] Let $m = \frac{e^2}{2}$ & $s = e^{-2}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and with value $2 \quad m$ at s: the corollary of the intermediate value theorem gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from 2 to m to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. [0.5 pt] Let $F(x) = 3\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{3}{x} + \frac{\ln x}{x} = f(x)$; F is an antiderivative on $]0, +\infty[$.

7. [0.5 pt] With this antiderivative, $F(e) = \frac{7}{2}$ & $F(1) = 0$. The integral therefore equals $\frac{7}{2}$, and the average value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{7}{2(e-1)}.$$

8. [1 pt] Domain $x > 2$; $x^2 - 4 = 32$, so $x = 6$, the negative root being excluded.

Exercise 2 · Study a period without guessing it

1. $f(x+\pi) = 2 + \sin(2x + 2\pi) = f(x)$. But $f(\pi/4) = 3$ and $f(3\pi/4) = 1$; so $\pi/2$ is not a period.

2. $f' = 2\cos 2x$. Increasing on $[0; \pi/4]$ from 2 to 3, decreasing on $[\pi/4; 3\pi/4]$ down to 1, then increasing up to 2.

3. $\sin(2x) \geq 1/2$, with $2 \quad x \in [0; 2\pi]$, therefore $x \in [\pi/12; 5\pi/12]$.

4. An antiderivative is $2 \quad x - \frac{1}{2}\cos 2x$; integral $2 \quad \pi$, mean 2. The integral of sine over a period is 0: the excesses and shortfalls around 2 cancel out.

Exercise 3 · Line, plane and projection

1. [0.5 pt] $(x, y, z) = (2 + t, 2t, 2t)$, $t \in \mathbb{R}$.

2. [0.5 pt] $2 + 9t = 12$ gives $t = \frac{10}{9}$; $H = (\frac{28}{9}, \frac{20}{9}, \frac{20}{9})$.

3. [0.5 pt] The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Since $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{10}{9}\vec{n}$, H is the orthogonal projection of M onto the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1 + 4 + 4} = 3$. Hence

$$d(M, \mathcal{P}) = MH = |\frac{10}{9}| 3 = \frac{10}{3}.$$

4. [1 pt] Let $y = z = s$; $x = 12 - 4s$. Hence $(x, y, z) = (12, 0, 0) + s(-4, 1, 1)$.

5. [0.5 pt] Yes, its last two coordinates are equal. H is on the line and $\overrightarrow{MH}$ is orthogonal to every direction in the plane, hence to this line. Distance $\frac{10}{3}$.

6. [0.5 pt] x $+ 2y + 2z = 2$; same normal vector and passing through M.

7. [0.5 pt] $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. [1 pt] $N = M + \vec{n}$ is on the same normal line as M ; its projection onto $\mathcal{P}$ is therefore also H. Now $H \in \mathcal{P} \cap \mathcal{Q}$. For every P of this line, $NP^2 = NH^2 + HP^2$, therefore its minimum distance to N is NH. Finally,

$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{10}{9} - 1| \, \|\vec{n}\| = \frac{1}{3}.$$

Exercise 4 · Stable mean or dependent copies

1. $E(X) = 2$; $V(X) = \frac{1}{2}(1 - 2)^2 + \frac{1}{2}(3 - 2)^2 = 1$.

2. $E(M_n) = 2$, $V(M_n) = 1/n$. Chebyshev gives a bound $25/n$.

3. Require $25/n \leq 0.01$ gives $n \geq 2500$. The complementary probability is then at least 0.99.

4. The mean is exactly X, with variance 1. Both its values are at distance 1 from the expectation, so the required probability is 0.

5. If B counts the values 3, $B \sim \mathcal{B}(n; 1/2)$ and $M_n = 1 + 2B/n$ (1 pt). The event is $0{,}4n < B < 0{,}6n$, i.e. $\lfloor 0{,}4n \rfloor + 1 \leq B \leq \lceil 0{,}6n \rceil - 1$ (1 pt). The binomial probabilities over these integers can be added.